

Data Analysis - EAMT Statistical Consulting

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Loading Required Packages and Dataset

```
# Required Packages
```

```
library(readr)
```

```
library(ggplot2)
```

```
## Warning: il pacchetto 'ggplot2' è stato creato con R versione 4.4.3
```

```
library(dplyr)
```

```
##
```

```
## Caricamento pacchetto: 'dplyr'
```

```
## I seguenti oggetti sono mascherati da 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## I seguenti oggetti sono mascherati da 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
library(see)
```

```
library(patchwork)
```

```
library(lme4)
```

```
## Warning: il pacchetto 'lme4' è stato creato con R versione 4.4.3
```

```
## Caricamento del pacchetto richiesto: Matrix
```

```
##
```

```
## Caricamento pacchetto: 'Matrix'
```

```
## I seguenti oggetti sono mascherati da 'package:tidyr':
```

```
##
```

```
##      expand, pack, unpack
```

```
library(lmerTest)
```

```
## Warning: il pacchetto 'lmerTest' è stato creato con R versione 4.4.3
```

```
##
```

```
## Caricamento pacchetto: 'lmerTest'
```

```
## Il seguente oggetto è mascherato da 'package:lme4':
```

```
##
```

```
##      lmer
```

```
## Il seguente oggetto è mascherato da 'package:stats':
```

```
##
```

```
##      step
```

```
library(lattice)
```

```
library(sjPlot)
```

```
## Warning: il pacchetto 'sjPlot' è stato creato con R versione 4.4.3
```

```
library(mice)
```

```
## Warning: il pacchetto 'mice' è stato creato con R versione 4.4.3
```

```
##
```

```
## Caricamento pacchetto: 'mice'
```

```
## Il seguente oggetto è mascherato da 'package:stats':
```

```
##
```

```
##      filter
```

```
## I seguenti oggetti sono mascherati da 'package:base':
```

```
##
```

```
##      cbind, rbind
```

```
library(performance)
```

```
## Warning: il pacchetto 'performance' è stato creato con R versione 4.4.3
```

```
library(ggeffects)
```

```
## Warning: il pacchetto 'ggeffects' è stato creato con R versione 4.4.3
```

```
library(glmmTMB)
```

```
## Warning: il pacchetto 'glmmTMB' è stato creato con R versione 4.4.3
```

```
library(DHARMA)
```

```
## Warning: il pacchetto 'DHARMA' è stato creato con R versione 4.4.3
```

```
## This is DHARMA 0.4.7. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')
```

```
library(forcats)
```

```
# Loading dataset
```

```
data <- read_csv("data/data.csv")
```

```
## Rows: 496 Columns: 20
```

```
## -- Column specification -----  
## Delimiter: ","  
## chr (11): TEXT, text_name, domain, prc_sugg_accepted, tags_edited, How diffi...  
## dbl (9): PET, condition, Sent_id, keystrokes, time, num_major, num_minor, n...  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Data Wrangling

```
# Factorization and Ordering of Categorical Variables
```

```
data$`How good was the quality of MT?` <- factor(  
  data$`How good was the quality of MT?`,  
  levels = c("Insufficient", "Fair", "Good", "Very good"),  
  ordered = TRUE  
)
```

```
data$`How difficult to translate was the source text?` <- factor(  
  data$`How difficult to translate was the source text?`,  
  levels = c("Very difficult", "Somewhat difficult", "Neutral",  
             "Somewhat easy", "Very easy"),  
  ordered = TRUE  
)
```

```
data$`How useful were the error annotations?` <- factor(  
  data$`How useful were the error annotations?`,  
  levels = c("nan", "Extremely not useful", "Somewhat not useful",  
             "Neutral", "Somewhat useful", "Extremely useful"),  
  ordered = TRUE  
)
```

```
data$`How useful were the error annotations?` <- factor(  
  data$`How useful were the error annotations?`,  
  levels = c("nan", "Extremely not useful", "Somewhat not useful",  
             "Neutral", "Somewhat useful", "Extremely useful"),  
  ordered = TRUE  
)
```

```

    ordered = TRUE
  )

data$`How accurate were the error annotations?` <- factor(
  data$`How accurate were the error annotations?`,
  levels = c("nan", "Somewhat inaccurate", "Not accurate nor inaccurate",
             "Somewhat accurate"),
  ordered = TRUE
)

data$`How useful were the translation suggestions?` <- factor(
  data$`How useful were the translation suggestions?`,
  levels = c("nan", "Neutral", "Somewhat useful",
             "Extremely useful"),
  ordered = TRUE
)

data$`How accurate were the translation suggestions?` <- factor(
  data$`How accurate were the translation suggestions?`,
  levels = c("nan", "Somewhat inaccurate", "Not accurate nor inaccurate",
             "Somewhat accurate", "Very accurate"),
  ordered = TRUE
)

# Creating tables per variable
data_categorical <- data[,-c(7,8,9,10,11,12,13)]
lapply(data_categorical[, -7], table)

```

```

## $PET
##
##  1  2  3  4  5  6  7  8
## 62 62 62 62 62 62 62 62
##
## $TEXT
##
## Text 1 Text 2 Text 3 Text 4 Text 5 Text 6 Text 7 Text 8
##    58    66    58    66    58    66    58    66
##
## $text_name
##
##                doc13                doc18                doc20
##                64                56                80
##                doc34                news_pa.5274                news_scotsman.87462
##                64                48                56
## news_seattle_times.799809 news_seattle_times.800119
##                64                64
##
## $condition
##
##  1  2  3  4
## 124 124 124 124
##
## $domain

```

```

##
## medical      news
##      264      232
##
## $Sent_id
##
##  1  2  3  4  5  6  7  8  9 10
## 64 64 64 64 64 64 56 40  8  8
##
## $'How difficult to translate was the source text?'
##
##      Very difficult Somewhat difficult          Neutral          Somewhat easy
##              59              140              129              142
##      Very easy
##              26
##
## $'How good was the quality of MT?'
##
## Insufficient          Fair          Good          Very good
##              28              188              214              66
##
## $'How useful were the error annotations?'
##
##              nan Extremely not useful Somewhat not useful
##              124              18              90
##              Neutral          Somewhat useful          Extremely useful
##              54              204              6
##
## $'How accurate were the error annotations?'
##
##              nan          Somewhat inaccurate
##              124              89
## Not accurate nor inaccurate          Somewhat accurate
##              76              207
##
## $'How useful were the translation suggestions?'
##
##              nan          Neutral Somewhat useful Extremely useful
##              372              8              94              22
##
## $'How accurate were the translation suggestions?'
##
##              nan          Somewhat inaccurate
##              372              10
## Not accurate nor inaccurate          Somewhat accurate
##              31              69
##              Very accurate
##              14

```

```

data$PET <- as.factor(data$PET)
data$condition <- as.factor(data$condition)
data$domain <- as.factor(data$domain)
data$text_name <- as.factor(data$text_name)
data$Sent_id <- as.factor(data$Sent_id)

```

```
data$MT_Quality <- as.numeric(data$`How good was the quality of MT?`)
data$Difficulty <- as.numeric(data$`How difficult to translate was the source text?`)

data[data == "N/A"] <- NA
```

Tables For Insights

```
table(data$condition, data$text_name)
```

```
##
##      doc13 doc18 doc20 doc34 news_pa.5274 news_scotsman.87462
##  1      16      14      20      16              12              14
##  2      16      14      20      16              12              14
##  3      16      14      20      16              12              14
##  4      16      14      20      16              12              14
##
##      news_seattle_times.799809 news_seattle_times.800119
##  1                      16                      16
##  2                      16                      16
##  3                      16                      16
##  4                      16                      16
```

```
table(data$PET, data$`How accurate were the translation suggestions?`)
```

```
##
##      nan Somewhat inaccurate Not accurate nor inaccurate Somewhat accurate
##  1  46                      0                      0                      16
##  2  44                      0                      0                      18
##  3  48                      0                      7                      7
##  4  48                      0                      0                      0
##  5  46                      0                     16                      0
##  6  44                     10                      8                      0
##  7  48                      0                      0                      14
##  8  48                      0                      0                      14
##
##      Very accurate
##  1              0
##  2              0
##  3              0
##  4             14
##  5              0
##  6              0
##  7              0
##  8              0
```

```
table(data$PET, data$`How useful were the translation suggestions?`)
```

```
##
```

```
##      nan Neutral Somewhat useful Extremely useful
## 1 46      0      16      0
## 2 44      0      18      0
## 3 48      0      14      0
## 4 48      0       6      8
## 5 46      0      16      0
## 6 44      8      10      0
## 7 48      0      14      0
## 8 48      0       0     14
```

```
table(data$PET, data$`How good was the quality of MT?`)
```

```
##
##      Insufficient Fair Good Very good
## 1      7 23 14 18
## 2      0 36 26  0
## 3      0 36 26  0
## 4      7  0 17 38
## 5     14  6 32 10
## 6      0 33 29  0
## 7      0 16 46  0
## 8      0 38 24  0
```

```
table(data$`How good was the quality of MT?`, data$condition)
```

```
##
##           1  2  3  4
## Insufficient 7 21  0  0
## Fair        38 45 43 62
## Good        79 50 37 48
## Very good   0  8 44 14
```

```
table(data$text_name, data$`How good was the quality of MT?`)
```

```
##
##           Insufficient Fair Good Very good
## doc13           0 24 32 8
## doc18          21 14 21 0
## doc20           0 10 50 20
## doc34           0  8 48 8
## news_pa.5274    0 24 18 6
## news_scotsman.87462 7 28 21 0
## news_seattle_times.799809 0 40 8 16
## news_seattle_times.800119 0 40 16 8
```

```
data %>%
  group_by(`How good was the quality of MT?`) %>%
  summarise(mean_time = mean(time, na.rm = TRUE),
            mean_keys = mean(keystrokes, na.rm = TRUE),
            n = n())
```

```
## # A tibble: 4 x 4
##   'How good was the quality of MT?' mean_time mean_keys    n
##   <ord>                        <dbl>      <dbl> <int>
## 1 Insufficient                94.4        52     28
## 2 Fair                       93.5        59.9    188
## 3 Good                       76.5        38.0    214
## 4 Very good                   47.3        33.2     66
```

```
data %>%
  mutate(prc_sugg_accepted = as.numeric(as.character(prc_sugg_accepted))) %>%
  group_by(condition) %>%
  summarise(mean_time = mean(time),
            mean_keys = mean(keystrokes),
            mean_prc_sugg_acc = mean(prc_sugg_accepted, na.rm = TRUE),
            n = n())
```

```
## # A tibble: 4 x 5
##   condition mean_time mean_keys mean_prc_sugg_acc    n
##   <fct>      <dbl>      <dbl>          <dbl> <int>
## 1 1          81.9        37.9            NaN    124
## 2 2          80.5        49.4            NaN    124
## 3 3          76.4        50.2             0    124
## 4 4          81.4        48.2           49.9    124
```

```
table(data$condition, data$`How difficult to translate was the source text?`)
```

```
##
##   Very difficult Somewhat difficult Neutral Somewhat easy Very easy
## 1           22           46          32           8          16
## 2            8           36          40          40           0
## 3            7           35          25          57           0
## 4           22           23          32          37          10
```

```
table(data$condition, data$num_major)
```

```
##
##      0  1  2  3  4  5  8
## 1 124  0  0  0  0  0  0
## 2  76 28 14  2  0  2  2
## 3 114  8  0  0  2  0  0
## 4 114  8  0  0  2  0  0
```

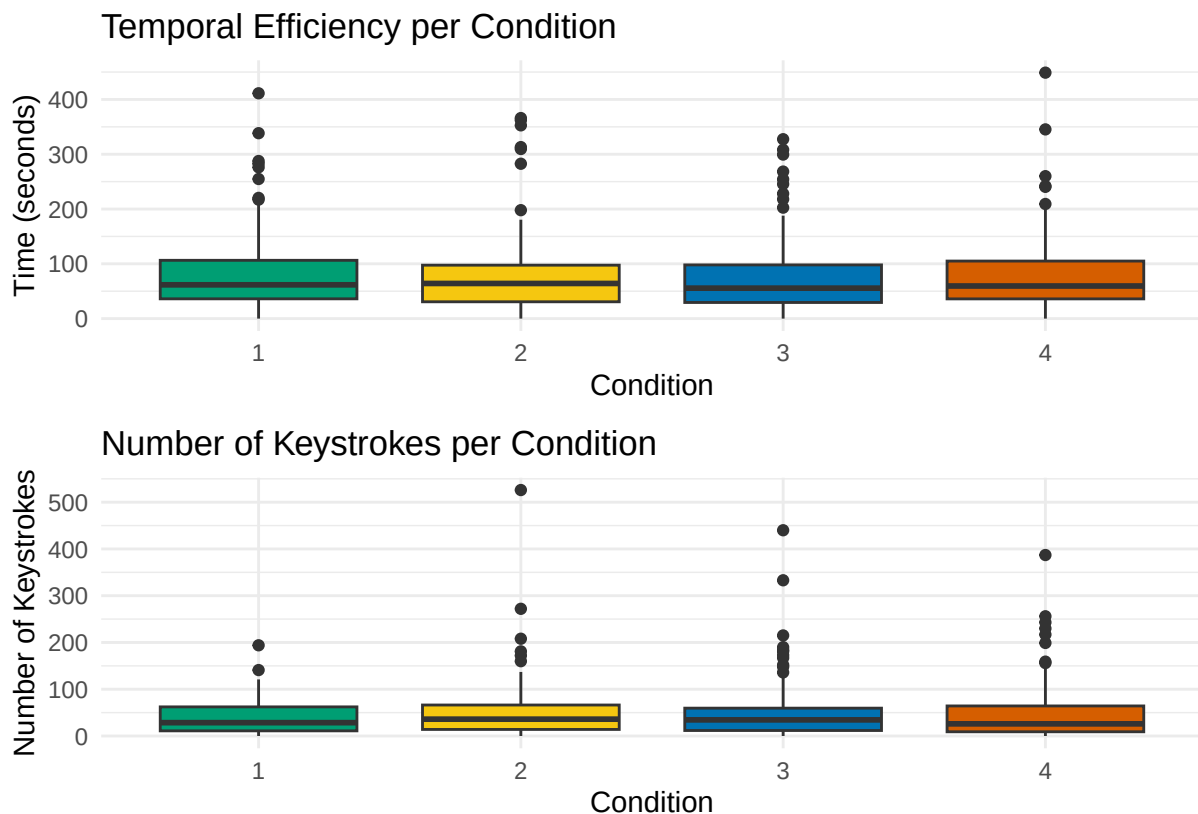
```
table(data$condition, data$num_minor)
```

```
##
##      0  1  2  3  4  5  6  7
## 1 124  0  0  0  0  0  0  0
## 2  30 24 28  8 18  8  2  6
## 3  30 18 30 12 12 12  4  6
## 4  30 18 30 12 12 12  4  6
```

Figures

Boxplots of Keystrokes and Time per Condition

```
timebp <- ggplot(data = data, mapping = aes(x = as.factor(condition),  
                                             y = time, fill = as.factor(condition))) +  
  geom_boxplot() +  
  theme_minimal() +  
  scale_fill_okabeito(order = 3:6) +  
  labs(title = "Temporal Efficiency per Condition",  
       x = "Condition",  
       y = "Time (seconds)") +  
  theme(legend.position = "none")  
  
keybp <- ggplot(data = data, mapping = aes(x = as.factor(condition),  
                                           y = keystrokes, fill = as.factor(condition))) +  
  geom_boxplot() +  
  theme_minimal() +  
  scale_fill_okabeito(order = 3:6) +  
  labs(title = "Number of Keystrokes per Condition",  
       x = "Condition",  
       y = "Number of Keystrokes") +  
  theme(legend.position = "none")  
  
timebp/keybp
```



Boxplots of Keystrokes and Time per Condition for each Professional Translator

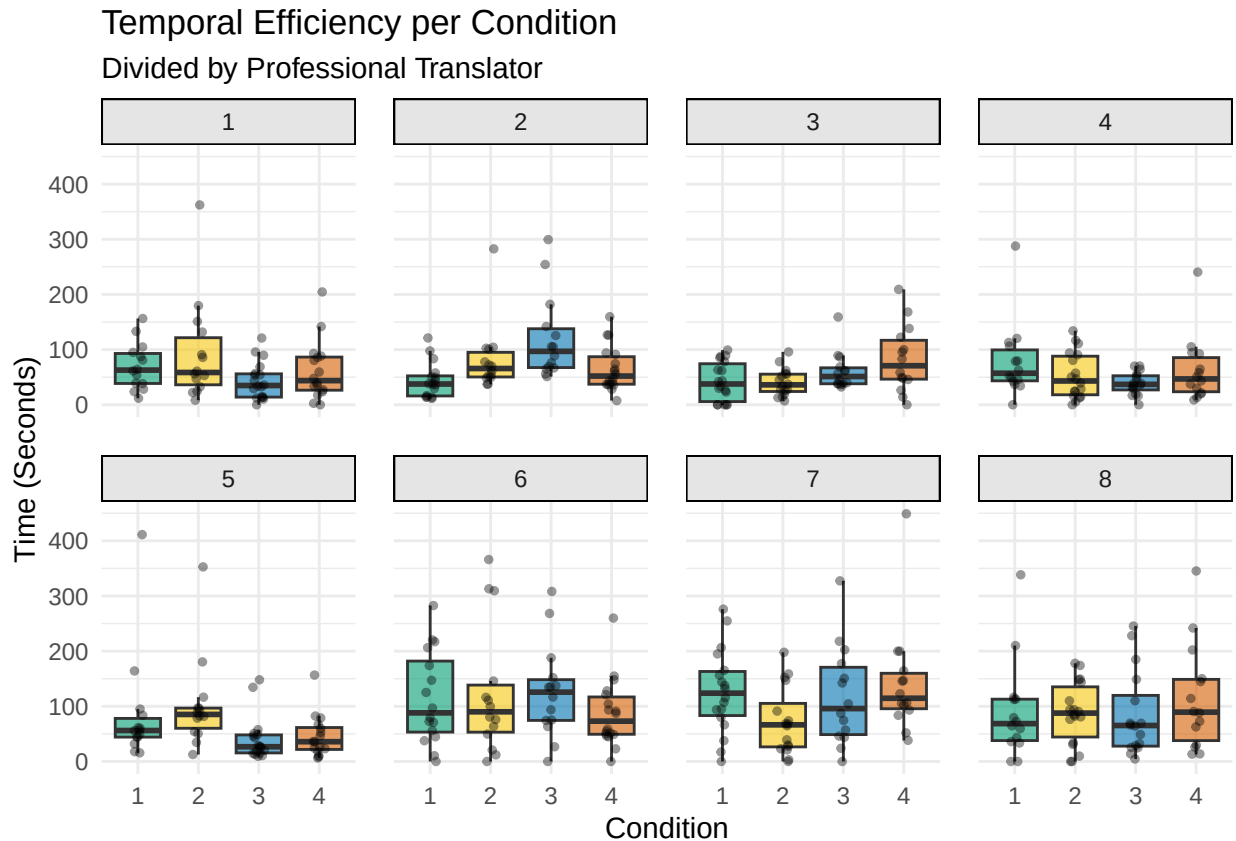
```
timebp_faceted <- ggplot(data = data, aes(x = as.factor(condition), y = time)) +
  geom_boxplot(aes(fill = as.factor(condition)), alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.4, size = 1) +
  facet_wrap(~ PET, ncol = 4) +

  theme_minimal() +
  scale_fill_okabeito(order = 3:6) +

  labs(title = "Temporal Efficiency per Condition",
       subtitle = "Divided by Professional Translator",
       x = "Condition",
       y = "Time (Seconds)") +

  theme(legend.position = "none",
       strip.background = element_rect(fill = "gray90"),
       panel.spacing = unit(1, "lines"))

print(timebp_faceted)
```



```
keybp_faceted <- ggplot(data = data, aes(x = as.factor(condition), y = keystrokes)) +
  geom_boxplot(aes(fill = as.factor(condition)), alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.4, size = 1) +
```

```

facet_wrap(~ PET, ncol = 4) +

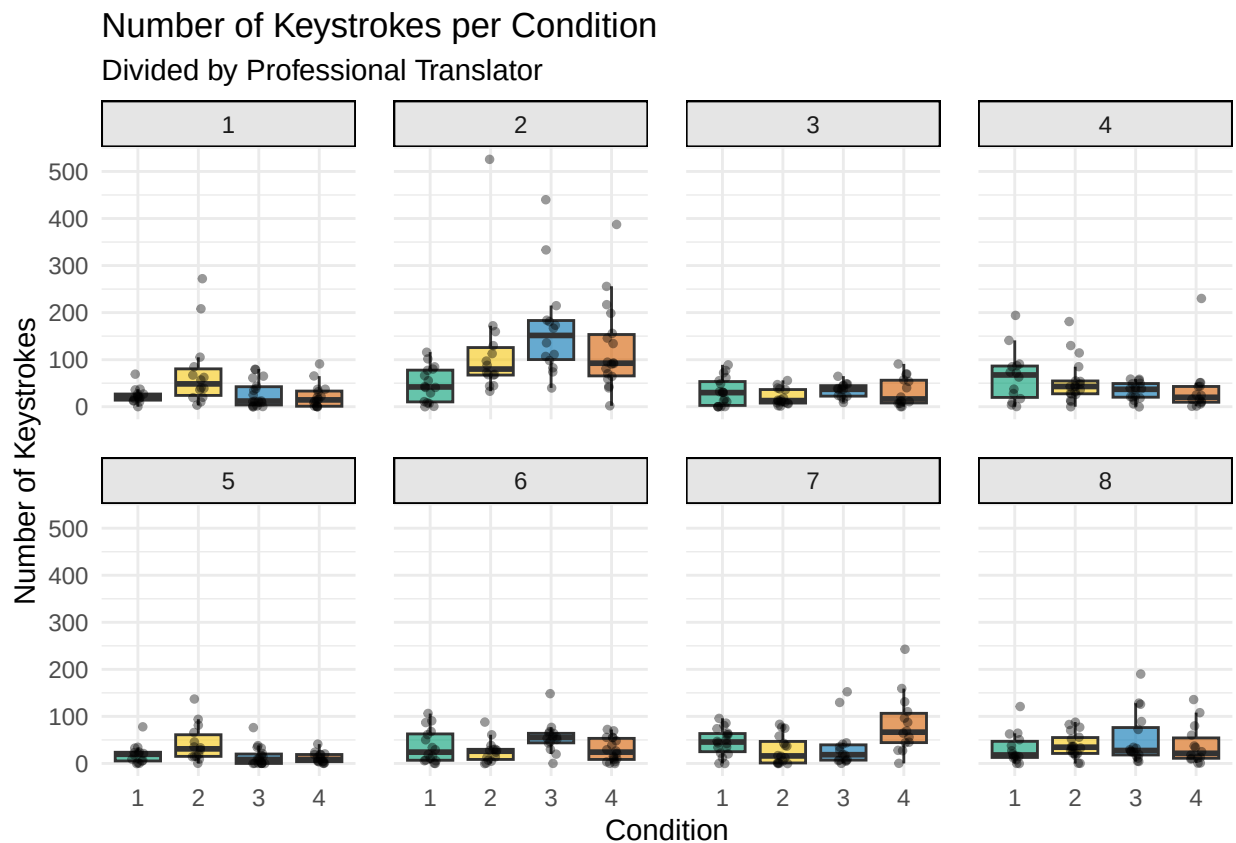
theme_minimal() +
scale_fill_okabeito(order = 3:6) +

labs(title = "Number of Keystrokes per Condition",
      subtitle = "Divided by Professional Translator",
      x = "Condition",
      y = "Number of Keystrokes") +

theme(legend.position = "none",
      strip.background = element_rect(fill = "gray90"),
      panel.spacing = unit(1, "lines"))

print(keybp_faceted)

```



Boxplots of Keystrokes and Time per Domain

```

domainboxplots <- ggplot(data = data, mapping = aes(x = as.factor(domain),
                                                    y = log(time), fill = as.factor(domain))) +

geom_boxplot() +
theme_minimal() +
scale_fill_okabeito(order = 2:3) +

```

```

labs(title = "Temporal Efficiency per Domain",
     x = "Domain",
     y = "Log Time (seconds)") +
theme(legend.position = "none")

domainboxplots2 <- ggplot(data = data, mapping = aes(x = as.factor(domain),
                                                    y = log(keystrokes), fill = as.factor(domain))) +

  geom_boxplot() +
  theme_minimal() +
  scale_fill_okabeito(order = 2:3) +
  labs(title = "Number of Keystrokes per Domain",
       x = "Domain",
       y = "Log Number of Keystrokes") +
  theme(legend.position = "none")

domainboxplots + domainboxplots2

```

```

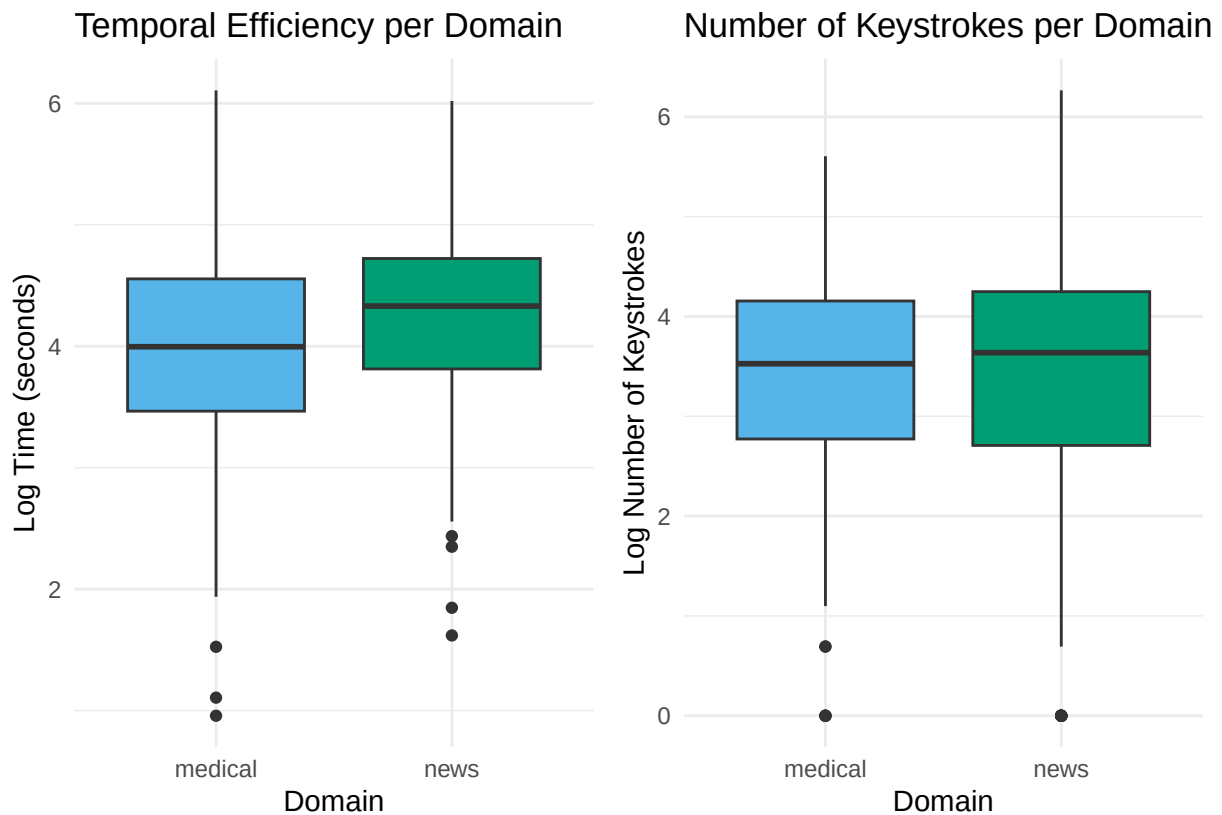
## Warning: Removed 22 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```

## Warning: Removed 45 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



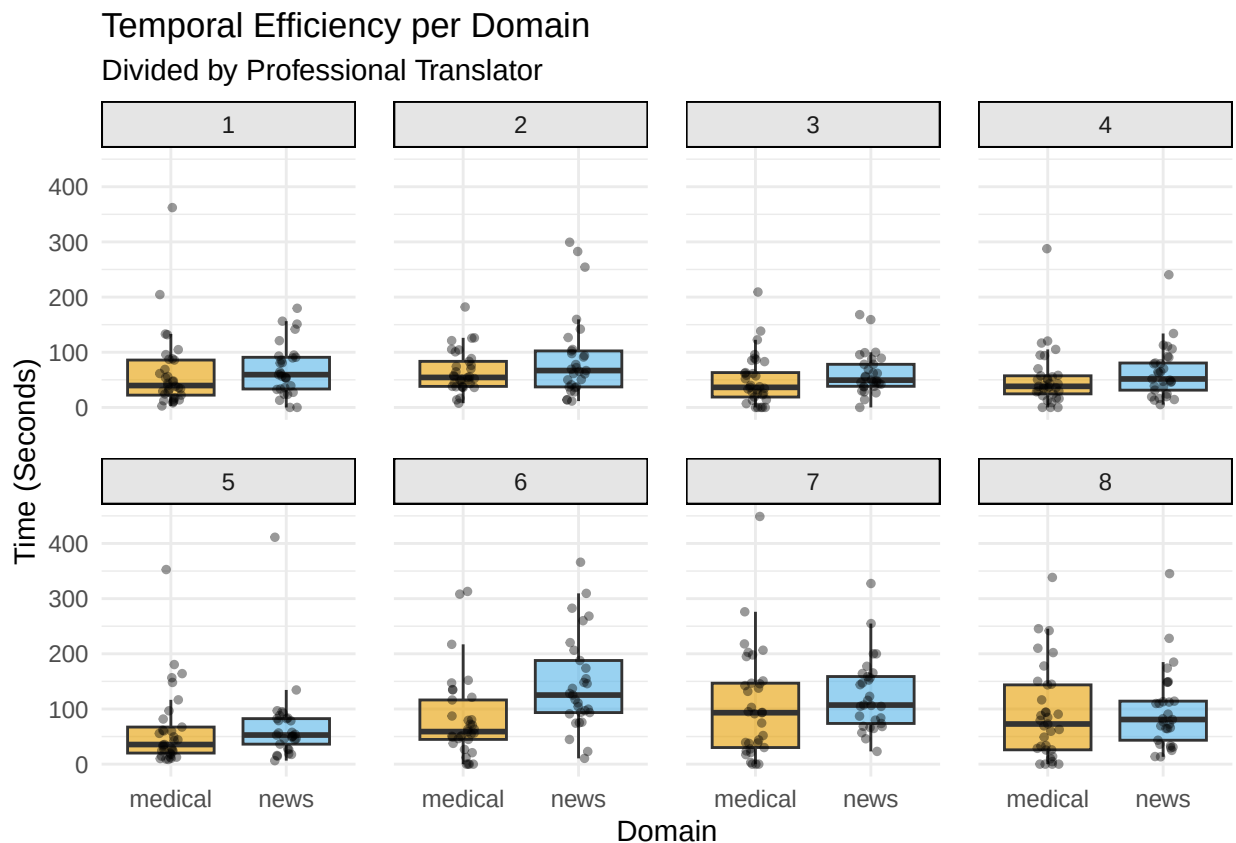
Boxplots of Keystrokes and Time per Domain per Professional Translator

```
domain_time_faceted <- ggplot(data = data, aes(x = as.factor(domain), y = time)) +
  geom_boxplot(aes(fill = as.factor(domain)), alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.4, size = 1) +
  facet_wrap(~ PET, ncol = 4) +

  theme_minimal() +
  scale_fill_okabeito(order = 1:2) +

  labs(title = "Temporal Efficiency per Domain",
       subtitle = "Divided by Professional Translator",
       x = "Domain",
       y = "Time (Seconds)") +

  theme(legend.position = "none",
       strip.background = element_rect(fill = "gray90"),
       panel.spacing = unit(1, "lines"))
domain_time_faceted
```



```
domain_key_faceted <- ggplot(data = data, aes(x = as.factor(domain), y = keystrokes)) +
  geom_boxplot(aes(fill = as.factor(domain)), alpha = 0.6, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.4, size = 1) +
  facet_wrap(~ PET, ncol = 4) +
```

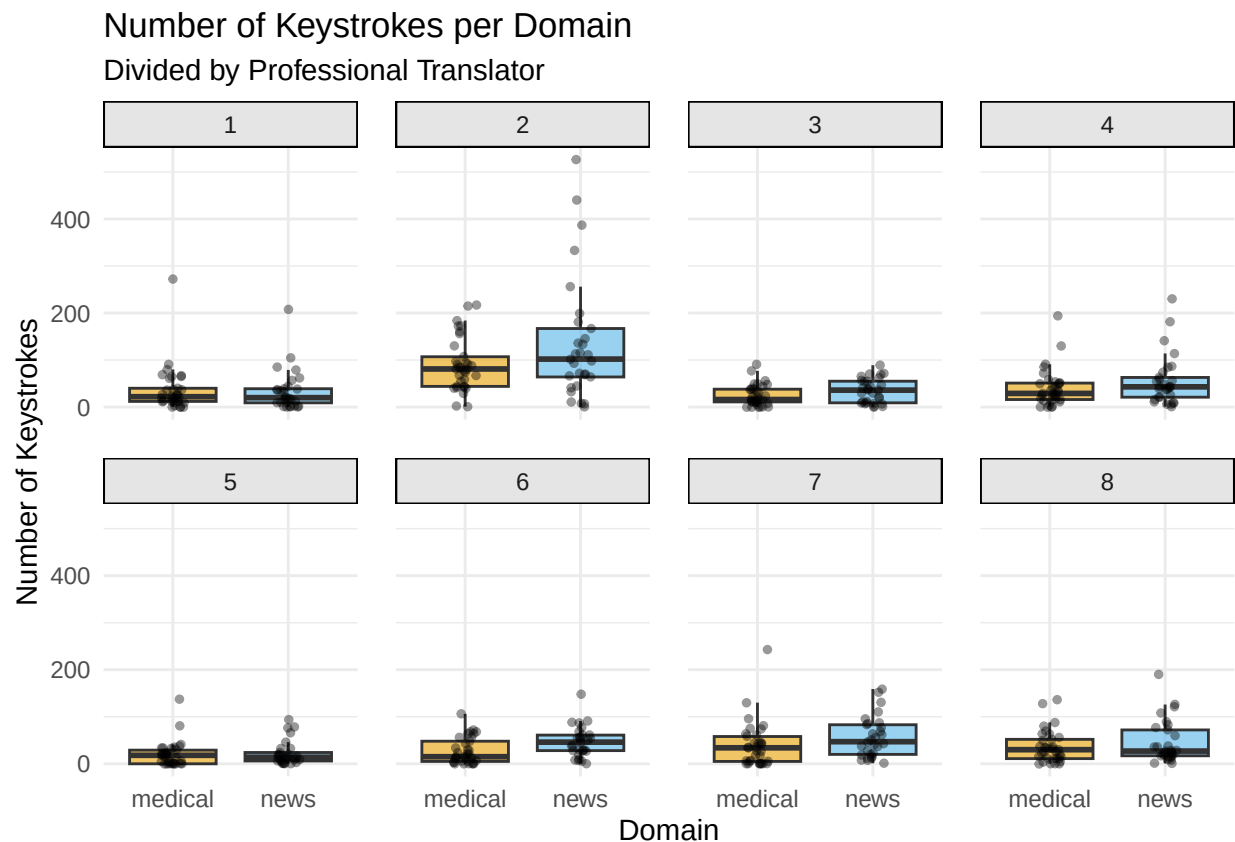
```

theme_minimal() +
scale_fill_okabeito(order = 1:2) +

labs(title = "Number of Keystrokes per Domain",
      subtitle = "Divided by Professional Translator",
      x = "Domain",
      y = "Number of Keystrokes") +

theme(legend.position = "none",
      strip.background = element_rect(fill = "gray90"),
      panel.spacing = unit(1, "lines"))
domain_key_faceted

```



Boxplots for each text

```

texts_boxplots_time <- ggplot(data, aes(x = text_name, y = log(time + 1), fill = as.factor(condition)))
  geom_boxplot(alpha = 0.7, outlier.size = 1) +
  facet_wrap(~ domain, scales = "free_x") +

  theme_minimal() +
  scale_fill_okabeito(order = 3:6) +

  labs(title = "Time distribution divided by Document and Condition",

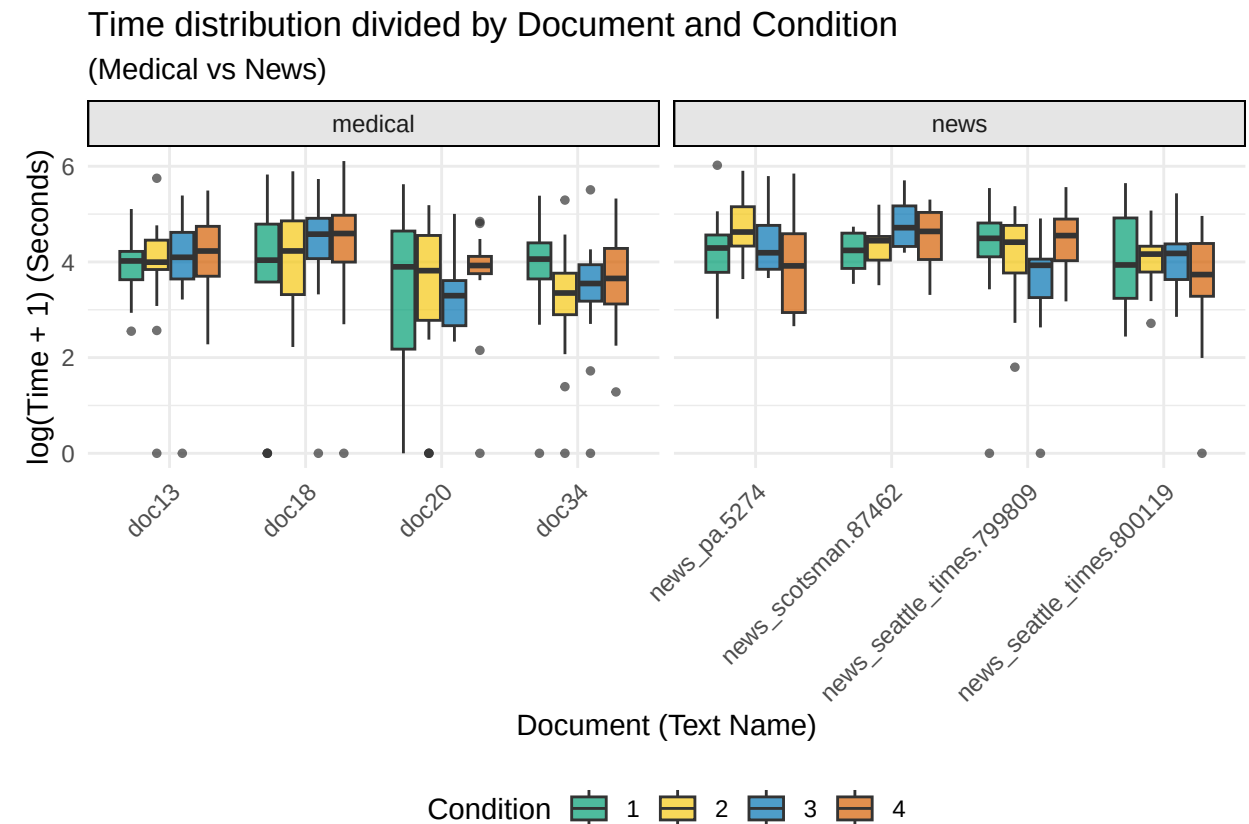
```

```

    subtitle = "(Medical vs News)",
    x = "Document (Text Name)",
    y = "log(Time + 1) (Seconds)",
    fill = "Condition") +

    theme(axis.text.x = element_text(angle = 45, hjust = 1),
          legend.position = "bottom",
          strip.background = element_rect(fill = "gray90"))
print(texts_boxplots_time)

```



```

texts_boxplots_keys <- ggplot(data, aes(x = text_name, y = log(keystrokes+1), fill = as.factor(condition))) +
  geom_boxplot(alpha = 0.7, outlier.size = 1) +
  facet_wrap(~ domain, scales = "free_x") +

  theme_minimal() +
  scale_fill_okabeito(order = 3:6) +

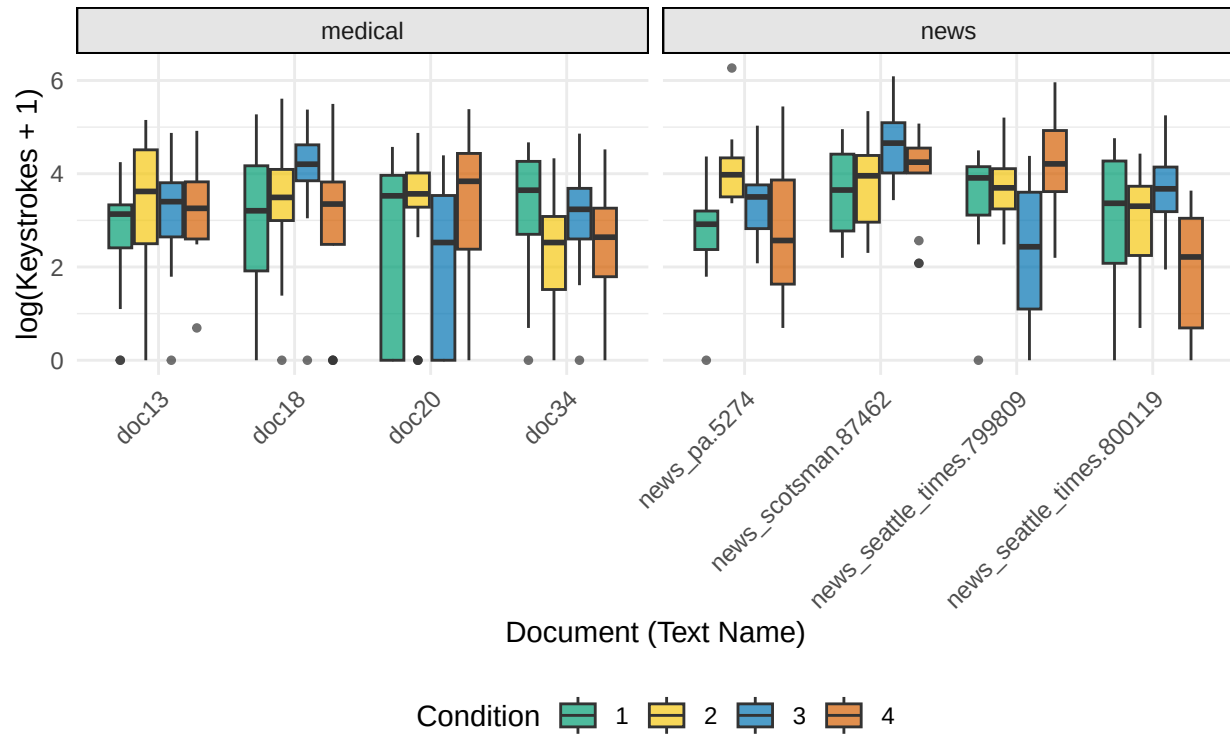
  labs(title = "Keystrokes distribution divided by Document and Condition",
       subtitle = "(Medical vs News)",
       x = "Document (Text Name)",
       y = "log(Keystrokes + 1)",
       fill = "Condition") +

  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "bottom",

```

```
strip.background = element_rect(fill = "gray90"))
print(texts_boxplots_keys)
```

Keystrokes distribution divided by Document and Condition (Medical vs News)

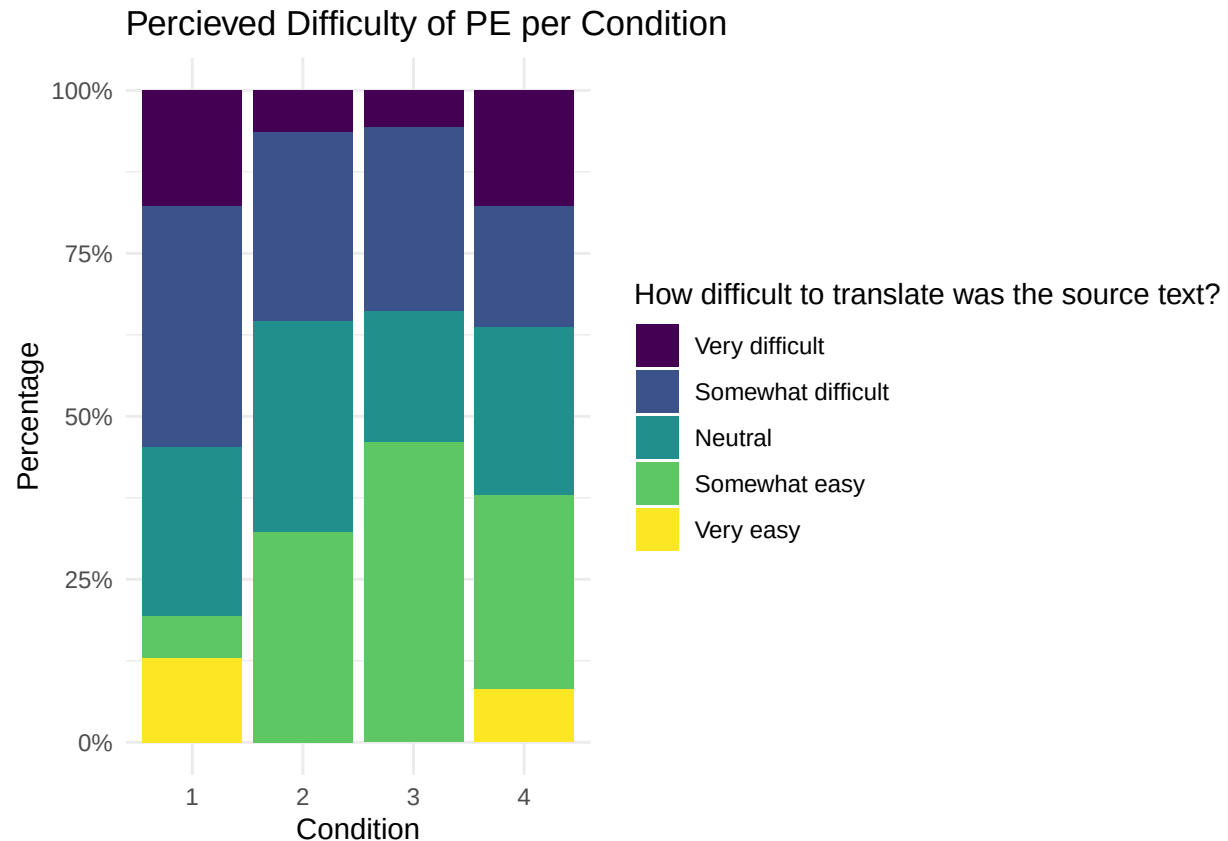


Stacked Bar Chart of Perceived Quality of MT and Perceived Difficulty

```
ggplot(data, aes(x = condition, fill = `How good was the quality of MT?`)) +
  geom_bar(position = "fill") +
  scale_y_reverse(labels = scales::percent) +
  scale_fill_manual(values = c(
    "Insufficient" = "firebrick",
    "Fair"         = "#FFF176",
    "Good"         = "#81C784",
    "Very good"    = "navy"
  )) +
  theme_minimal() +
  guides(fill = guide_legend(reverse = TRUE)) +
  labs(title = "Perceived Quality of MT per Condition",
       y = "Percentage", x = "Condition")
```



```
ggplot(data, aes(x = condition, fill = `How difficult to translate was the source text?`)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  labs(title = "Percieved Difficulty of PE per Condition",
       y = "Percentage", x = "Condition")
```



Scatter Plots of Time or Keystrokes vs Highlight ratio or Number of Characters

```
plot_data <- data %>%
  mutate(
    log_time = log(time+1),
    log_keystrokes = log(keystrokes+1),
    highlight_ratio = as.numeric(as.character(highlight_ratio)),
    condition = as.factor(condition)
  ) %>%
  filter(!is.na(highlight_ratio), !is.na(num_characters))

# 1. Time vs Highlight Ratio
p1 <- ggplot(plot_data, aes(x = highlight_ratio, y = log_time)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() + labs(title = "Time vs Highlight Ratio", y = "log(Time)")

# 2. Keystrokes vs Highlight Ratio
p2 <- ggplot(plot_data, aes(x = highlight_ratio, y = log_keystrokes)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() + labs(title = "Keystrokes vs Highlight Ratio", y = "log(Keystrokes)")

# 3. Time vs Num Characters
```

```

p3 <- ggplot(plot_data, aes(x = num_characters, y = log_time)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() + labs(title = "Time vs Sentence Length", x = "Num Characters", y = "log(Time)")

# 4. Keystrokes vs Num Characters
p4 <- ggplot(plot_data, aes(x = num_characters, y = log_keystrokes)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() + labs(title = "Keystrokes vs Sentence Length", x = "Num Characters", y = "log(Keystrokes)")

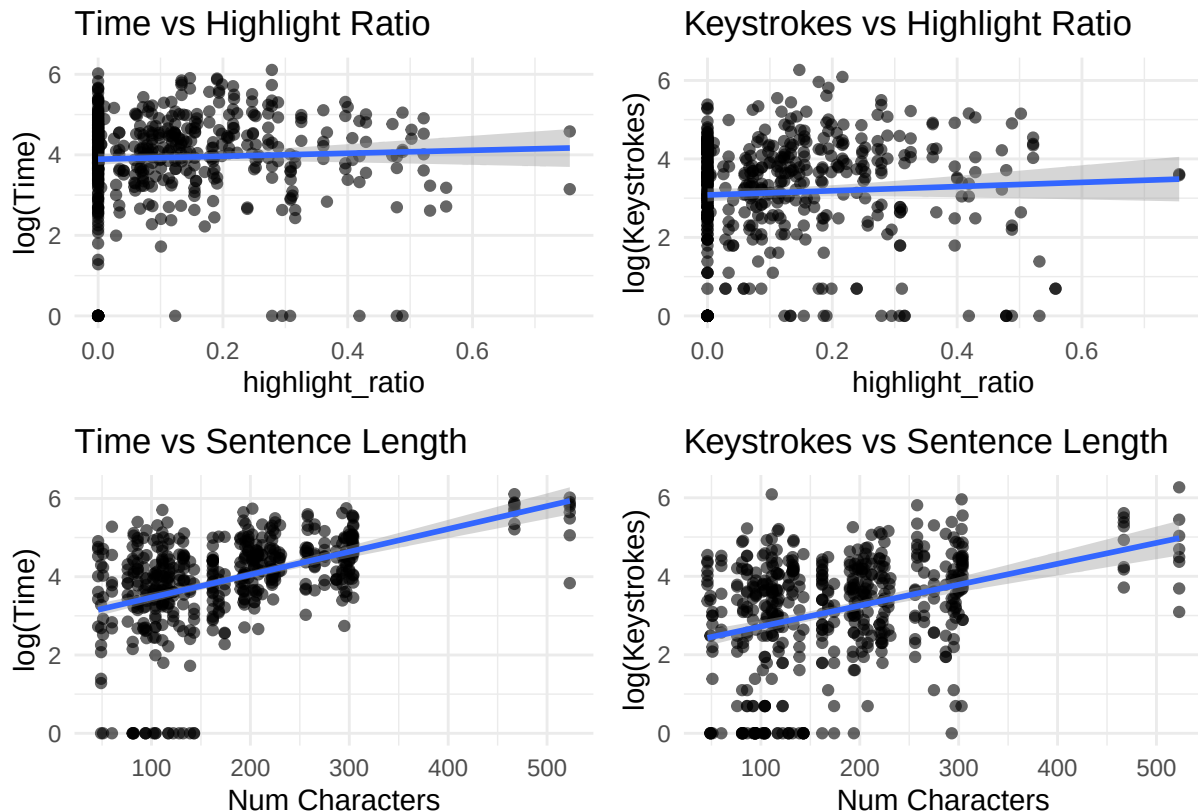
(p1 | p2) / (p3 | p4) + plot_layout(guides = 'collect')

```

```

## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'

```



Number of major and Minor errors compared to time and Keystrokes

```

# 1. Time vs Major Errors
p11 <- ggplot(plot_data, aes(x = num_major, y = log_time)) +
  geom_point(alpha = 0.6) +

```

```

geom_smooth(method = "lm", se = TRUE, col = "firebrick1") +
theme_minimal() + labs(title = "Time vs Major Errors", x = "Num of Minor Errors", y = "log(Time)")

# 2. Keystrokes vs Major Errors
p12 <- ggplot(plot_data, aes(x = num_major, y = log_keystrokes)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, col = "firebrick1") +
  theme_minimal() + labs(title = "Keystrokes vs Major Errors", x = "Num of Minor Errors", y = "log(Keys

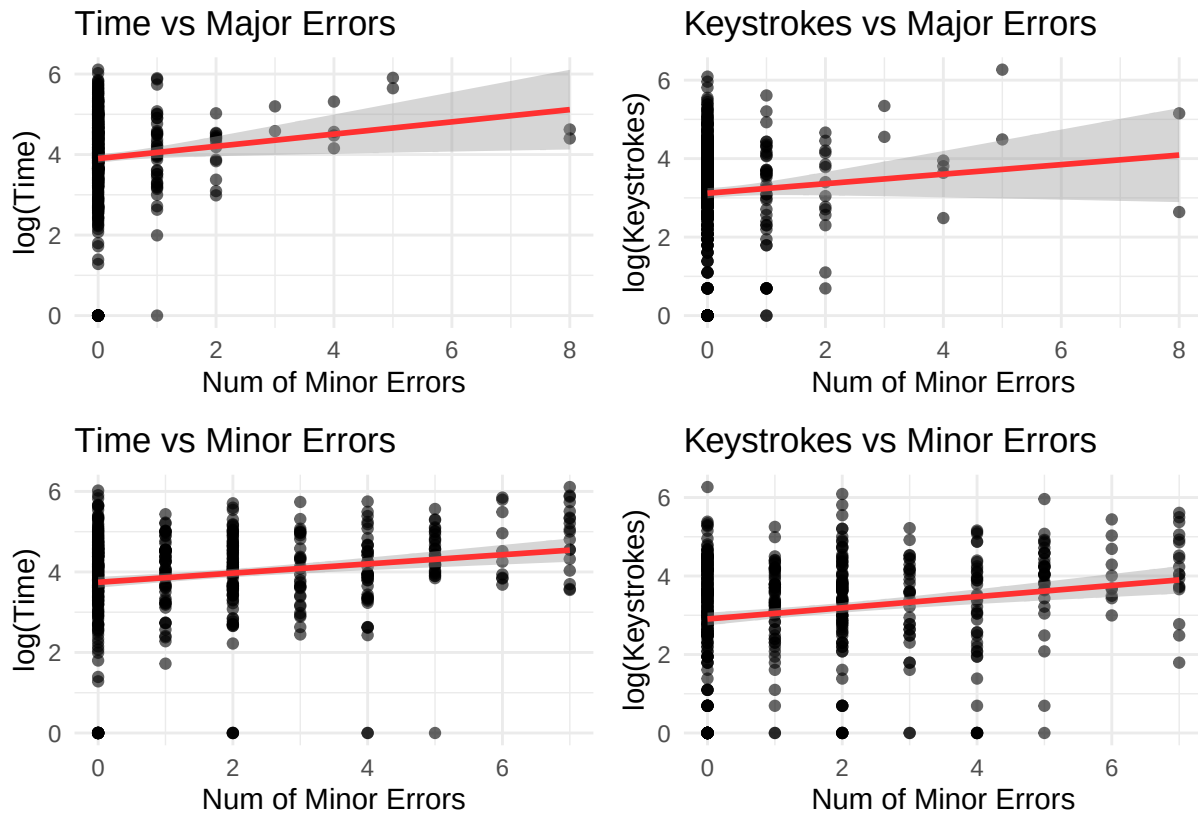
# 3. Time vs Minor Errors
p13 <- ggplot(plot_data, aes(x = num_minor, y = log_time)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, col = "firebrick1") +
  theme_minimal() + labs(title = "Time vs Minor Errors", x = "Num of Minor Errors", y = "log(Time)")

# 4. Keystrokes vs Minor Errors
p14 <- ggplot(plot_data, aes(x = num_minor, y = log_keystrokes)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, col = "firebrick1") +
  theme_minimal() + labs(title = "Keystrokes vs Minor Errors", x = "Num of Minor Errors", y = "log(Keys

(p11 | p12) / (p13 | p14) + plot_layout(guides = 'collect')

## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'

```

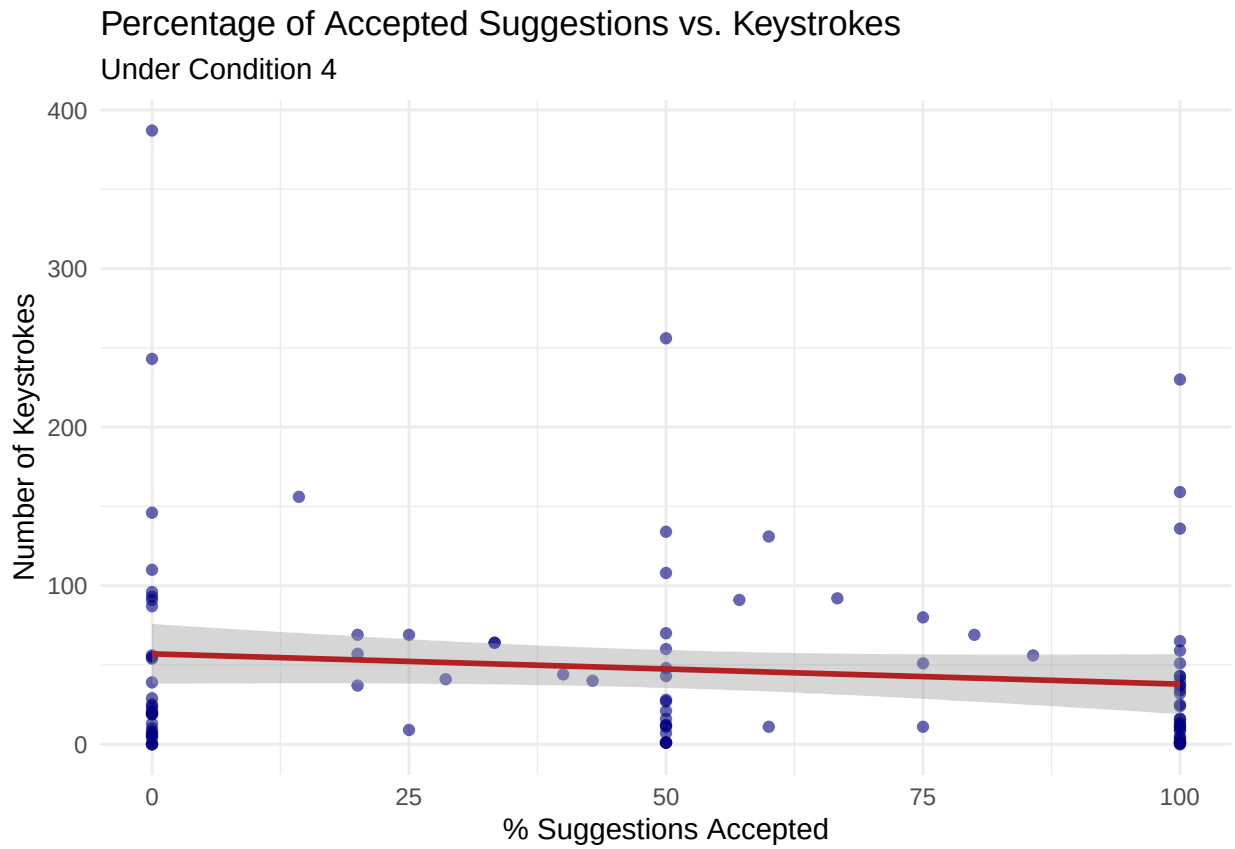


Percentage of suggestions accepted vs Keystrokes condition 4

```
data_plot <- data %>%
  filter(condition == 4) %>%
  mutate(
    prc_sugg_accepted = as.numeric(as.character(prc_sugg_accepted))) %>%
  filter(!is.na(prc_sugg_accepted))

ggplot(data_plot, aes(x = prc_sugg_accepted, y = keystrokes)) +
  geom_point(alpha = 0.6, color = "navy") +
  geom_smooth(method = "lm", color = "firebrick", se = TRUE) +
  theme_minimal() +
  labs(
    title = "Percentage of Accepted Suggestions vs. Keystrokes",
    subtitle = "Under Condition 4",
    x = "% Suggestions Accepted",
    y = "Number of Keystrokes"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Model Building - Time

Analysis of outcome variable distribution

```
zeros<-data[data$time == 0,]  
table(zeros$condition, zeros$domain)
```

```
##  
##      medical news  
##    1      9      1  
##    2      5      0  
##    3      3      1  
##    4      2      1
```

```
table(data$condition, data$domain)
```

```
##  
##      medical news  
##    1     66    58  
##    2     66    58  
##    3     66    58  
##    4     66    58
```

```
View(subset(data, condition == 1 & domain == "medical"))
```

```
data_clean <- data %>%  
  filter(time > 1)
```

```
p1 <- ggplot(data_clean, aes(x = time)) +  
  geom_histogram(aes(y = ..density..), bins = 30,  
                 fill = "#69b3a2", color = "white", alpha = 0.7) +  
  geom_density(color = "#404040", size = 1) +  
  theme_modern() +  
  labs(title = "Histogram of Elapsed Time",  
       x = "Time (seconds)", y = "Density")
```

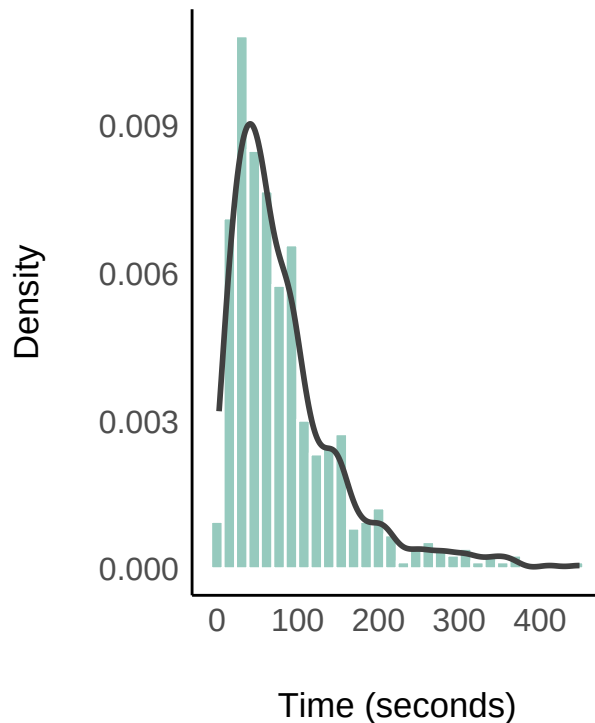
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

```
p2 <- ggplot(data_clean, aes(x = sqrt(time))) +  
  geom_histogram(aes(y = ..density..), bins = 30,  
                 fill = "#40739e", color = "white", alpha = 0.7) +  
  geom_density(color = "#404040", size = 1) +  
  theme_modern() +  
  labs(title = "Histogram of Elapsed Time after log-transformation",  
       x = "Time (log-seconds)", y = "Density")
```

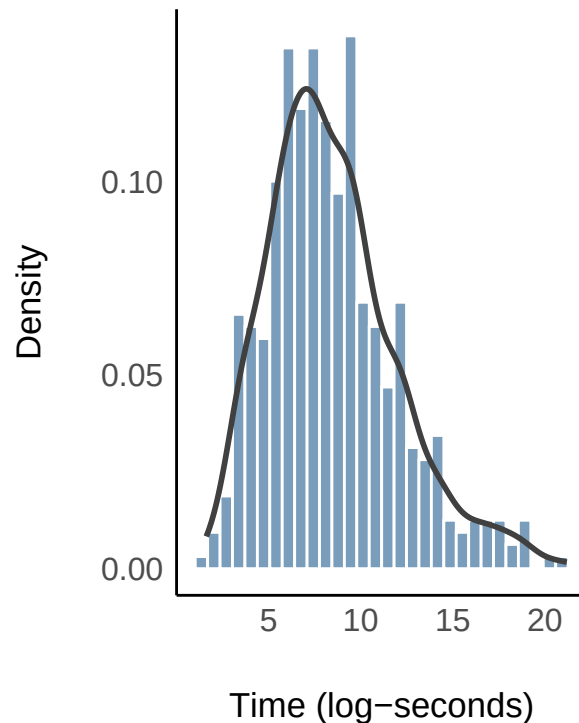
```
p1 + p2
```

```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(density)' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Histogram of Elapsed Time



Histogram of Elapsed Time after log-transformation



The Outcome variable is going to be log-transformed to address the skewness distribution. In the context of a linear mixed-effects model, this transformation shifts the modeling assumptions from additive to multiplicative, therefore making interpretation centered on proportional changes rather than absolute differences.

Model Building test 1 - raw data

```
model_full <- lmer(log(time+1) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality + Difficulty
  (1 | PET) + (1 | text_name / Sent_id),
  data = data,
  REML = FALSE)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(model_full)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
```

```

## method [lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + num_minor +
##          num_major + MT_Quality + Difficulty + (1 | PET) + (1 | text_name/Sent_id)
## Data: data
##
##          AIC          BIC    logLik deviance df.resid
##    1405.5    1477.0   -685.7   1371.5      479
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6914 -0.3465  0.1250  0.5156  2.6746
##
## Random effects:
## Groups              Name                Variance Std.Dev.
## Sent_id:text_name (Intercept) 0.1333    0.3651
## text_name          (Intercept) 0.0000    0.0000
## PET                (Intercept) 0.0491    0.2216
## Residual                    0.8180    0.9044
## Number of obs: 496, groups: Sent_id:text_name, 62; text_name, 8; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.340e+00  2.732e-01  1.794e+02  12.225 < 2e-16 ***
## condition2     -2.120e-01  1.976e-01  4.938e+02  -1.073  0.28379
## condition3      7.996e-02  1.811e-01  4.782e+02   0.442  0.65893
## condition4      2.063e-01  1.768e-01  4.759e+02   1.167  0.24381
## domainnews      3.812e-01  1.923e-01  2.618e+02   1.983  0.04844 *
## num_characters   5.185e-03  6.992e-04  6.760e+01   7.415  2.6e-10 ***
## num_minor        4.373e-02  3.288e-02  3.293e+02   1.330  0.18445
## num_major        6.075e-02  6.829e-02  3.309e+02   0.890  0.37430
## MT_Quality      -2.197e-01  6.638e-02  4.376e+02  -3.310  0.00101 **
## Difficulty      -4.418e-03  5.295e-02  2.643e+02  -0.083  0.93357
## condition2:domainnews  2.079e-01  2.362e-01  4.372e+02   0.880  0.37919
## condition3:domainnews -9.686e-02  2.309e-01  4.243e+02  -0.420  0.67501
## condition4:domainnews -3.709e-01  2.308e-01  4.242e+02  -1.607  0.10877
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

# Singular Fit, must simplify the random part

model_2 <- lmer(log(time+1) ~
                condition * domain + num_characters + num_minor + num_major + MT_Quality +
                (1 | PET) + (1 | text_name : Sent_id),
                data = data,
                REML = FALSE)

```

```
summary(model_2)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + num_minor +
## num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
## Data: data
##
##      AIC      BIC    logLik deviance df.resid
## 1401.5    1464.6   -685.7   1371.5      481
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6888 -0.3500  0.1225  0.5159  2.6756
##
## Random effects:
## Groups              Name                Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.13352   0.3654
## PET                    (Intercept) 0.04864   0.2205
## Residual                      0.81798   0.9044
## Number of obs: 496, groups: text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.333e+00  2.582e-01  1.878e+02  12.908 < 2e-16 ***
## condition2     -2.145e-01  1.952e-01  4.926e+02  -1.099 0.272178
## condition3      7.812e-02  1.797e-01  4.756e+02   0.435 0.663888
## condition4      2.050e-01  1.761e-01  4.762e+02   1.165 0.244758
## domainnews      3.782e-01  1.887e-01  2.529e+02   2.004 0.046149 *
## num_characters   5.190e-03  6.971e-04  6.739e+01   7.444 2.34e-10 ***
## num_minor        4.387e-02  3.284e-02  3.274e+02   1.336 0.182535
## num_major        6.100e-02  6.821e-02  3.294e+02   0.894 0.371838
## MT_Quality      -2.211e-01  6.435e-02  4.609e+02  -3.435 0.000645 ***
## condition2:domainnews 2.087e-01  2.360e-01  4.371e+02   0.884 0.376971
## condition3:domainnews -9.671e-02  2.309e-01  4.241e+02  -0.419 0.675492
## condition4:domainnews -3.714e-01  2.307e-01  4.240e+02  -1.610 0.108070
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2   -0.370
## condition3   -0.199  0.597
## condition4   -0.312  0.616  0.595
## domainnews   -0.337  0.323  0.340  0.358
## num_chrctrs  -0.467  0.194  0.140  0.144 -0.092
## num_minor     0.133 -0.556 -0.436 -0.439  0.031 -0.319
## num_major     0.029 -0.344 -0.212 -0.204  0.019 -0.174  0.312
## MT_Quality    -0.674  0.051 -0.200 -0.042  0.062  0.017  0.022  0.078
## cndtn2:dmnn  0.230 -0.631 -0.370 -0.379 -0.591 -0.055  0.195 -0.028  0.000
## cndtn3:dmnn  0.163 -0.264 -0.602 -0.299 -0.605  0.006 -0.023  0.027  0.064
## cndtn4:dmnn  0.189 -0.269 -0.297 -0.605 -0.609  0.005 -0.022  0.028  0.026
## cndt2: cndt3:
```

```
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn 0.479
## cndtn4:dmnn 0.481 0.501
```

```
model_2.1 <- lmer(log(time+1) ~
                  condition * domain + num_characters + MT_Quality +
                  (1 | PET) + (1 | text_name : Sent_id),
                  data = data,
                  REML = FALSE)

summary(model_2.1)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + MT_Quality +
## (1 | PET) + (1 | text_name:Sent_id)
## Data: data
##
##      AIC      BIC   logLik deviance df.resid
## 1399.4    1454.1   -686.7   1373.4      483
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.7106 -0.3445  0.1116  0.5288  2.7039
##
## Random effects:
## Groups              Name                Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.15027   0.3877
## PET                  (Intercept) 0.05019   0.2240
## Residual                        0.81345   0.9019
## Number of obs: 496, groups: text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   3.284e+00  2.590e-01 1.900e+02  12.680 < 2e-16 ***
## condition2    -5.150e-02  1.579e-01 4.285e+02  -0.326  0.744501
## condition3     1.891e-01  1.606e-01 4.319e+02   1.177  0.239737
## condition4     3.142e-01  1.573e-01 4.271e+02   1.998  0.046378 *
## domainnews     3.696e-01  1.911e-01 2.458e+02   1.934  0.054291 .
## num_characters  5.515e-03  6.807e-04 6.144e+01   8.102  2.8e-11 ***
## MT_Quality     -2.235e-01  6.428e-02 4.609e+02  -3.477  0.000554 ***
## condition2:domainnews 1.583e-01  2.297e-01 4.267e+02   0.689  0.491183
## condition3:domainnews -9.354e-02  2.300e-01 4.269e+02  -0.407  0.684433
## condition4:domainnews -3.686e-01  2.298e-01 4.267e+02  -1.604  0.109504
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##          (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr MT_Qlt cndt2: cndt3:
## condition2 -0.366
## condition3 -0.156  0.467
## condition4 -0.282  0.495  0.495
## domainnews -0.342  0.415  0.389  0.409
## num_chrctrs -0.463  0.003 -0.006 -0.001 -0.088
## MT_Quality -0.677  0.095 -0.207 -0.031  0.060  0.030
## cndtn2:dmnn  0.206 -0.681 -0.334 -0.343 -0.601  0.000  0.002
## cndtn3:dmnn  0.165 -0.333 -0.680 -0.342 -0.596  0.002  0.062  0.499
## cndtn4:dmnn  0.191 -0.339 -0.339 -0.684 -0.599  0.001  0.024  0.501  0.500
```

```
anova(model_2, model_2.1)
```

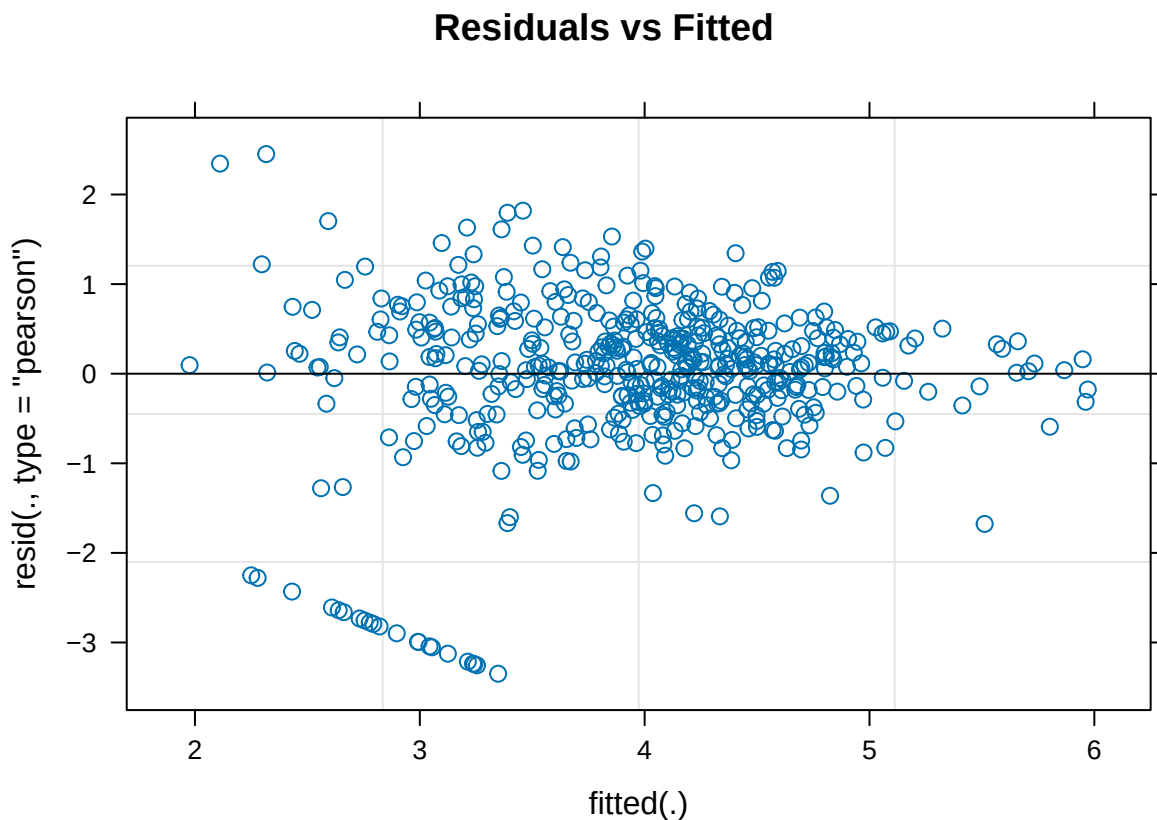
```
## Data: data
## Models:
## model_2.1: log(time + 1) ~ condition * domain + num_characters + MT_Quality + (1 | PET) + (1 | text_name)
## model_2: log(time + 1) ~ condition * domain + num_characters + num_minor + num_major + MT_Quality + (1 | PET) + (1 | text_name)
##          npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## model_2.1   13 1399.4 1454.1 -686.69   1373.4
## model_2     15 1401.5 1464.6 -685.75   1371.5 1.8905  2     0.3886
```

```
model_3 <- lmer(log(time+1) ~
  condition * domain + num_characters + MT_Quality +
  (1 | PET) + (1 | text_name : Sent_id),
  data = data,
  REML = TRUE)
summary(model_3)
```

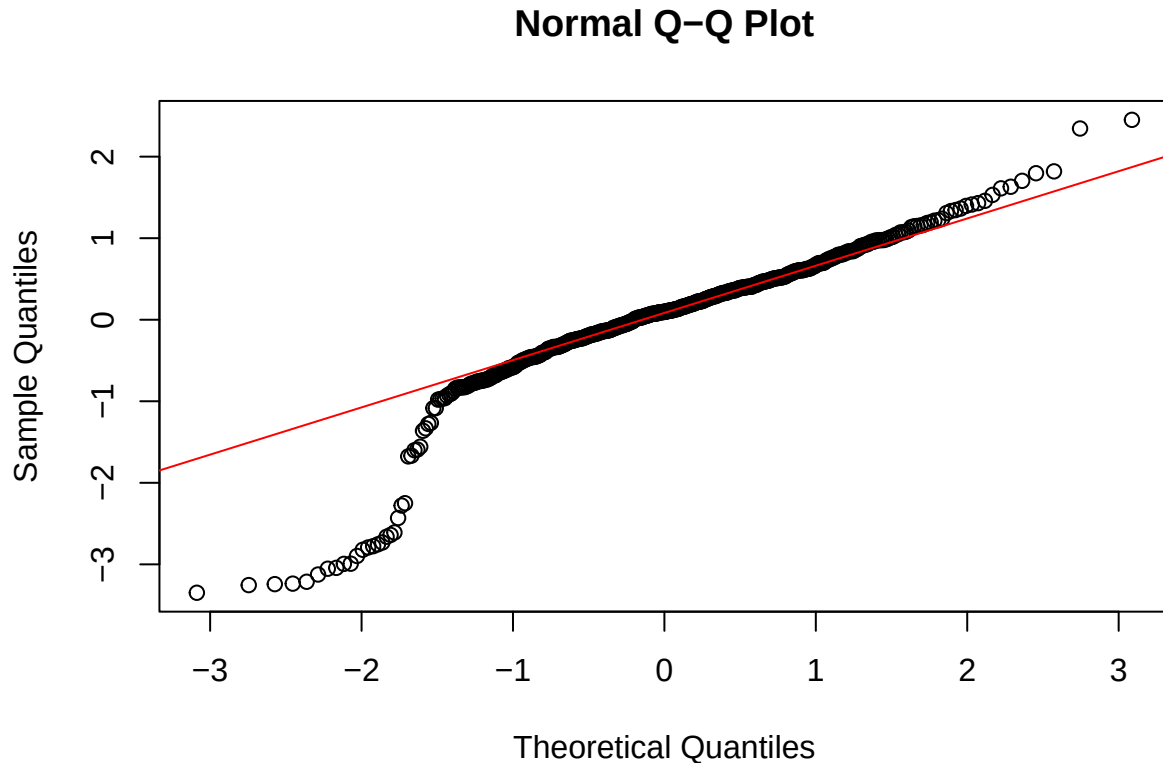
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + MT_Quality +
##          (1 | PET) + (1 | text_name:Sent_id)
## Data: data
##
## REML criterion at convergence: 1406.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6833 -0.3380  0.1065  0.5216  2.6965
##
## Random effects:
## Groups              Name                Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.15938  0.3992
## PET                  (Intercept) 0.05742  0.2396
## Residual                0.82638  0.9091
## Number of obs: 496, groups: text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   3.281488   0.264040 169.444641  12.428 < 2e-16 ***
## condition2    -0.051347   0.159192 421.941113  -0.323  0.747197
```

```
## condition3          0.189014    0.161937 425.252934    1.167 0.243781
## condition4          0.313963    0.158542 420.632669    1.980 0.048320 *
## domainnews          0.369548    0.193800 231.750942    1.907 0.057776 .
## num_characters      0.005515    0.000695  58.820634    7.935 7.18e-11 ***
## MT_Quality          -0.222757    0.065020 453.560379   -3.426 0.000668 ***
## condition2:domainnews 0.158694    0.231568 420.291964    0.685 0.493532
## condition3:domainnews -0.093338    0.231833 420.483480   -0.403 0.687441
## condition4:domainnews -0.368138    0.231635 420.347973   -1.589 0.112744
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr MT_Qlt cndt2: cndt3:
## condition2 -0.363
## condition3 -0.154  0.467
## condition4 -0.279  0.495  0.495
## domainnews -0.339  0.412  0.386  0.407
## num_chrctrs -0.463  0.003 -0.006 -0.001 -0.089
## MT_Quality  -0.672  0.095 -0.208 -0.031  0.060  0.030
## cndtn2:dmnn  0.204 -0.681 -0.334 -0.343 -0.597  0.000  0.002
## cndtn3:dmnn  0.163 -0.333 -0.680 -0.342 -0.592  0.002  0.062  0.499
## cndtn4:dmnn  0.189 -0.339 -0.339 -0.684 -0.596  0.001  0.024  0.501  0.500
```

```
plot(model_3, which = 1, main = "Residuals vs Fitted")
```



```
qqnorm(residuals(model_3))
qqline(residuals(model_3), col = "red")
```



There is a systematic pattern of errors due to the presence of time = 0 observations violating the assumption of homoscedasticity (constant variance of errors)

The QQplot is completely skewed in lower values aswell residuals are NOT normally distributed.

Therefore results are unreliable

Model Building test 2 - Dataset with removed 0 time instances

```
model_full <- lmer(log(time+1) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality + Difficulty
  (1 | PET) + (1 | text_name / Sent_id),
  data = data_clean,
  REML = FALSE)

summary(model_full)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + num_minor +
## num_major + MT_Quality + Difficulty + (1 | PET) + (1 | text_name/Sent_id)
```

```

## Data: data_clean
##
##      AIC      BIC   logLik deviance df.resid
##    941.7   1012.4   -453.8   907.7     457
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2124 -0.5915  0.0179  0.6334  2.6118
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## Sent_id:text_name (Intercept) 0.075744 0.27522
## text_name         (Intercept) 0.009839 0.09919
## PET               (Intercept) 0.058932 0.24276
## Residual              0.332375 0.57652
## Number of obs: 474, groups: Sent_id:text_name, 62; text_name, 8; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.867e+00  2.088e-01  8.726e+01  18.520 < 2e-16 ***
## condition2     -5.016e-01  1.359e-01  4.660e+02  -3.691 0.000250 ***
## condition3     -3.528e-01  1.237e-01  4.556e+02  -2.853 0.004526 **
## condition4     -2.290e-01  1.197e-01  4.526e+02  -1.913 0.056331 .
## domainnews     -5.170e-02  1.501e-01  1.935e+01  -0.344 0.734323
## num_characters  3.831e-03  5.019e-04  6.398e+01   7.633 1.43e-10 ***
## num_minor       5.208e-02  2.218e-02  3.435e+02   2.347 0.019475 *
## num_major       7.898e-02  4.566e-02  3.563e+02   1.730 0.084556 .
## MT_Quality     -1.563e-01  4.524e-02  4.001e+02  -3.455 0.000609 ***
## Difficulty      1.998e-02  3.788e-02  2.403e+02   0.527 0.598372
## condition2:domainnews 4.146e-01  1.562e-01  4.189e+02   2.655 0.008244 **
## condition3:domainnews 2.955e-01  1.519e-01  4.052e+02   1.945 0.052509 .
## condition4:domainnews 3.880e-02  1.514e-01  4.049e+02   0.256 0.797931
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

model_2 <- lmer(log(time+1) ~
                condition * domain + num_characters + num_minor + num_major + MT_Quality +
                (1 | PET) + (1 | text_name / Sent_id),
                data = data_clean,
                REML = FALSE)

summary(model_2)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + num_minor +
##          num_major + MT_Quality + (1 | PET) + (1 | text_name/Sent_id)
## Data: data_clean

```

```

##
##      AIC      BIC    logLik deviance df.resid
##    939.9    1006.5   -454.0    907.9      458
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2190 -0.5847  0.0347  0.6160  2.6392
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
##   Sent_id:text_name (Intercept) 0.075560 0.27488
##   text_name          (Intercept) 0.007986 0.08937
##   PET                 (Intercept) 0.061051 0.24708
##   Residual                        0.332877 0.57696
## Number of obs: 474, groups:  Sent_id:text_name, 62; text_name, 8; PET, 8
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.905e+00  1.976e-01  8.241e+01  19.758 < 2e-16 ***
## condition2     -4.918e-01  1.344e-01  4.670e+02  -3.659 0.000283 ***
## condition3     -3.437e-01  1.226e-01  4.535e+02  -2.803 0.005282 **
## condition4     -2.239e-01  1.193e-01  4.531e+02  -1.876 0.061230 .
## domainnews     -3.715e-02  1.445e-01  2.095e+01  -0.257 0.799567
## num_characters    3.815e-03  4.999e-04  6.460e+01   7.630 1.37e-10 ***
## num_minor        5.204e-02  2.217e-02  3.438e+02   2.347 0.019503 *
## num_major        7.846e-02  4.562e-02  3.558e+02   1.720 0.086359 .
## MT_Quality      -1.526e-01  4.417e-02  3.550e+02  -3.455 0.000618 ***
## condition2:domainnews 4.105e-01  1.561e-01  4.189e+02   2.630 0.008850 **
## condition3:domainnews 2.929e-01  1.520e-01  4.052e+02   1.927 0.054660 .
## condition4:domainnews 4.036e-02  1.515e-01  4.052e+02   0.266 0.790065
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2   -0.350
## condition3   -0.193  0.620
## condition4   -0.292  0.642  0.627
## domainnews   -0.350  0.301  0.319  0.335
## num_chrctrs  -0.445  0.196  0.149  0.155 -0.073
## num_minor     0.135 -0.568 -0.454 -0.457  0.014 -0.303
## num_major     0.042 -0.364 -0.231 -0.225  0.006 -0.168  0.329
## MT_Quality    -0.589  0.057 -0.206 -0.055  0.039  0.006  0.011  0.066
## cndtn2:dmnn  0.221 -0.653 -0.401 -0.411 -0.520 -0.064  0.214 -0.008  0.002
## cndtn3:dmnn  0.152 -0.290 -0.620 -0.331 -0.532 -0.007 -0.005  0.040  0.083
## cndtn4:dmnn  0.179 -0.291 -0.323 -0.617 -0.537 -0.008 -0.013  0.038  0.039
##
##          cndt2: cndt3:
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major

```

```
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn 0.492
## cndtn4:dmnn 0.494 0.515
```

```
anova(model_full, model_2)
```

```
## Data: data_clean
## Models:
## model_2: log(time + 1) ~ condition * domain + num_characters + num_minor + num_major + MT_Quality +
## model_full: log(time + 1) ~ condition * domain + num_characters + num_minor + num_major + MT_Quality
##          npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## model_2      16 939.92 1006.5 -453.96   907.92
## model_full    17 941.67 1012.4 -453.83   907.67 0.2534 1      0.6147
```

```
model_3 <- lmer(log(time+1) ~
                condition * domain + num_characters + num_minor + num_major + MT_Quality +
                (1 | PET) + (1 | text_name / Sent_id),
                data = data_clean,
                REML = TRUE)
summary(model_3)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(time + 1) ~ condition * domain + num_characters + num_minor +
##          num_major + MT_Quality + (1 | PET) + (1 | text_name/Sent_id)
##          Data: data_clean
##
## REML criterion at convergence: 958.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.15972 -0.57416  0.02962  0.60064  2.60235
##
## Random effects:
##   Groups                Name                Variance Std.Dev.
##   Sent_id:text_name (Intercept) 0.07812   0.2795
##   text_name         (Intercept) 0.01386   0.1177
##   PET                (Intercept) 0.06894   0.2626
##   Residual                    0.33966   0.5828
## Number of obs: 474, groups:  Sent_id:text_name, 62; text_name, 8; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   3.894e+00  2.060e-01  6.475e+01  18.902 < 2e-16 ***
## condition2    -4.882e-01  1.360e-01  4.558e+02  -3.589 0.000368 ***
## condition3    -3.439e-01  1.240e-01  4.437e+02  -2.774 0.005768 **
## condition4    -2.224e-01  1.206e-01  4.436e+02  -1.844 0.065901 .
## domainnews    -3.778e-02  1.557e-01  1.432e+01  -0.243 0.811688
## num_characters  3.823e-03  5.098e-04  6.224e+01   7.498 2.89e-10 ***
## num_minor      5.103e-02  2.246e-02  3.364e+02   2.271 0.023748 *
## num_major      7.798e-02  4.625e-02  3.478e+02   1.686 0.092707 .
## MT_Quality    -1.486e-01  4.498e-02  3.565e+02  -3.303 0.001054 **
```

```

## condition2:domainnews 4.098e-01 1.577e-01 4.110e+02 2.599 0.009693 **
## condition3:domainnews 2.948e-01 1.536e-01 3.976e+02 1.920 0.055606 .
## condition4:domainnews 4.144e-02 1.530e-01 3.977e+02 0.271 0.786711
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2 -0.341
## condition3 -0.187 0.620
## condition4 -0.284 0.643 0.627
## domainnews -0.362 0.282 0.299 0.314
## num_chrctrs -0.435 0.195 0.149 0.154 -0.070
## num_minor   0.134 -0.570 -0.454 -0.458 0.013 -0.302
## num_major   0.043 -0.366 -0.232 -0.227 0.006 -0.167 0.331
## MT_Quality  -0.573 0.060 -0.206 -0.053 0.035 0.004 0.006 0.062
## cndtn2:dmnn 0.215 -0.652 -0.401 -0.411 -0.488 -0.064 0.214 -0.008 0.002
## cndtn3:dmnn 0.146 -0.290 -0.620 -0.330 -0.499 -0.007 -0.005 0.040 0.084
## cndtn4:dmnn 0.173 -0.291 -0.323 -0.616 -0.503 -0.008 -0.013 0.038 0.040
##      cndt2: cndt3:
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn 0.492
## cndtn4:dmnn 0.494 0.515

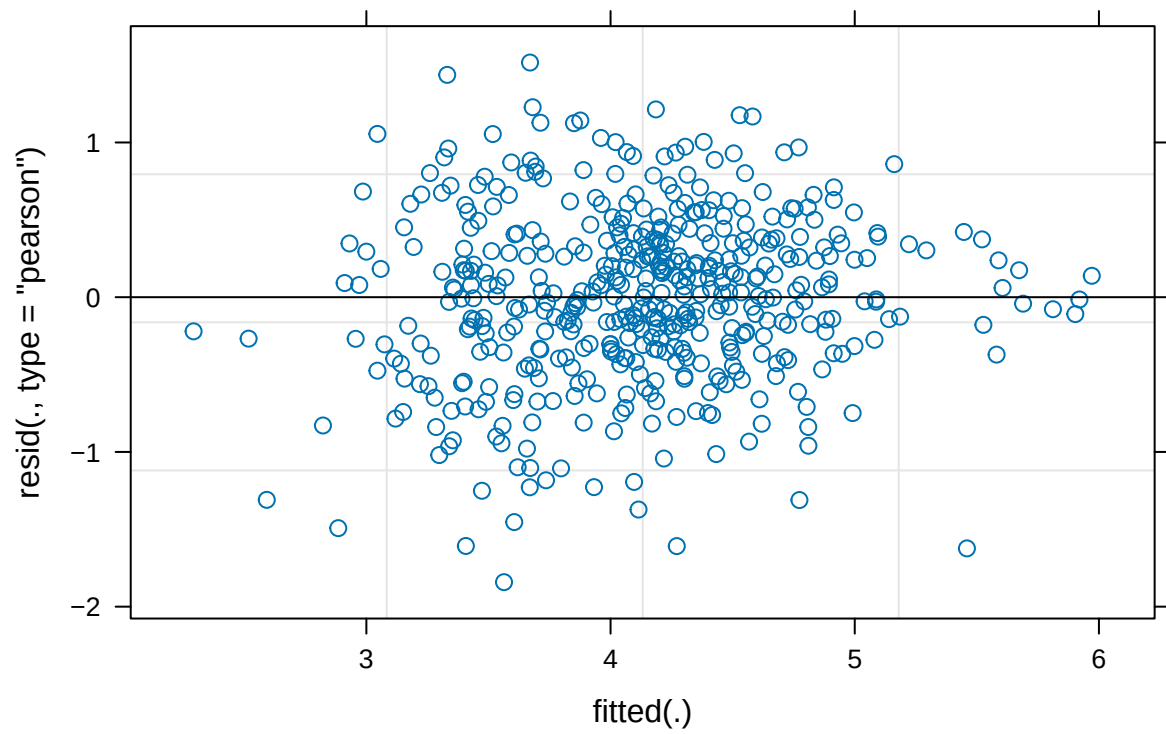
```

```

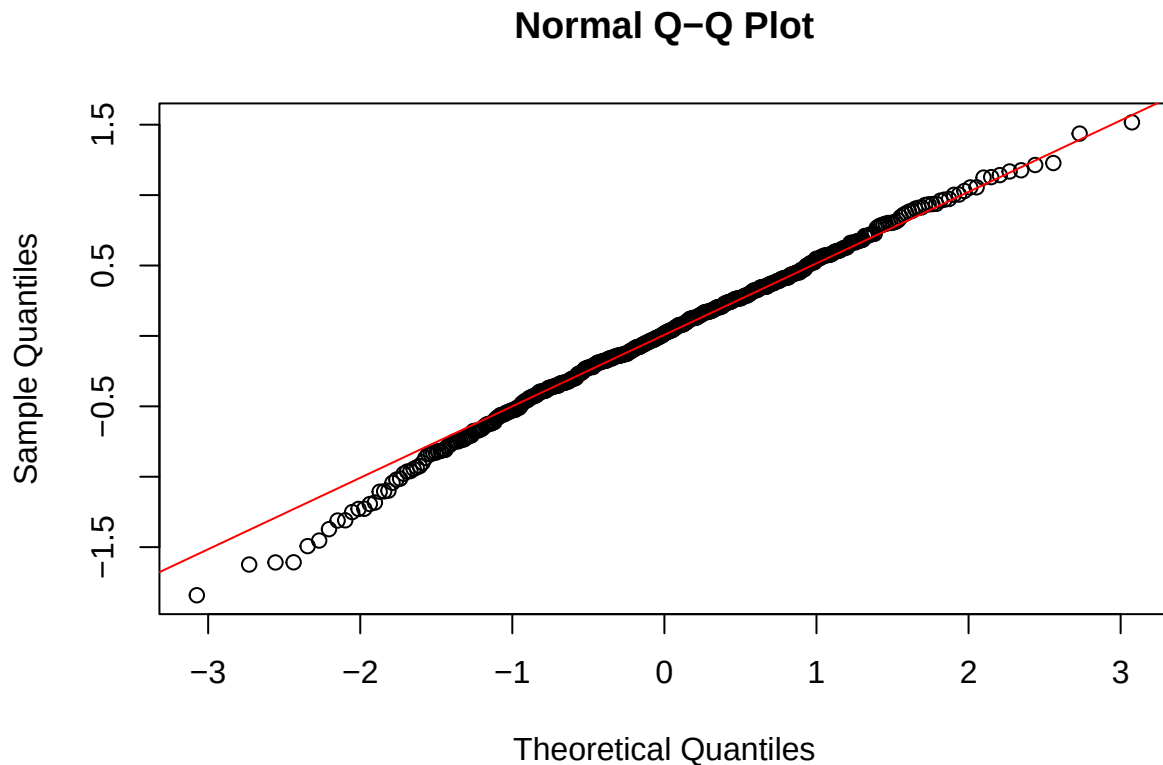
plot(model_3, which = 1, main = "Residuals vs Fitted")

```

Residuals vs Fitted



```
qqnorm(residuals(model_3))  
qqline(residuals(model_3), col = "red")
```



Assumptions checked, residuals are normally distributed

limit of this model is: removing zeros is **CONCEPTUALLY WRONG** as they are not MCAR (missing completely at random) this causes **SELECTION BIAS**

Model Building test 3 (Option 1) - dataset with 0 time instances substituted with number of characters/20

Under the assumption a professional translator takes around 1 second to make sure 20 characters contain no errors and should not be edited.

```
data_impute <- data
data_impute$time_fixed <- ifelse(data_impute$time == 0, data_impute$num_characters / 20, data_impute$time)
View(data_impute[data_impute$time == 0,])

model_full <- lmer(log(time_fixed) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality + Difficulty
  (1 | PET) + (1 | text_name / Sent_id),
  data = data_impute,
  REML = FALSE)
```

boundary (singular) fit: see help('isSingular')

```

# Singular Fit

model_2 <- lmer(log(time_fixed) ~
                condition * domain + num_characters + num_minor + num_major + MT_Quality +
                (1 | PET) + (1 | text_name : Sent_id),
                data = data_impute,
                REML = FALSE)

summary(model_2)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time_fixed) ~ condition * domain + num_characters + num_minor +
## num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
## Data: data_impute
##
##      AIC      BIC   logLik deviance df.resid
##  1173.0   1236.1   -571.5   1143.0     481
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2114 -0.5007  0.1237  0.5827  2.9960
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.10706  0.3272
## PET                (Intercept) 0.05395  0.2323
## Residual              0.50204  0.7085
## Number of obs: 496, groups: text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.539e+00  2.164e-01  1.473e+02  16.354 < 2e-16 ***
## condition2     -3.169e-01  1.546e-01  4.913e+02  -2.050 0.040909 *
## condition3     -1.066e-01  1.416e-01  4.725e+02  -0.753 0.451990
## condition4      1.699e-02  1.388e-01  4.735e+02   0.122 0.902571
## domainnews      1.944e-01  1.534e-01  2.236e+02   1.268 0.206144
## num_characters  4.747e-03  5.886e-04  6.705e+01   8.066 1.82e-11 ***
## num_minor       4.624e-02  2.646e-02  3.604e+02   1.747 0.081459 .
## num_major       6.528e-02  5.493e-02  3.629e+02   1.188 0.235452
## MT_Quality     -1.953e-01  5.134e-02  4.854e+02  -3.804 0.000161 ***
## condition2:domainnews 2.885e-01  1.851e-01  4.364e+02   1.559 0.119814
## condition3:domainnews 8.195e-02  1.809e-01  4.242e+02   0.453 0.650731
## condition4:domainnews -1.937e-01  1.807e-01  4.240e+02  -1.072 0.284349
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2   -0.352
## condition3   -0.187  0.601
## condition4   -0.295  0.622  0.599
## domainnews   -0.322  0.307  0.325  0.342

```

```
## num_chrctrs -0.470  0.189  0.137  0.141 -0.096
## num_minor    0.134 -0.567 -0.445 -0.449  0.030 -0.305
## num_major    0.032 -0.353 -0.217 -0.210  0.019 -0.167  0.315
## MT_Quality   -0.643  0.057 -0.198 -0.039  0.060  0.018  0.014  0.073
## cndtn2:dmnn  0.217 -0.629 -0.372 -0.381 -0.568 -0.054  0.200 -0.027 -0.002
## cndtn3:dmnn  0.150 -0.260 -0.598 -0.297 -0.583  0.006 -0.024  0.028  0.065
## cndtn4:dmnn  0.176 -0.266 -0.295 -0.601 -0.587  0.005 -0.023  0.029  0.027
##
##          cndt2: cndt3:
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn  0.478
## cndtn4:dmnn  0.480  0.501
```

```
model_fixed <- lmer(log(time_fixed) ~
                    condition * domain + num_characters +
                    num_minor + num_major + MT_Quality +
                    (1 | PET) + (1 | text_name : Sent_id),
                    data = data_impute, REML = TRUE)
```

```
summary(model_fixed)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(time_fixed) ~ condition * domain + num_characters + num_minor +
##          num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
## Data: data_impute
##
## REML criterion at convergence: 1190.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.1780 -0.4955  0.1155  0.5775  2.9801
##
## Random effects:
##   Groups                Name                Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.11493  0.3390
## PET                  (Intercept) 0.06149  0.2480
## Residual                        0.51166  0.7153
## Number of obs: 496, groups: text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)   3.535e+00  2.217e-01 1.264e+02  15.943 < 2e-16 ***
## condition2    -3.127e-01  1.564e-01  4.798e+02  -1.999  0.046134 *
## condition3    -1.041e-01  1.431e-01  4.627e+02  -0.727  0.467505
## condition4     1.956e-02  1.402e-01  4.638e+02   0.139  0.889117
## domainnews     1.943e-01  1.561e-01  2.087e+02   1.245  0.214594
```

```
## num_characters      4.756e-03  6.034e-04  6.406e+01  7.881  5.2e-11 ***
## num_minor          4.512e-02  2.686e-02  3.538e+02  1.680  0.093921 .
## num_major          6.402e-02  5.575e-02  3.563e+02  1.148  0.251560
## MT_Quality         -1.944e-01  5.199e-02  4.742e+02 -3.739  0.000207 ***
## condition2:domainnews 2.874e-01  1.869e-01  4.286e+02  1.538  0.124906
## condition3:domainnews 8.231e-02  1.826e-01  4.169e+02  0.451  0.652410
## condition4:domainnews -1.933e-01  1.825e-01  4.168e+02 -1.059  0.290065
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2 -0.348
## condition3 -0.184  0.602
## condition4 -0.292  0.623  0.600
## domainnews -0.318  0.304  0.322  0.339
## num_chrctrs -0.471  0.188  0.136  0.141 -0.096
## num_minor   0.133 -0.570 -0.447 -0.451  0.030 -0.302
## num_major   0.033 -0.354 -0.218 -0.211  0.019 -0.165  0.316
## MT_Quality  -0.636  0.058 -0.198 -0.038  0.060  0.018  0.012  0.072
## cndtn2:dmnn  0.214 -0.629 -0.372 -0.381 -0.564 -0.054  0.201 -0.027 -0.002
## cndtn3:dmnn  0.148 -0.260 -0.597 -0.296 -0.578  0.006 -0.024  0.028  0.065
## cndtn4:dmnn  0.173 -0.265 -0.294 -0.600 -0.582  0.005 -0.023  0.029  0.027
##      cndtn2: cndtn3:
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn  0.477
## cndtn4:dmnn  0.479  0.501
```

Condition 2 effect is statistically significant in medical domain and lowers elapsed time by 27% compared to baseline.

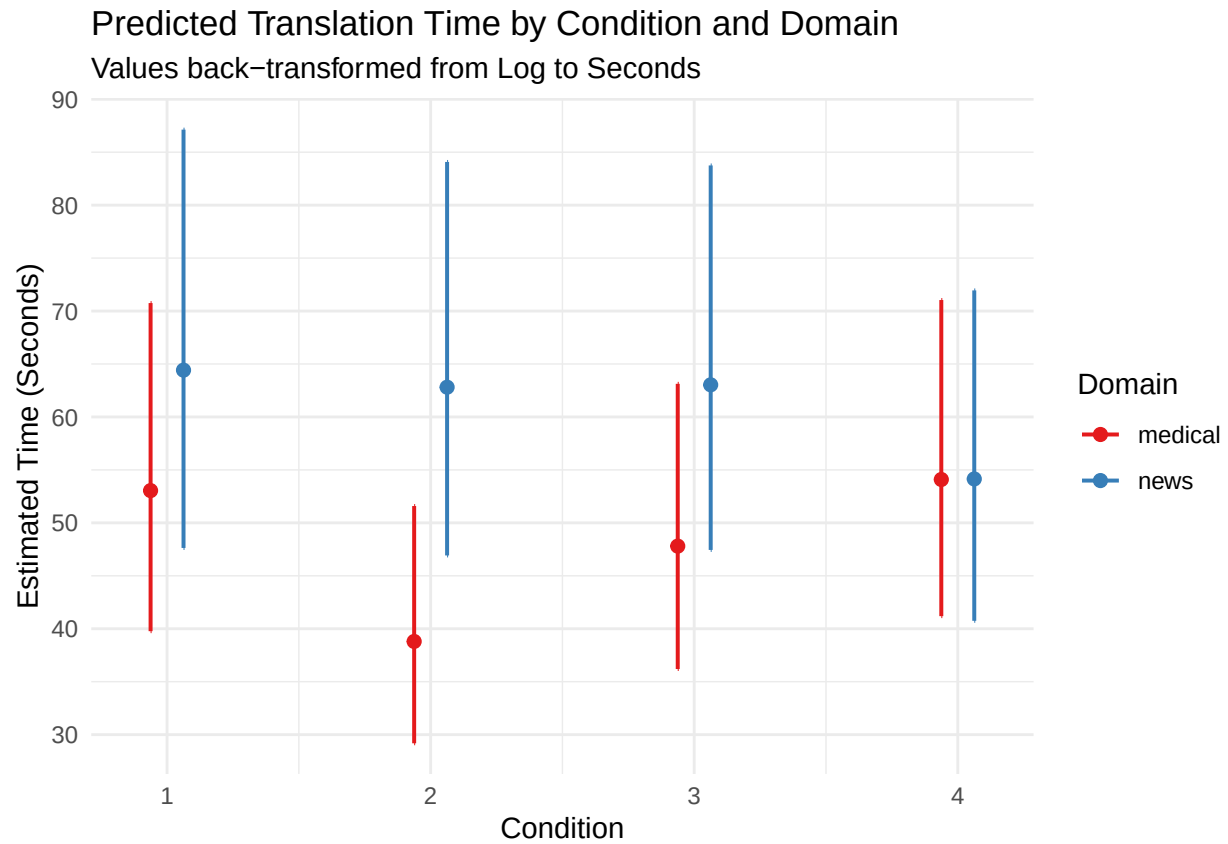
Effects Plots

```
eff_int <- predict_response(model_fixed, terms = c("condition", "domain"))
```

```
## Model has log-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.
```

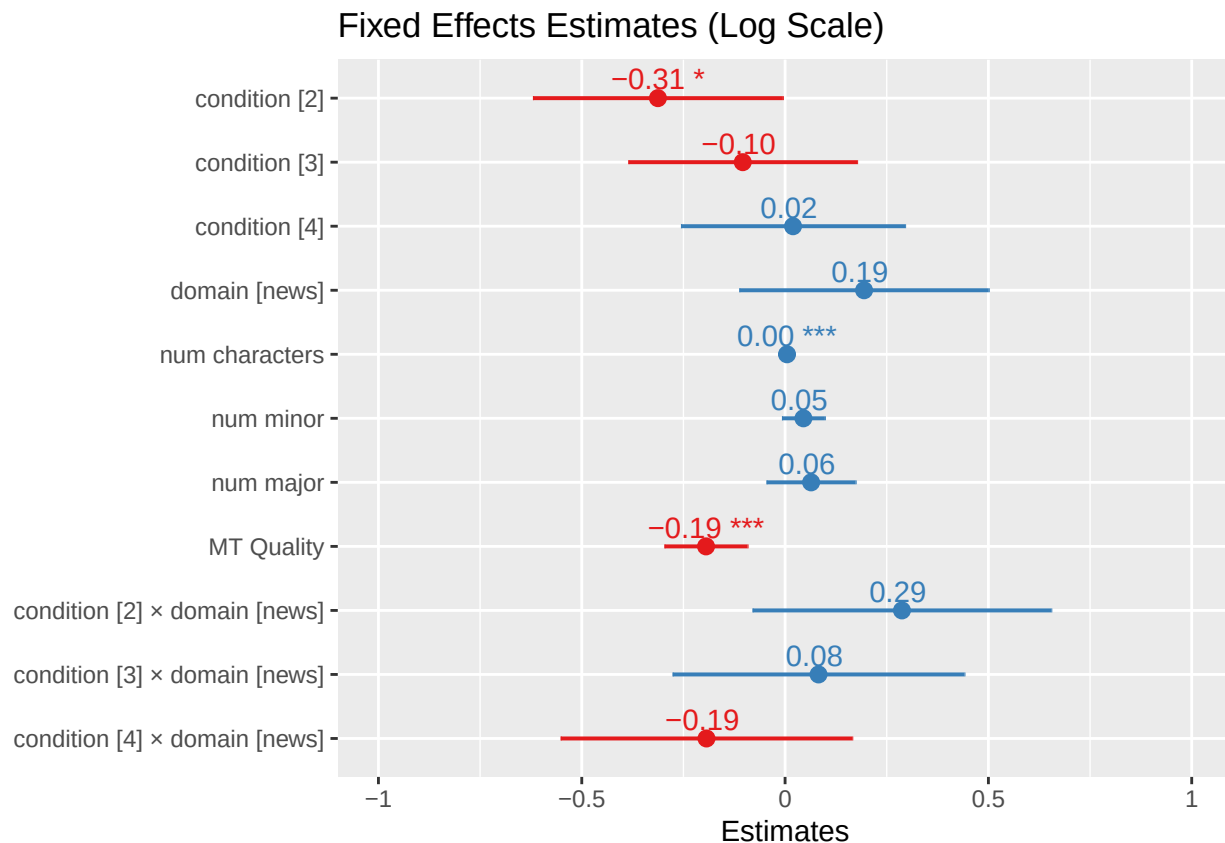
```
plot(eff_int) +
  labs(
    title = "Predicted Translation Time by Condition and Domain",
    subtitle = "Values back-transformed from Log to Seconds",
```

```
x = "Condition",
y = "Estimated Time (Seconds)",
colour = "Domain"
) +
theme_minimal()
```



```
plot_model(model_fixed,
            type = "est",
            show.values = TRUE,
            value.offset = .3,
            title = "Fixed Effects Estimates (Log Scale)")
```

```
## Registered S3 methods overwritten by 'parameters':
##   method                                from
##   display.parameters_distribution        datawizard
##   plot.parameters_distribution          datawizard
##   print_md.parameters_distribution      datawizard
```



```
icc_results1 <- icc(model_fixed)

print(icc_results1) # Random effects account for 18% of total variance
```

```
## # Intraclass Correlation Coefficient
##
##   Adjusted ICC: 0.256
##   Unadjusted ICC: 0.177
```

```
r2_results1 <- r2(model_fixed)

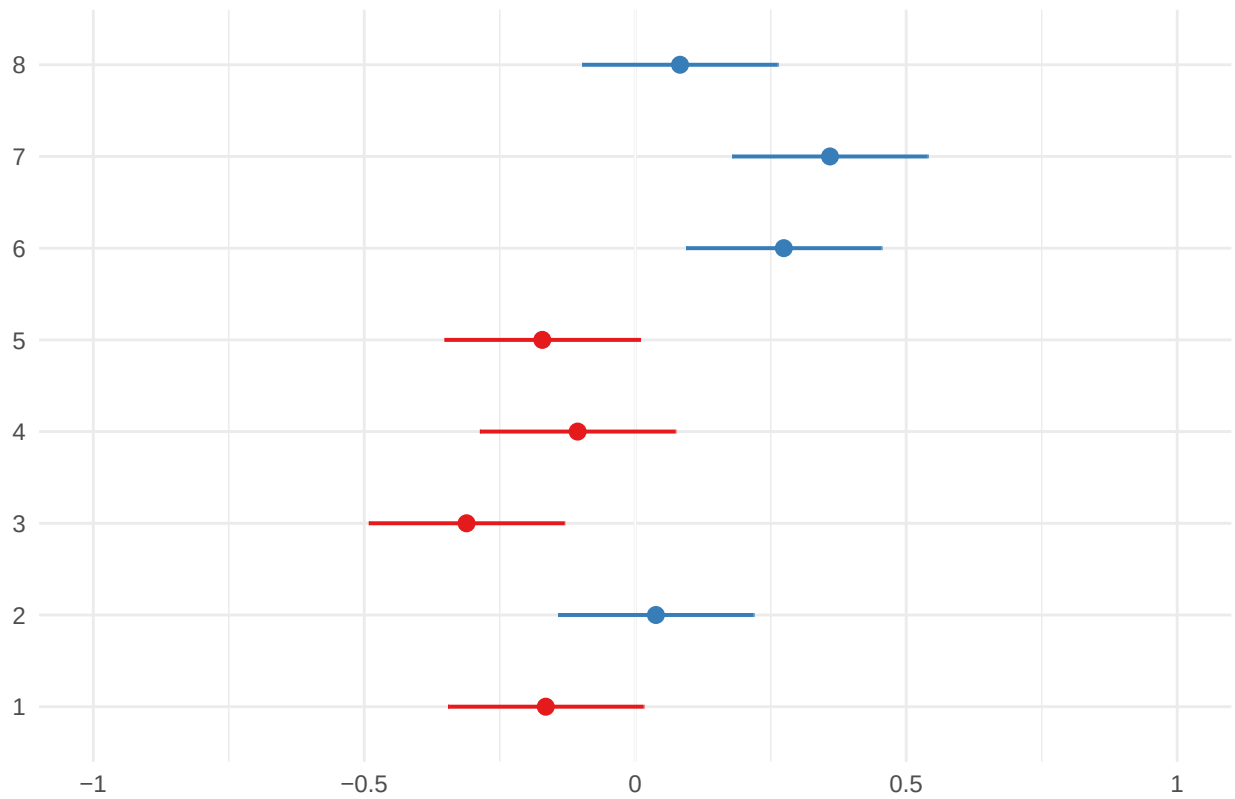
print(r2_results1) # 31% of total variance comes from fixed effects
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.486
##   Marginal R2: 0.309
```

```
# R2 is 49%
```

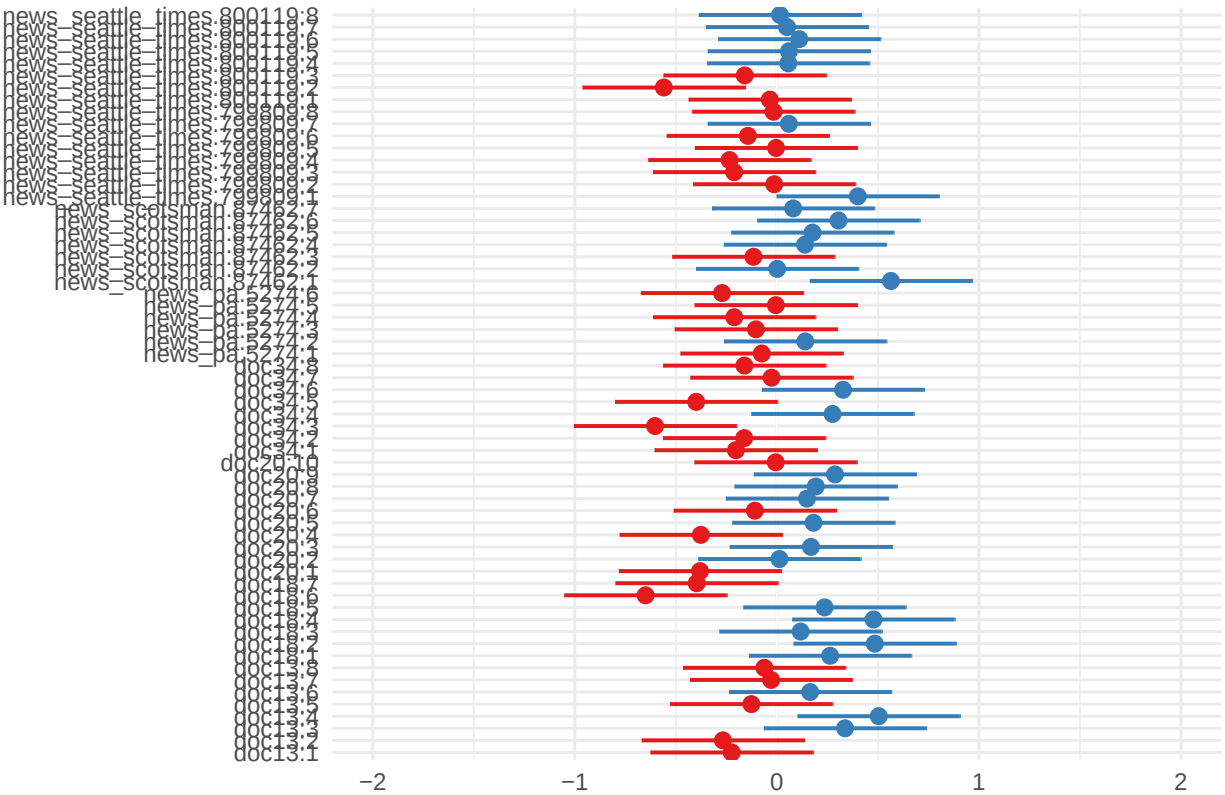
```
random_effects <- ranef(model_fixed)
plot_model(model_fixed, type = "re", grid = FALSE)[[2]] +
  theme_minimal() +
  labs(title = "Random Effects: PET (Translators)")
```

Random Effects: PET (Translators)



```
plot_model(model_fixed, type = "re", grid = FALSE)[[1]] +  
  theme_minimal() +  
  labs(title = "Random Effects: Sentence ID")
```

Random Effects: Sentence ID



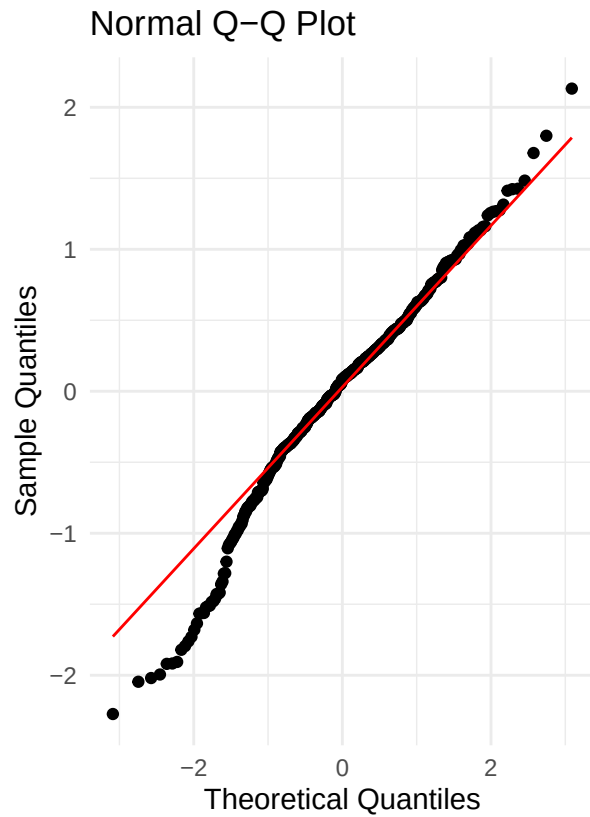
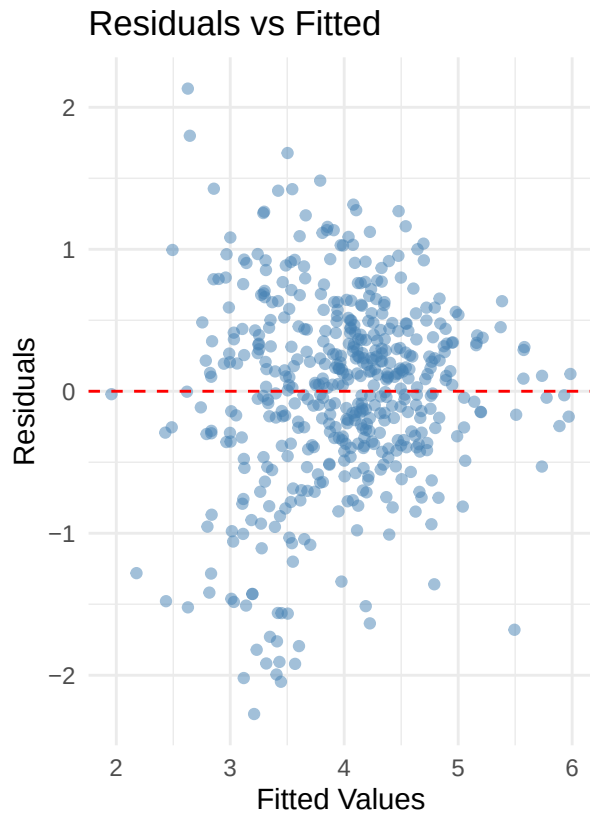
Assumptions Check

```
diag_data <- data.frame(
  Fitted = fitted(model_fixed),
  Residuals = residuals(model_fixed)
)

d1 <- ggplot(diag_data, aes(x = Fitted, y = Residuals)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted",
       x = "Fitted Values",
       y = "Residuals") +
  theme_minimal()

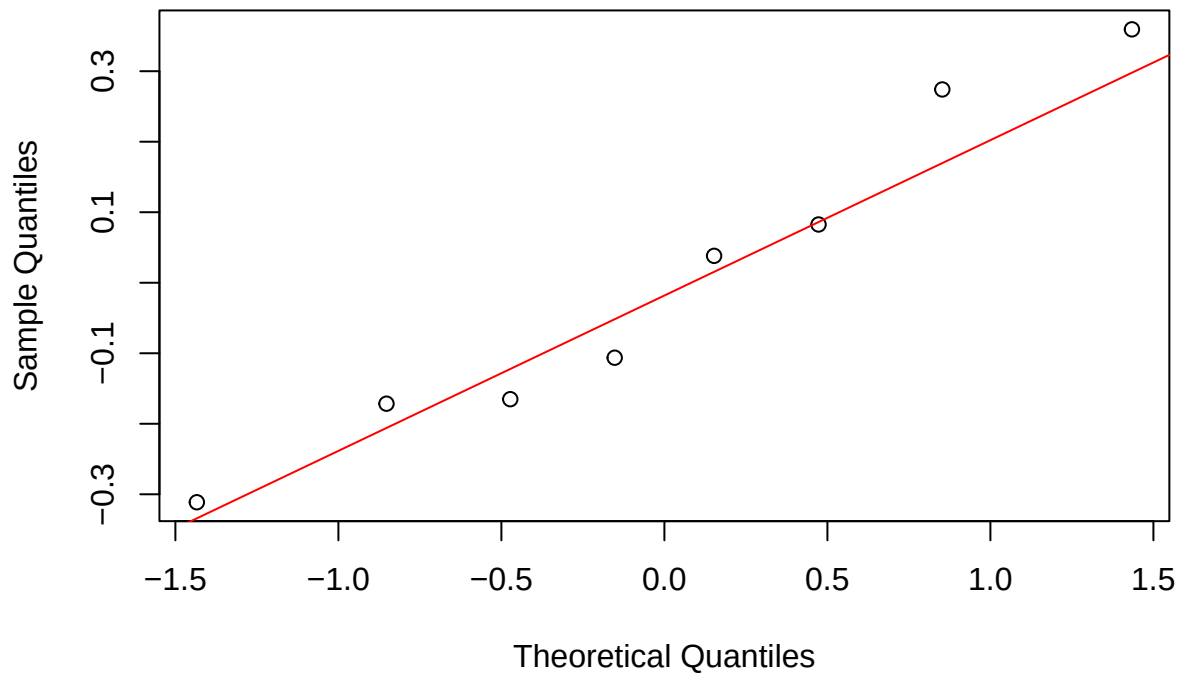
d2 <- ggplot(diag_data, aes(sample = Residuals)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "Normal Q-Q Plot",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()

d1+d2
```



```
pet_re <- ranef(model_fixed)$PET[,1]
qqnorm(pet_re, main = "Q-Q Plot: Random Effects (PET)")
qqline(pet_re, col = "red")
```

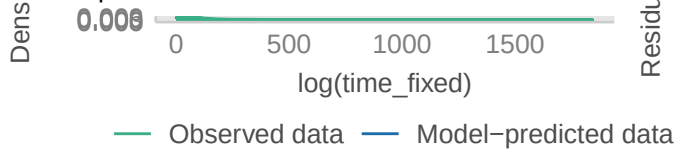
Q-Q Plot: Random Effects (PET)



```
check_model(model_fixed) # The high VIF does not distort the model's standard errors or the significance
```

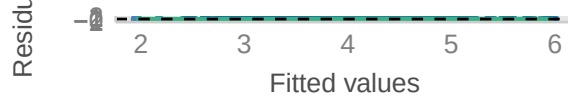
Posterior Predictive Check

Model-predicted lines should resemble observed data



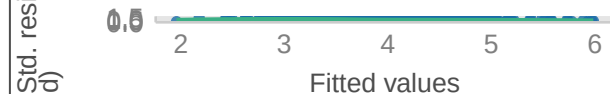
Linearity

Reference line should be flat and horizontal



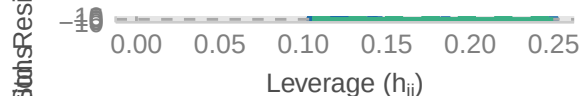
Homogeneity of Variance

Reference line should be flat and horizontal



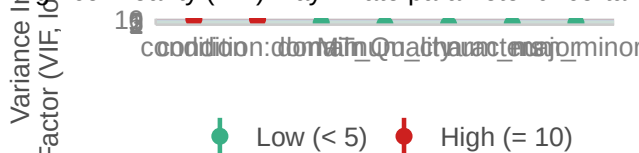
Influential Observations

Points should be inside the contour lines



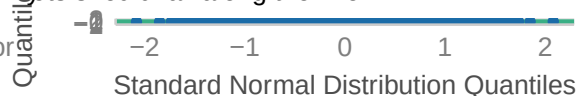
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



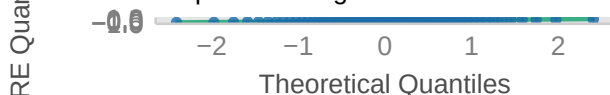
Normality of Residuals

Dots should fall along the line



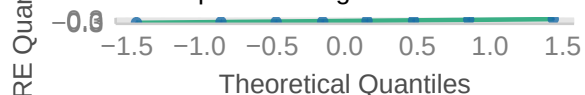
Normality of Random Effects (text_name: Sent_id)

Dots should be plotted along the line



Normality of Random Effects (PET)

Dots should be plotted along the line



Contrast Analysis - Switch of Baseline

```
data_impute$condition <- relevel(as.factor(data_impute$condition), ref = "2")
model_fixed <- lmer(log(time_fixed) ~
  condition * domain + num_characters +
  num_minor + num_major + MT_Quality +
  (1 | PET) + (1 | text_name : Sent_id),
  data = data_impute, REML = TRUE)

summary(model_fixed)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(time_fixed) ~ condition * domain + num_characters + num_minor +
##      num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
##      Data: data_impute
##
## REML criterion at convergence: 1190.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.1780 -0.4955  0.1155  0.5775  2.9801
##
## Random effects:
```

```

## Groups          Name          Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.11493  0.3390
## PET              (Intercept) 0.06149  0.2480
## Residual                0.51166  0.7153
## Number of obs: 496, groups:  text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.222e+00  2.224e-01  1.319e+02  14.486 < 2e-16 ***
## condition1     3.127e-01  1.564e-01  4.798e+02   1.999 0.046134 *
## condition3     2.087e-01  1.342e-01  4.432e+02   1.555 0.120573
## condition4     3.323e-01  1.297e-01  4.366e+02   2.563 0.010722 *
## domainnews     4.817e-01  1.625e-01  2.189e+02   2.964 0.003369 **
## num_characters  4.756e-03  6.034e-04  6.406e+01   7.881 5.2e-11 ***
## num_minor      4.512e-02  2.686e-02  3.538e+02   1.680 0.093921 .
## num_major      6.402e-02  5.575e-02  3.563e+02   1.148 0.251560
## MT_Quality     -1.944e-01  5.199e-02  4.742e+02  -3.739 0.000207 ***
## condition1:domainnews -2.874e-01  1.869e-01  4.286e+02  -1.538 0.124906
## condition3:domainnews -2.051e-01  1.890e-01  4.336e+02  -1.085 0.278320
## condition4:domainnews -4.807e-01  1.885e-01  4.327e+02  -2.550 0.011120 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndtn1 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition1   -0.356
## condition3   -0.159  0.524
## condition4   -0.270  0.533  0.538
## domainnews   -0.363  0.431  0.375  0.398
## num_chrctrs -0.337 -0.188 -0.074 -0.075 -0.154
## num_minor    -0.268  0.570  0.187  0.199  0.261 -0.302
## num_major    -0.217  0.354  0.181  0.199 -0.013 -0.165  0.316
## MT_Quality   -0.593 -0.058 -0.278 -0.111  0.055  0.018  0.012  0.072
## cndtn1:dmnn  0.229 -0.629 -0.336 -0.346 -0.609  0.054 -0.201  0.027  0.002
## cndtn3:dmnn  0.192 -0.371 -0.655 -0.349 -0.608  0.059 -0.222  0.054  0.065
## cndtn4:dmnn  0.213 -0.367 -0.337 -0.662 -0.611  0.058 -0.222  0.055  0.028
##
##          cndt1: cndt3:
## condition1
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn1:dmnn
## cndtn3:dmnn  0.528
## cndtn4:dmnn  0.528  0.533

```

Model Building test 4 (Option 2) - dataset with instances with time = 0 substituted with newly proposed times

Dataset Loading + Merging

```
data_time <- read_csv("data/data_time.csv")
```

```
## Rows: 22 Columns: 28
## -- Column specification -----
## Delimiter: ","
## chr (14): TEXT, text_name, domain, prc_sugg_accepted, How difficult to trans...
## dbl (13): PET, condition, Sent_id, keystrokes, time, productivity (chr/s), n...
## lgl (1): tags_edited
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(data_time)
```

```
indices_zeros <- which(data$time == 0)
```

```
data$time_hybrid <- data$time
data$time_hybrid[indices_zeros] <- data_time$time
```

```
data_clean2 <- data[data$time_hybrid>1,] # There are only 3 instances still with unrealistic values, al
```

```
model_hyb <- lmer(log(time_hybrid) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality + Difficu
  (1 | PET) + (1 | text_name / Sent_id),
  data = data_clean2,
  REML = FALSE)
```

```
summary(model_hyb)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time_hybrid) ~ condition * domain + num_characters + num_minor +
## num_major + MT_Quality + Difficulty + (1 | PET) + (1 | text_name/Sent_id)
## Data: data_clean2
##
##      AIC      BIC    logLik deviance df.resid
## 1033.6   1105.0   -499.8    999.6      476
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.3262 -0.5700  0.0267  0.6238  2.5081
##
## Random effects:
## Groups              Name              Variance Std.Dev.
## Sent_id:text_name (Intercept) 0.07683  0.2772
## text_name         (Intercept) 0.01040  0.1020
```

```
## PET (Intercept) 0.05578 0.2362
## Residual 0.37734 0.6143
## Number of obs: 493, groups: Sent_id:text_name, 62; text_name, 8; PET, 8
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 3.795158 0.211945 87.060993 17.906 < 2e-16 ***
## condition2 -0.537575 0.137378 485.468933 -3.913 0.000104 ***
## condition3 -0.391636 0.124993 471.718067 -3.133 0.001836 **
## condition4 -0.332399 0.121492 470.126147 -2.736 0.006455 **
## domainnews -0.108422 0.153322 19.090983 -0.707 0.488015
## num_characters 0.003894 0.000513 65.595071 7.591 1.48e-10 ***
## num_minor 0.051112 0.023059 347.515301 2.217 0.027300 *
## num_major 0.083338 0.047836 359.638300 1.742 0.082335 .
## MT_Quality -0.142150 0.046609 418.306522 -3.050 0.002435 **
## Difficulty 0.038625 0.039239 221.839859 0.984 0.326021
## condition2:domainnews 0.453757 0.161404 435.067810 2.811 0.005157 **
## condition3:domainnews 0.346781 0.157453 422.437874 2.202 0.028174 *
## condition4:domainnews 0.153573 0.157053 422.230509 0.978 0.328712
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
## vcov(x) if you need it
```

```
model_hyb2 <- lmer(log(time_hybrid) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality +
  (1 | PET) + (1 | text_name : Sent_id),
  data = data_clean2,
  REML = FALSE)

summary(model_hyb2)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log(time_hybrid) ~ condition * domain + num_characters + num_minor +
## num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
## Data: data_clean2
##
## AIC BIC logLik deviance df.resid
## 1030.9 1093.9 -500.4 1000.9 478
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -3.3673 -0.5801 0.0145 0.6177 2.5040
##
## Random effects:
## Groups Name Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.08269 0.2876
## PET (Intercept) 0.05830 0.2415
## Residual 0.37883 0.6155
```

```

## Number of obs: 493, groups:  text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.889e+00  1.949e-01  1.159e+02  19.953 < 2e-16 ***
## condition2     -5.233e-01  1.361e-01  4.873e+02  -3.846 0.000136 ***
## condition3     -3.734e-01  1.242e-01  4.706e+02  -3.007 0.002780 **
## condition4     -3.233e-01  1.213e-01  4.718e+02  -2.666 0.007938 **
## domainnews     -8.237e-02  1.340e-01  2.237e+02  -0.615 0.539442
## num_characters    3.845e-03  5.151e-04  6.786e+01   7.465 2.07e-10 ***
## num_minor        5.152e-02  2.314e-02  3.611e+02   2.226 0.026601 *
## num_major        8.143e-02  4.790e-02  3.647e+02   1.700 0.089968 .
## MT_Quality      -1.411e-01  4.490e-02  4.882e+02  -3.143 0.001773 **
## condition2:domainnews  4.495e-01  1.616e-01  4.352e+02   2.781 0.005654 **
## condition3:domainnews  3.423e-01  1.577e-01  4.224e+02   2.170 0.030547 *
## condition4:domainnews  1.567e-01  1.573e-01  4.223e+02   0.996 0.319696
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2 -0.345
## condition3 -0.185  0.603
## condition4 -0.289  0.625  0.602
## domainnews -0.314  0.308  0.327  0.345
## num_chrctrs -0.458  0.191  0.137  0.143 -0.094
## num_minor    0.134 -0.573 -0.448 -0.453  0.027 -0.305
## num_major    0.034 -0.356 -0.219 -0.213  0.017 -0.167  0.318
## MT_Quality   -0.623  0.060 -0.195 -0.039  0.059  0.018  0.010  0.070
## cndtn2:dmnn  0.214 -0.633 -0.376 -0.386 -0.567 -0.057  0.207 -0.022 -0.004
## cndtn3:dmnn  0.147 -0.262 -0.601 -0.300 -0.582  0.006 -0.020  0.029  0.065
## cndtn4:dmnn  0.172 -0.267 -0.297 -0.602 -0.586  0.004 -0.021  0.031  0.028
##
##      cndt2: cndt3:
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn  0.477
## cndtn4:dmnn  0.480  0.502

model_hyb3 <- lmer(log(time_hybrid) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality +
  (1 | PET) + (1 | text_name : Sent_id),
  data = data_clean2,
  REML = TRUE)
summary(model_hyb3)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]

```

```

## Formula: log(time_hybrid) ~ condition * domain + num_characters + num_minor +
##      num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
##      Data: data_clean2
##
## REML criterion at convergence: 1051
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -3.3263 -0.5659  0.0167  0.6132  2.4830
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.08849  0.2975
## PET                (Intercept) 0.06643  0.2577
## Residual              0.38621  0.6215
## Number of obs: 493, groups:  text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.885e+00  2.001e-01  9.629e+01  19.414 < 2e-16 ***
## condition2     -5.206e-01  1.377e-01  4.759e+02  -3.782 0.000176 ***
## condition3     -3.719e-01  1.255e-01  4.608e+02  -2.963 0.003203 **
## condition4     -3.217e-01  1.226e-01  4.619e+02  -2.624 0.008984 **
## domainnews     -8.244e-02  1.363e-01  2.095e+02  -0.605 0.546038
## num_characters   3.851e-03  5.278e-04  6.494e+01   7.297 5.19e-10 ***
## num_minor       5.076e-02  2.348e-02  3.542e+02   2.162 0.031323 *
## num_major       8.088e-02  4.860e-02  3.579e+02   1.664 0.096955 .
## MT_Quality     -1.401e-01  4.545e-02  4.766e+02  -3.082 0.002175 **
## condition2:domainnews  4.487e-01  1.632e-01  4.273e+02   2.749 0.006237 **
## condition3:domainnews  3.427e-01  1.593e-01  4.150e+02   2.152 0.031984 *
## condition4:domainnews  1.571e-01  1.588e-01  4.149e+02   0.989 0.323077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndtn2 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition2  -0.341
## condition3  -0.182  0.603
## condition4  -0.285  0.626  0.603
## domainnews  -0.310  0.305  0.324  0.342
## num_chrctrs -0.457  0.189  0.136  0.142 -0.095
## num_minor   0.134 -0.575 -0.450 -0.454  0.027 -0.302
## num_major   0.035 -0.358 -0.220 -0.214  0.017 -0.166  0.319
## MT_Quality  -0.615  0.061 -0.194 -0.039  0.058  0.018  0.008  0.069
## cndtn2:dmnn 0.211 -0.633 -0.376 -0.387 -0.563 -0.056  0.208 -0.022 -0.004
## cndtn3:dmnn 0.145 -0.262 -0.601 -0.300 -0.577  0.006 -0.021  0.029  0.065
## cndtn4:dmnn 0.169 -0.267 -0.297 -0.602 -0.582  0.004 -0.021  0.031  0.028
##
##      cndt2: cndt3:
## condition2
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor

```

```
## num_major
## MT_Quality
## cndtn2:dmnn
## cndtn3:dmnn 0.477
## cndtn4:dmnn 0.480 0.502
```

In medical texts all condition effects are statistically relevant compared to the baseline. - Condition 2 reduces time by 40.6% in medical texts. - Condition 3 by 31.1% in medical texts. - Condition 4 by 27.5% in medical and news texts.

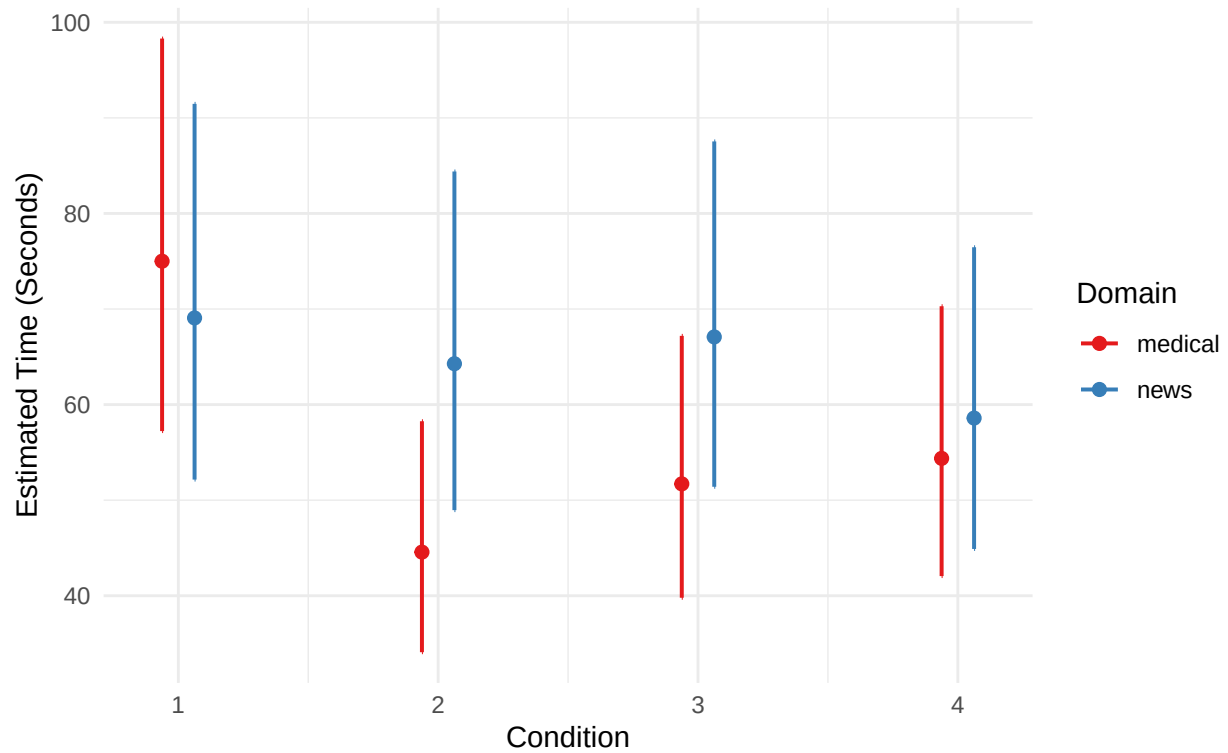
Effects Plots

```
eff_int_hyb <- predict_response(model_hyb3, terms = c("condition", "domain"))
```

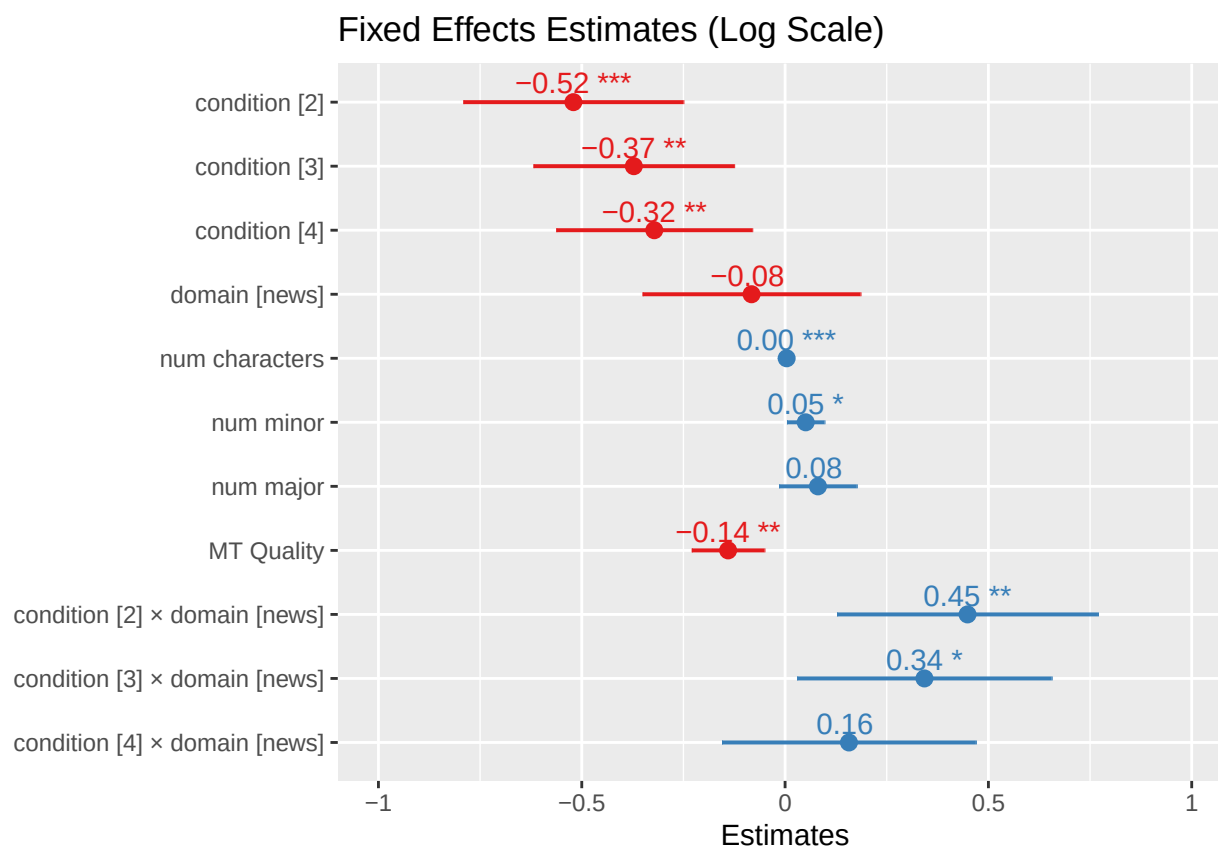
```
## Model has log-transformed response. Back-transforming predictions to
## original response scale. Standard errors are still on the transformed
## scale.
```

```
plot(eff_int_hyb) +
  labs(
    title = "Predicted Translation Time by Condition and Domain",
    subtitle = "Values back-transformed from Log to Seconds",
    x = "Condition",
    y = "Estimated Time (Seconds)",
    colour = "Domain"
  ) +
  theme_minimal()
```

Predicted Translation Time by Condition and Domain
Values back-transformed from Log to Seconds



```
plot_model(model_hyb3,  
  type = "est",  
  show.values = TRUE,  
  value.offset = .3,  
  title = "Fixed Effects Estimates (Log Scale)")
```



```
icc_results2 <- icc(model_hyb3)

print(icc_results2) #random part accounts for 21% of total variance

## # Intraclass Correlation Coefficient
##
##   Adjusted ICC: 0.286
##   Unadjusted ICC: 0.206

r2_results2 <- r2(model_hyb3) # fixed effects are 28% of total variance

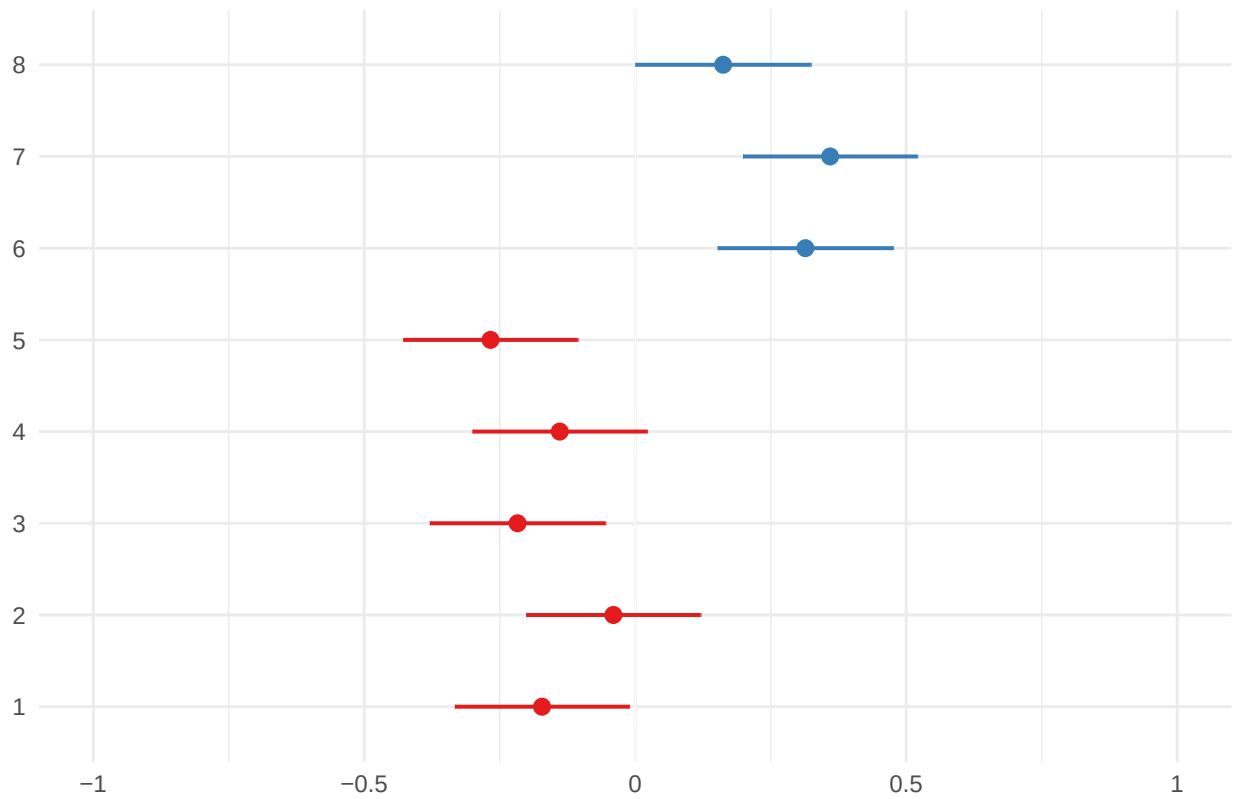
print(r2_results2)

## # R2 for Mixed Models
##
##   Conditional R2: 0.486
##   Marginal R2: 0.279

# R2 is 49%

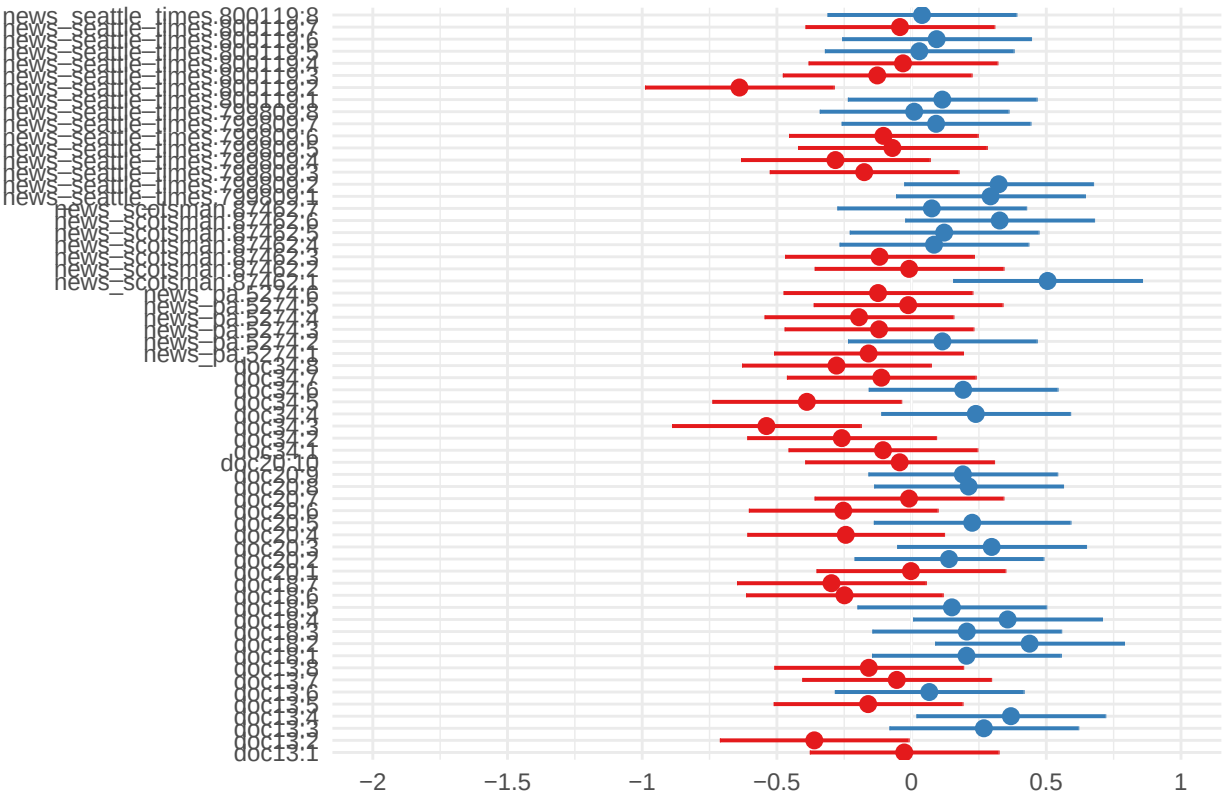
random_effects <- ranef(model_hyb3)
plot_model(model_hyb3, type = "re", grid = FALSE)[[2]] +
  theme_minimal() +
  labs(title = "Random Effects: PET (Translators)")
```

Random Effects: PET (Translators)



```
plot_model(model_hyb3, type = "re", grid = FALSE)[[1]] +  
  theme_minimal() +  
  labs(title = "Random Effects: Sentence ID")
```

Random Effects: Sentence ID



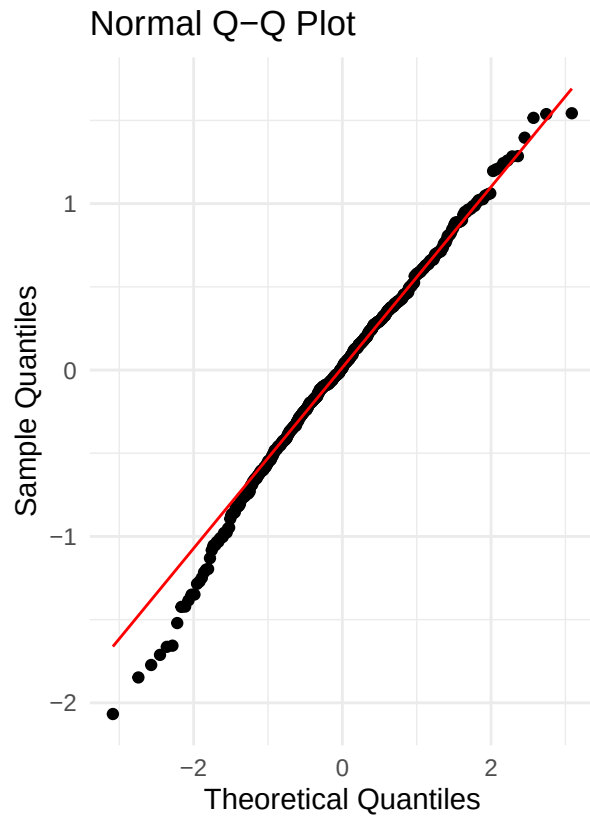
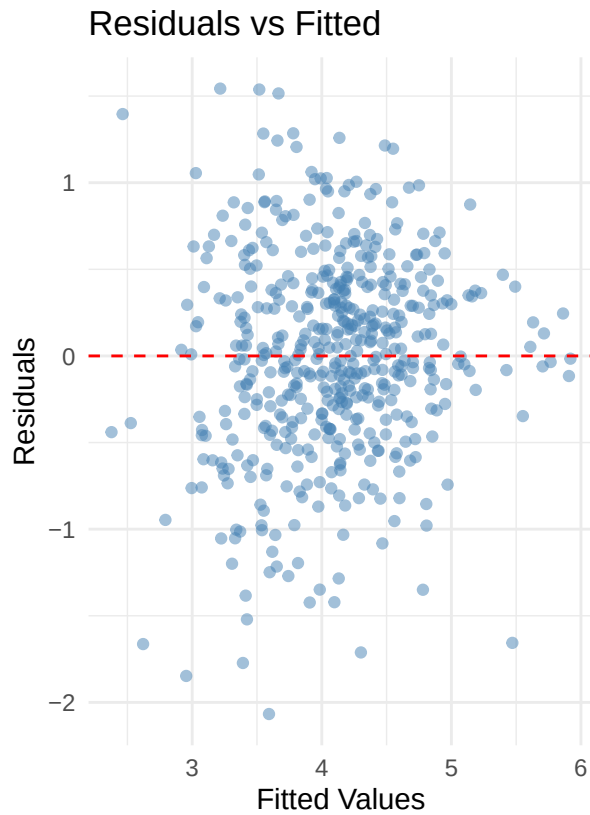
Assumptions Check

```
diag_data <- data.frame(
  Fitted = fitted(model_hyb3),
  Residuals = residuals(model_hyb3)
)

d1 <- ggplot(diag_data, aes(x = Fitted, y = Residuals)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted",
       x = "Fitted Values",
       y = "Residuals") +
  theme_minimal()

d2 <- ggplot(diag_data, aes(sample = Residuals)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(title = "Normal Q-Q Plot",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles") +
  theme_minimal()

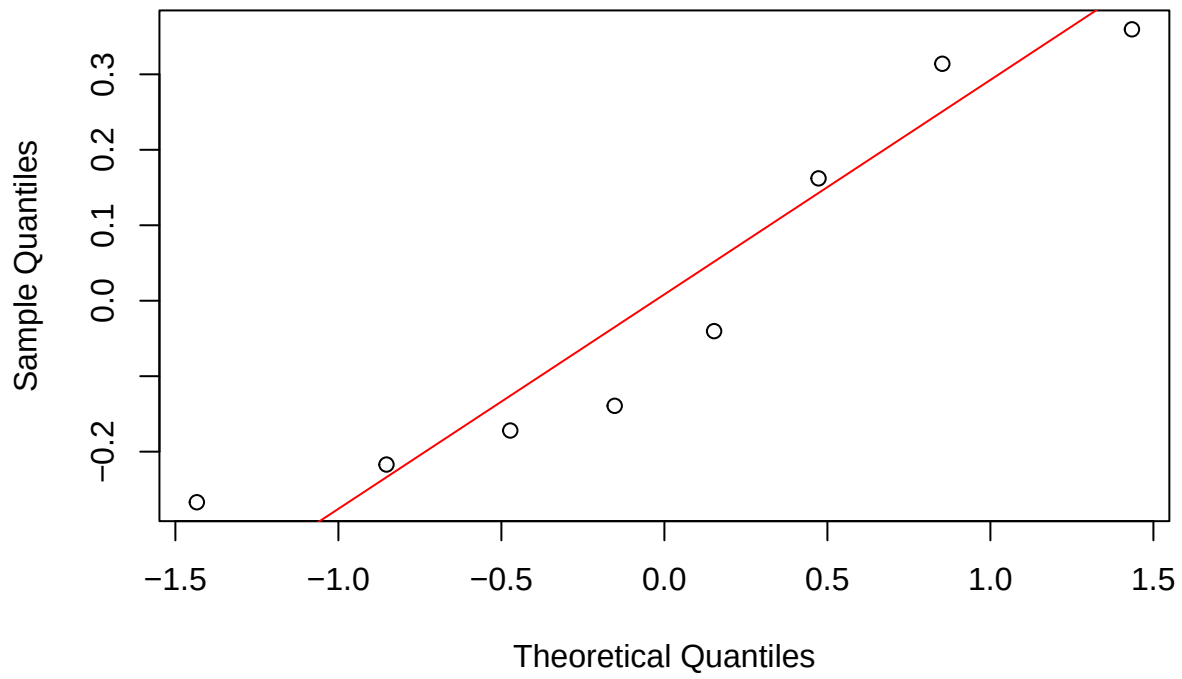
d1+d2
```



```
# residuals are normally distributed
```

```
pet_re <- ranef(model_hyb3)$PET[,1]  
qqnorm(pet_re, main = "Q-Q Plot: Random Effects (PET)")  
qqline(pet_re, col = "red")
```

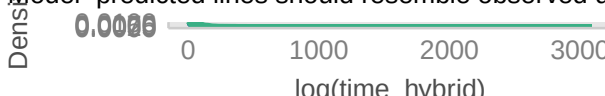
Q-Q Plot: Random Effects (PET)



```
check_model(model_hyb3) # The high VIF does not distort the model's standard errors or the significance
```

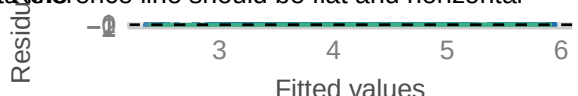
Posterior Predictive Check

Model-predicted lines should resemble observed data



Linearity

Reference line should be flat and horizontal



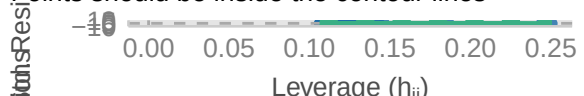
Homogeneity of Variance

Reference line should be flat and horizontal



Influential Observations

Points should be inside the contour lines



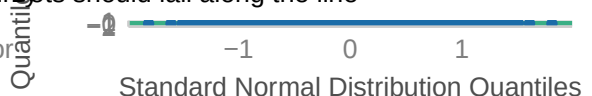
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



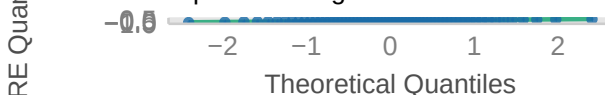
Normality of Residuals

Points should fall along the line



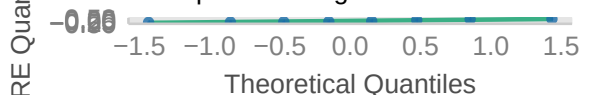
Normality of Random Effects (text_name: Sent_id)

Points should be plotted along the line



Normality of Random Effects (PET)

Points should be plotted along the line



Limit of this analysis is that the newly imputed times are measured differently than the originals they may be affected by external factors, such as translator taking breaks, and longer time to actually start the task.

Contrast Analysis - Switch of Baseline

```
data_clean2$condition <- relevel(as.factor(data_clean2$condition), ref = "2")
model_hyb3 <- lmer(log(time_hybrid) ~
  condition * domain + num_characters + num_minor + num_major + MT_Quality +
  (1 | PET) + (1 | text_name : Sent_id),
  data = data_clean2,
  REML = TRUE)
summary(model_hyb3)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(time_hybrid) ~ condition * domain + num_characters + num_minor +
## num_major + MT_Quality + (1 | PET) + (1 | text_name:Sent_id)
## Data: data_clean2
##
## REML criterion at convergence: 1051
##
## Scaled residuals:
##    Min      1Q  Median      3Q     Max
## -3.3263 -0.5659  0.0167  0.6132  2.4830
```

```

##
## Random effects:
##   Groups          Name          Variance Std.Dev.
## text_name:Sent_id (Intercept) 0.08849  0.2975
## PET                (Intercept) 0.06643  0.2577
## Residual                      0.38621  0.6215
## Number of obs: 493, groups: text_name:Sent_id, 62; PET, 8
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    3.365e+00  2.006e-01  9.932e+01  16.777 < 2e-16 ***
## condition1     5.206e-01  1.377e-01  4.759e+02   3.782 0.000176 ***
## condition3     1.486e-01  1.177e-01  4.393e+02   1.263 0.207335
## condition4     1.989e-01  1.134e-01  4.343e+02   1.754 0.080059 .
## domainnews     3.662e-01  1.421e-01  2.204e+02   2.577 0.010613 *
## num_characters  3.851e-03  5.278e-04  6.494e+01   7.297 5.19e-10 ***
## num_minor      5.076e-02  2.348e-02  3.542e+02   2.162 0.031323 *
## num_major      8.088e-02  4.860e-02  3.579e+02   1.664 0.096955 .
## MT_Quality     -1.401e-01  4.545e-02  4.766e+02  -3.082 0.002175 **
## condition1:domainnews -4.487e-01  1.632e-01  4.273e+02  -2.749 0.006237 **
## condition3:domainnews -1.059e-01  1.649e-01  4.309e+02  -0.642 0.521006
## condition4:domainnews -2.915e-01  1.643e-01  4.306e+02  -1.775 0.076675 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndtn1 cndtn3 cndtn4 dmnwns nm_chr nm_mnr nm_mjr MT_Qlt
## condition1 -0.346
## condition3 -0.157 0.526
## condition4 -0.264 0.537 0.542
## domainnews -0.353 0.434 0.379 0.402
## num_chrctrs -0.326 -0.189 -0.077 -0.076 -0.156
## num_minor   -0.261 0.575 0.193 0.206 0.265 -0.302
## num_major   -0.211 0.358 0.184 0.203 -0.010 -0.166 0.319
## MT_Quality  -0.571 -0.061 -0.279 -0.116 0.052 0.018 0.008 0.069
## cndtn1:dmnn 0.224 -0.633 -0.339 -0.351 -0.609 0.056 -0.208 0.022 0.004
## cndtn3:dmnn 0.188 -0.374 -0.659 -0.353 -0.608 0.061 -0.226 0.050 0.066
## cndtn4:dmnn 0.209 -0.371 -0.341 -0.664 -0.612 0.059 -0.227 0.052 0.031
##      cndt1: cndt3:
## condition1
## condition3
## condition4
## domainnews
## num_chrctrs
## num_minor
## num_major
## MT_Quality
## cndtn1:dmnn
## cndtn3:dmnn 0.529
## cndtn4:dmnn 0.530 0.535

```

Improved Plots

```
my_colors <- c("medical" = "#2c3e50", "news" = "#e67e22") # Blu notte e Arancione
my_labels <- c("PE", "H-QE", "H-APE", "S-APE")
```

```
p11 <- plot(eff_int) +
  scale_x_continuous(breaks = 1:4, labels = my_labels) +
  scale_color_manual(values = my_colors) +
  labs(
    title = "Option 1: Min Threshold",
    x = "Condition",
    y = "Estimated Time (Seconds)",
    colour = "Domain"
  ) +
  theme_minimal()
```

```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

```
p22 <- plot(eff_int_hyb) +
  scale_x_continuous(breaks = 1:4, labels = my_labels) +
  scale_color_manual(values = my_colors) +
  labs(
    title = "Option 2: Screen Time",
    x = "Condition",
    y = "Estimated Time (Seconds)",
    colour = "Domain"
  ) +
  theme_minimal()
```

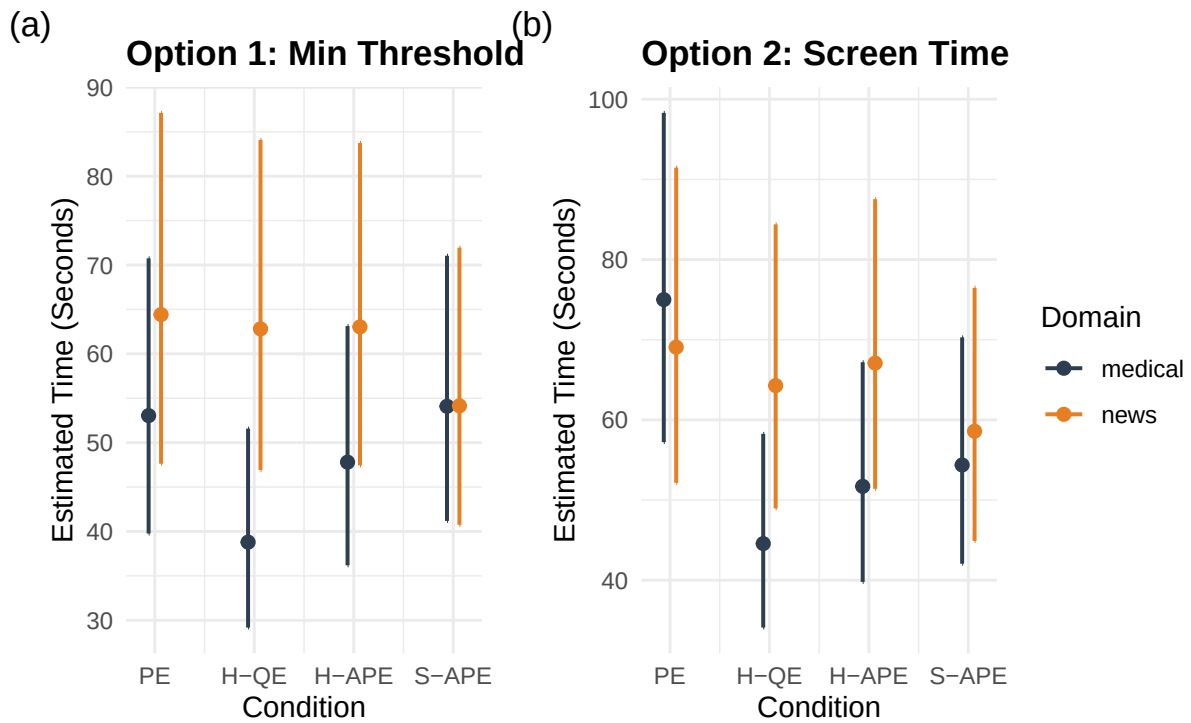
```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```

```
combined_plot <- p11 + p22 +
  plot_layout(guides = "collect") +
  plot_annotation(
    title = "Predicted Translation Time: Comparison of Imputation Scenarios",
    subtitle = "Values back-transformed from Log to Seconds",
    tag_levels = 'a',
    tag_prefix = '(',
    tag_suffix = ')'
  ) &
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )
```

```
combined_plot
```

Predicted Translation Time: Comparison of Imputation Scenarios

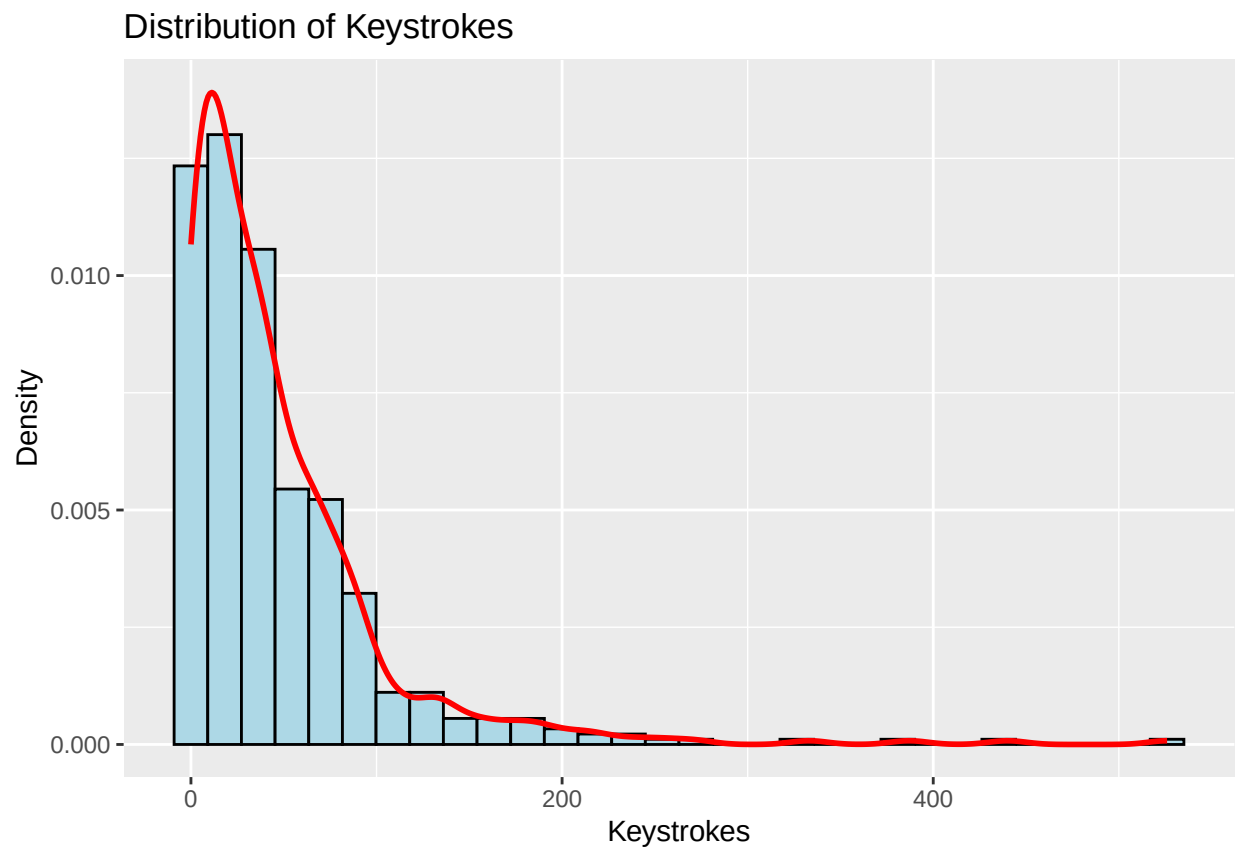
Values back-transformed from Log to Seconds



Model Building - Keystrokes

Analysis of outcome variable distribution

```
ggplot(data, aes(x = keystrokes)) +  
  geom_histogram(aes(y = ..density..),  
    bins = 30,  
    fill = "lightblue",  
    color = "black") +  
  geom_density(color = "red", size = 1) +  
  labs(title = "Distribution of Keystrokes",  
    x = "Keystrokes",  
    y = "Density")
```



Tweedie GLMM

The Tweedie distribution is a member of the Exponential Dispersion Family. Here, it is specifically a Compound Poisson-Gamma distribution.

It acts as a single mathematical framework that handles two processes at once: 1) The Discrete Part (Poisson) models the probability of an event occurring (e.g., whether the translator chooses to edit the sentence or leave it at 0 keystrokes); The Continuous Part (Gamma) models the magnitude of the event (e.g., if they do edit, how many keystrokes are required)

```
full_model <- glmmTMB(keystrokes ~ condition * domain + num_characters + num_minor + num_major + MT_Quality +
  (1 | PET) + (1 | text_name) + (1 | Sent_id),
  family = tweedie(), data = data, REML = FALSE)
```

```
## Warning in (function (start, objective, gradient = NULL, hessian = NULL, :
## NA/NaN function evaluation
```

```
summary(full_model)
```

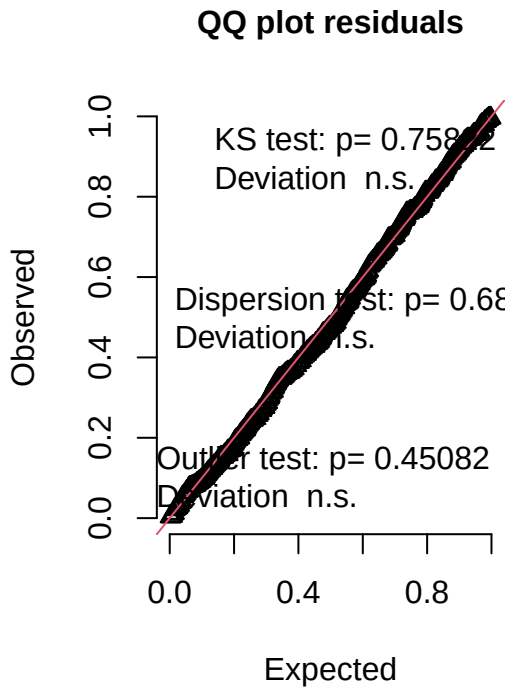
```
## Family: tweedie ( log )
## Formula:
## keystrokes ~ condition * domain + num_characters + num_minor +
##   num_major + MT_Quality + Difficulty + (1 | PET) + (1 | text_name) +
##   (1 | Sent_id)
## Data: data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  4540.6   4616.3   -2252.3    4504.6      478
##
## Random effects:
##
## Conditional model:
##   Groups   Name      Variance Std.Dev.
##   PET      (Intercept) 0.22448  0.4738
##   text_name (Intercept) 0.07371  0.2715
##   Sent_id   (Intercept) 0.01331  0.1154
## Number of obs: 496, groups: PET, 8; text_name, 8; Sent_id, 10
##
## Dispersion parameter for tweedie family (): 5.07
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.1923025  0.3131078  10.196 < 2e-16 ***
## condition2       -0.2059139  0.1724809  -1.194 0.232542
## condition3       -0.0322123  0.1629450  -0.198 0.843289
## condition4       -0.1364385  0.1593411  -0.856 0.391850
## domainnews       -0.0585156  0.2463233  -0.238 0.812226
## num_characters    0.0035341  0.0004367   8.092 5.85e-16 ***
## num_minor        0.0615679  0.0238404   2.583 0.009809 **
## num_major        0.0762211  0.0462809   1.647 0.099574 .
## MT_Quality       -0.2096219  0.0622165  -3.369 0.000754 ***
## Difficulty       0.0871202  0.0546948   1.593 0.111196
## condition2:domainnews 0.2543400  0.2077050   1.225 0.220754
## condition3:domainnews 0.1797527  0.2036987   0.882 0.377537
## condition4:domainnews 0.0474774  0.2064133   0.230 0.818083
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
full_model$sdr$pdHess
```

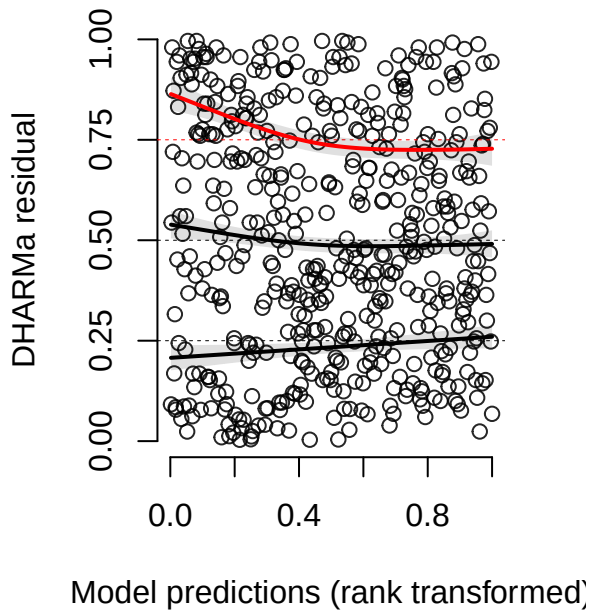
```
## [1] TRUE
```

```
res <- simulateResiduals(full_model)
plot(res)
```

DHARMA residual



DHARMA residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



```
# A more simplified model
refined_model <- glmmTMB(keystrokes ~ condition * domain +
  num_characters + num_minor + num_major +
  MT_Quality + Difficulty +
  (1 | PET) + (1 | text_name),
  family = tweedie(),
  data = data)
```

```
## Warning in (function (start, objective, gradient = NULL, hessian = NULL, :
## NA/NaN function evaluation
```

```
summary(refined_model)
```

```
## Family: tweedie ( log )
## Formula:
## keystrokes ~ condition * domain + num_characters + num_minor +
##   num_major + MT_Quality + Difficulty + (1 | PET) + (1 | text_name)
## Data: data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  4541.8   4613.3   -2253.9   4507.8      479
```

```
##
## Random effects:
##
## Conditional model:
## Groups      Name          Variance Std.Dev.
## PET          (Intercept) 0.2224   0.4716
## text_name    (Intercept) 0.0716   0.2676
## Number of obs: 496, groups: PET, 8; text_name, 8
##
## Dispersion parameter for tweedie family (): 5.08
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.2173385  0.3079667  10.447 < 2e-16 ***
## condition2       -0.2090329  0.1728145  -1.210 0.226440
## condition3       -0.0284707  0.1636996  -0.174 0.861928
## condition4       -0.1322924  0.1602395  -0.826 0.409036
## domainnews       -0.0433218  0.2445704  -0.177 0.859403
## num_characters    0.0033949  0.0003983   8.525 < 2e-16 ***
## num_minor         0.0587039  0.0233259   2.517 0.011847 *
## num_major         0.0970654  0.0447759   2.168 0.030174 *
## MT_Quality        -0.2068991  0.0626836  -3.301 0.000964 ***
## Difficulty         0.0846716  0.0550203   1.539 0.123825
## condition2:domainnews 0.2405202  0.2089792   1.151 0.249762
## condition3:domainnews 0.1911759  0.2051732   0.932 0.351451
## condition4:domainnews 0.0493353  0.2081055   0.237 0.812604
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

1. Scale variables to get numerical stability and Check stability

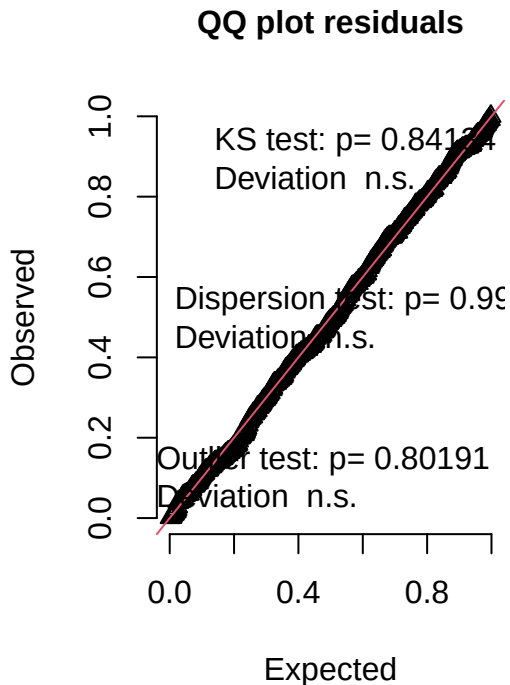
```
final_recovery_model <- glmmTMB(
  keystrokes ~ condition * domain +
    scale(num_characters) +
    scale(num_minor) +
    scale(num_major) +
    scale(MT_Quality) +
    scale(Difficulty) +
    (1 | PET) +
    (1 | text_name / Sent_id),
  family = tweedie(),
  data = data
)

final_recovery_model$sdr$pdHess
```

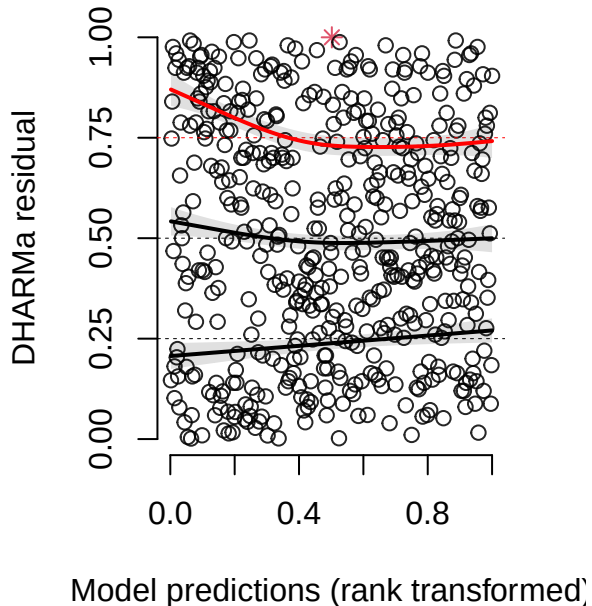
```
## [1] TRUE
```

```
res <- simulateResiduals(final_recovery_model)
plot(res)
```

DHARMa residual



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



```
summary(final_recovery_model)
```

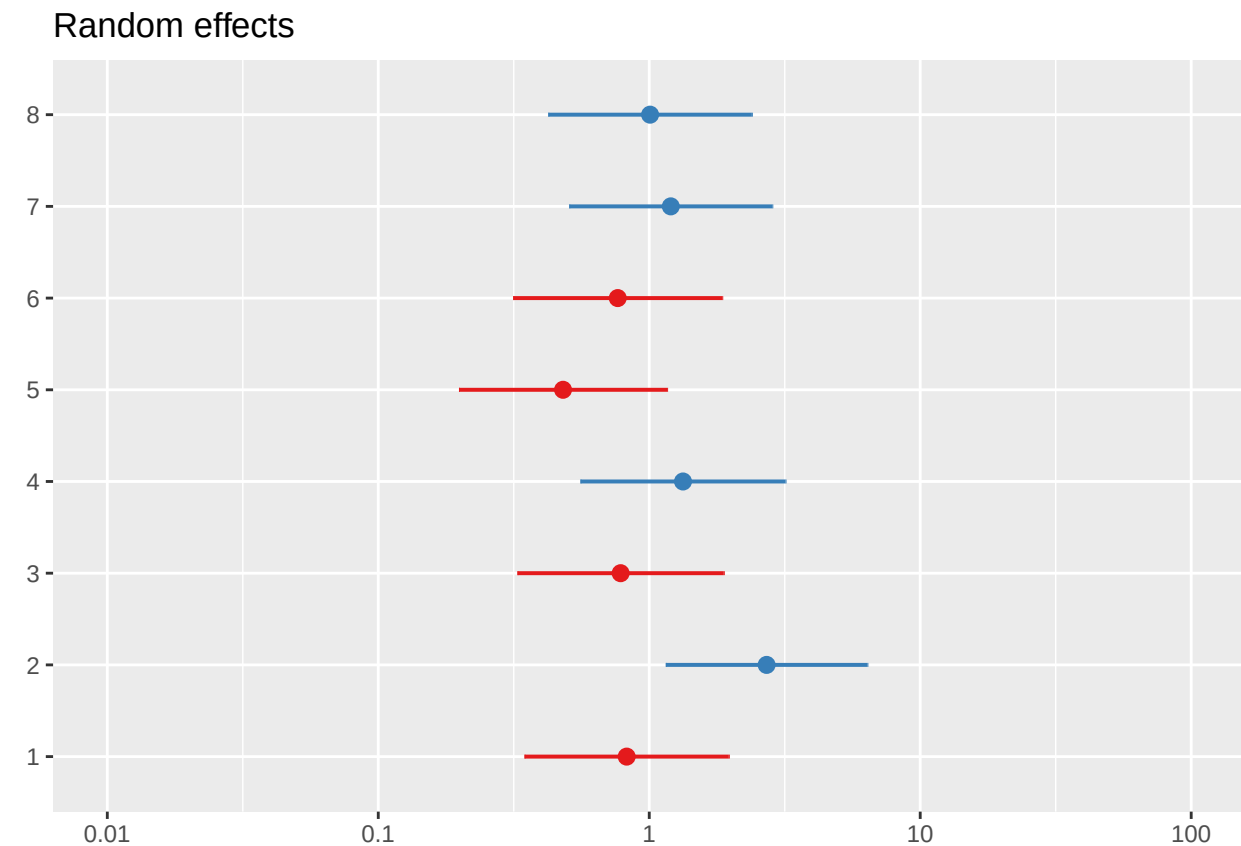
```
## Family: tweedie ( log )
## Formula:
## keystrokes ~ condition * domain + scale(num_characters) + scale(num_minor) +
##   scale(num_major) + scale(MT_Quality) + scale(Difficulty) +
##   (1 | PET) + (1 | text_name/Sent_id)
## Data: data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  4487.6   4563.3   -2225.8    4451.6     478
##
## Random effects:
##
## Conditional model:
## Groups          Name      Variance Std.Dev.
## PET              (Intercept) 0.24305  0.4930
## Sent_id:text_name (Intercept) 0.19266  0.4389
## text_name        (Intercept) 0.06014  0.2452
## Number of obs: 496, groups: PET, 8; Sent_id:text_name, 62; text_name, 8
##
## Dispersion parameter for tweedie family (): 4.7
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)          3.51186    0.25558  13.741 < 2e-16 ***
## condition2          -0.10173    0.17166   -0.593  0.553429
## condition3           0.04825    0.15716    0.307  0.758862
## condition4          -0.05910    0.15299   -0.386  0.699265
## domainnews          -0.07359    0.25094   -0.293  0.769312
## scale(num_characters) 0.36750    0.06919    5.311 1.09e-07 ***
## scale(num_minor)      0.05409    0.05554    0.974  0.330083
## scale(num_major)      0.06744    0.04258    1.584  0.113257
## scale(MT_Quality)     -0.16056    0.04422   -3.631  0.000282 ***
## scale(Difficulty)      0.11245    0.05597    2.009  0.044529 *
## condition2:domainnews 0.19956    0.19111    1.044  0.296385
## condition3:domainnews 0.19469    0.18488    1.053  0.292329
## condition4:domainnews 0.08572    0.18722    0.458  0.647066
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Visualising random intercepts

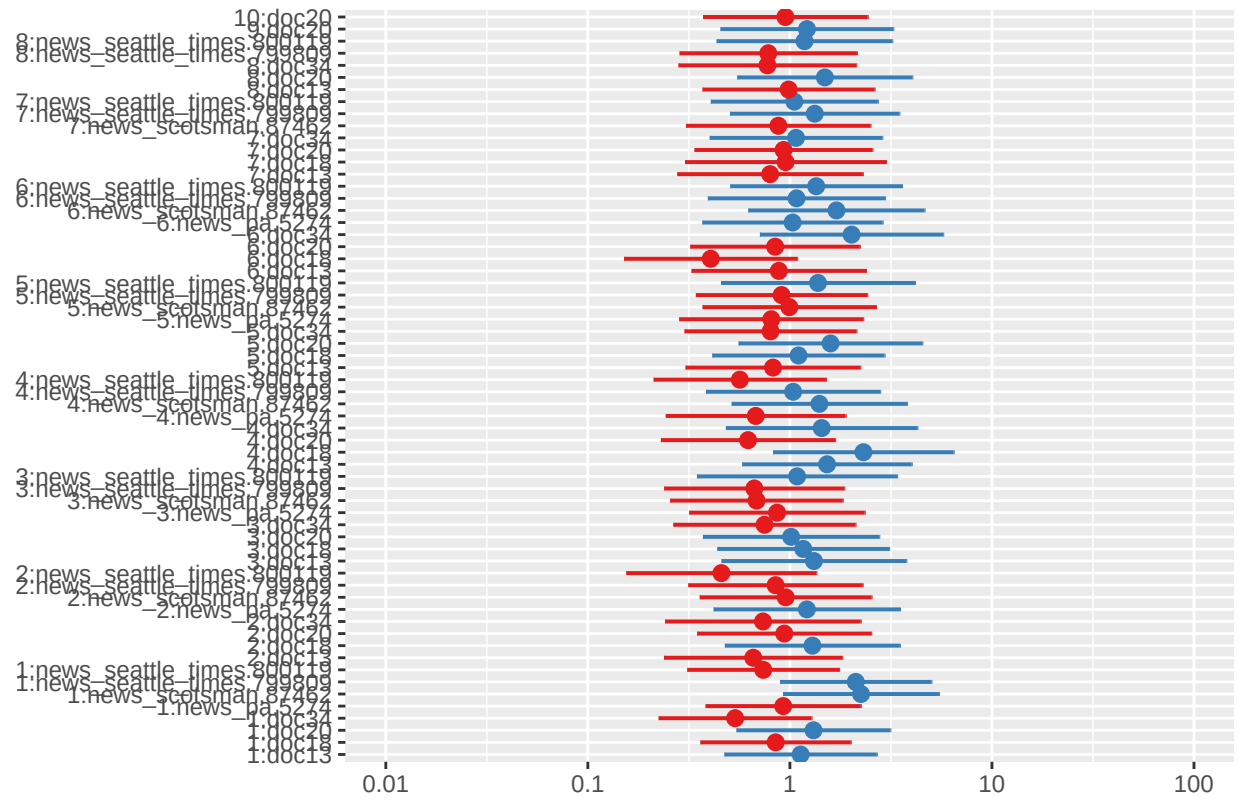
```
plot_model(final_recovery_model, type = "re")
```

```
## [[1]]
```



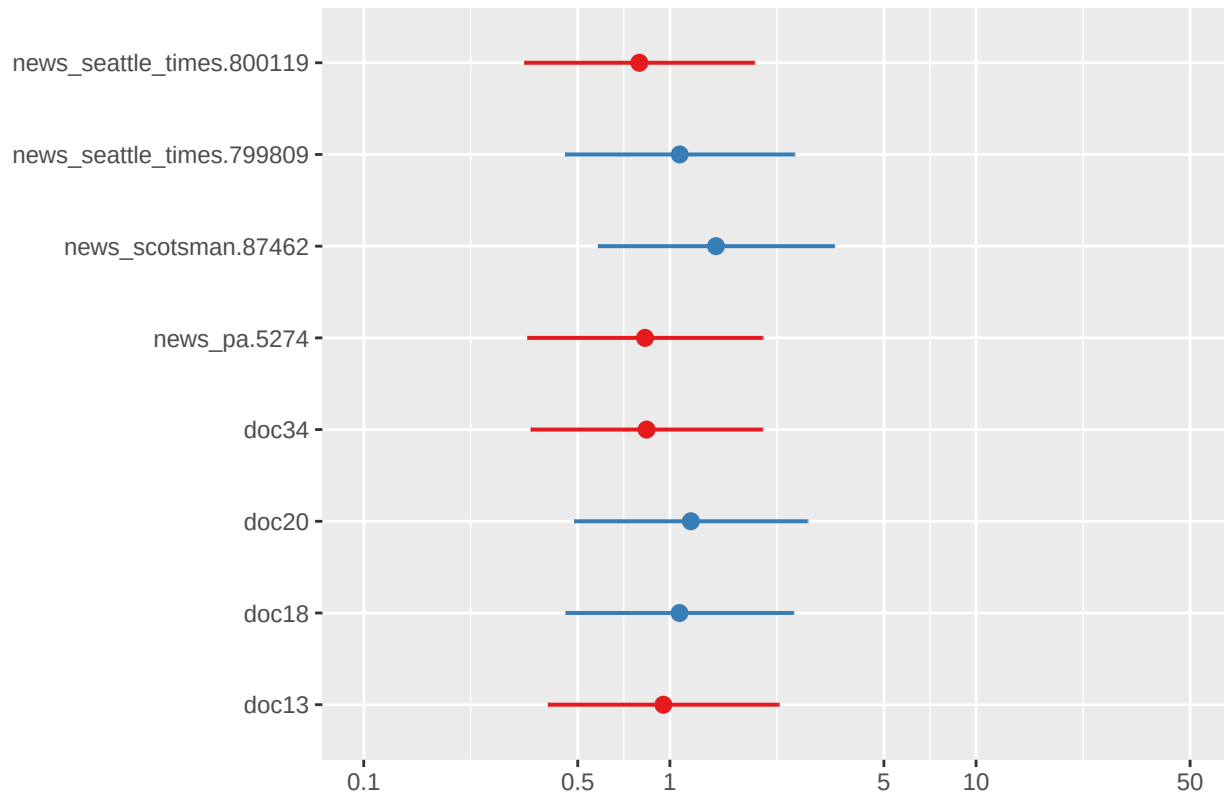
```
##
## [[2]]
```

Random effects



[[3]]

Random effects



Simplifying the model to make residuals homoschedastic

```
# Creating the simplified version: no interaction term
reduced_model <- update(final_recovery_model, . ~ . - condition:domain)
```

```
# Compare them
anova(final_recovery_model, reduced_model)
```

```
## Data: data
## Models:
## reduced_model: keystrokes ~ condition + domain + scale(num_characters) + scale(num_minor) + , zi=~0,
## reduced_model:      scale(num_major) + scale(MT_Quality) + scale(Difficulty) + , zi=~0, disp=~1
## reduced_model:      (1 | PET) + (1 | text_name/Sent_id), zi=~0, disp=~1
## final_recovery_model: keystrokes ~ condition * domain + scale(num_characters) + scale(num_minor) + ,
## final_recovery_model:      scale(num_major) + scale(MT_Quality) + scale(Difficulty) + , zi=~0, disp=~1
## final_recovery_model:      (1 | PET) + (1 | text_name/Sent_id), zi=~0, disp=~1
##
##           Df      AIC      BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
## reduced_model      15 4483.1 4546.2 -2226.6    4453.1
## final_recovery_model 18 4487.6 4563.3 -2225.8    4451.6 1.5615      3    0.6682
```

```
# no significant differences in predictive power and AIC and BIC are lower, we keep the reduced
```

```
# Check Variance Inflation Factors
```

```
performance::check_collinearity(final_recovery_model) # since the full model has interaction terms. VIF
```

```
## Model has interaction terms. VIFs might be inflated.
## Try to center the variables used for the interaction, or check
## multicollinearity among predictors of a model without interaction terms.

## # Check for Multicollinearity
##
## Low Correlation
##
##      Term   VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      domain 1.33 [ 1.22,  1.51]    1.15      0.75      [0.66, 0.82]
## scale(num_characters) 1.10 [ 1.04,  1.28]    1.05      0.91      [0.78, 0.97]
##      scale(num_minor) 2.34 [ 2.06,  2.70]    1.53      0.43      [0.37, 0.48]
##      scale(num_major) 1.61 [ 1.44,  1.83]    1.27      0.62      [0.55, 0.69]
##      scale(MT_Quality) 1.21 [ 1.12,  1.38]    1.10      0.82      [0.73, 0.89]
##      scale(Difficulty) 1.21 [ 1.11,  1.37]    1.10      0.83      [0.73, 0.90]
##
## High Correlation
##
##      Term   VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      condition 16.79 [14.25, 19.82]    4.10      0.06      [0.05, 0.07]
## condition:domain 11.57 [ 9.84, 13.63]    3.40      0.09      [0.07, 0.10]
```

performance::check_collinearity(reduced_model) *# here no highly correlated variables*

```
## # Check for Multicollinearity
##
## Low Correlation
##
##      Term   VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      condition 2.95 [2.57, 3.42]    1.72      0.34      [0.29, 0.39]
##      domain 1.05 [1.01, 1.33]    1.02      0.95      [0.75, 0.99]
## scale(num_characters) 1.09 [1.03, 1.28]    1.05      0.91      [0.78, 0.97]
##      scale(num_minor) 2.14 [1.89, 2.46]    1.46      0.47      [0.41, 0.53]
##      scale(num_major) 1.59 [1.43, 1.80]    1.26      0.63      [0.55, 0.70]
##      scale(MT_Quality) 1.18 [1.10, 1.35]    1.09      0.85      [0.74, 0.91]
##      scale(Difficulty) 1.17 [1.08, 1.33]    1.08      0.86      [0.75, 0.92]
```

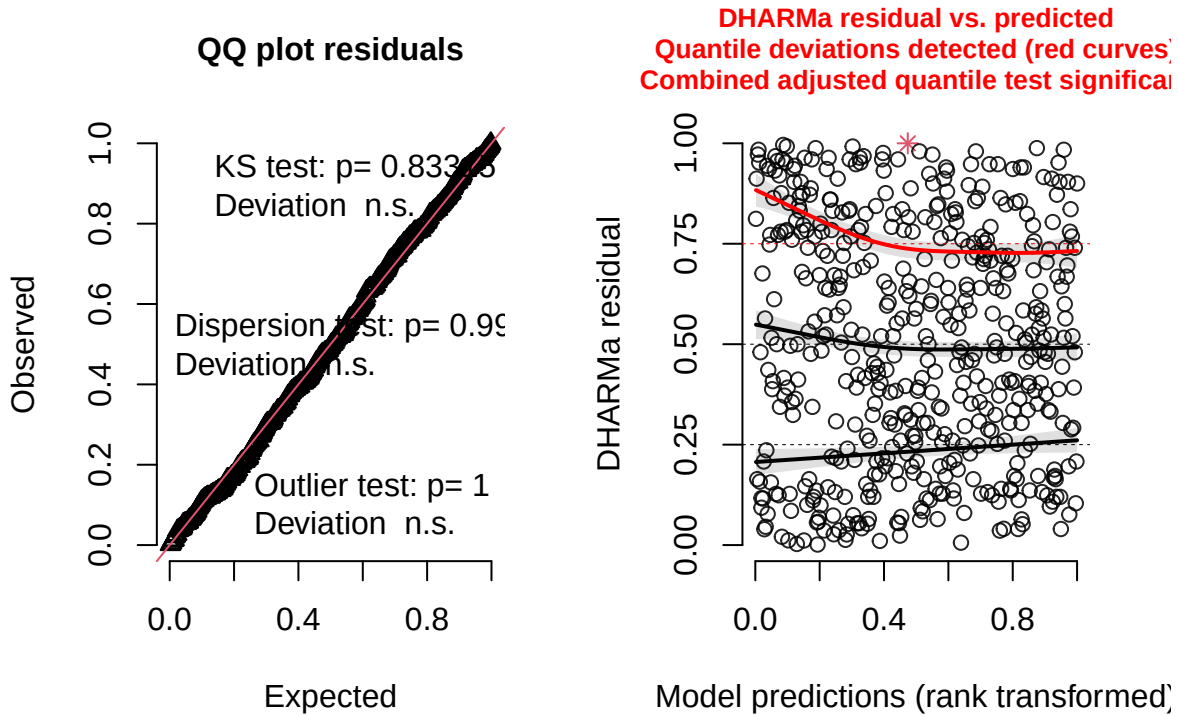
Check stability

```
final_recovery_model$sdr$pdHess
```

```
## [1] TRUE
```

```
res <- simulateResiduals(reduced_model)
plot(res)
```

DHARMa residual



Further Simplification

```
summary(reduced_model)
```

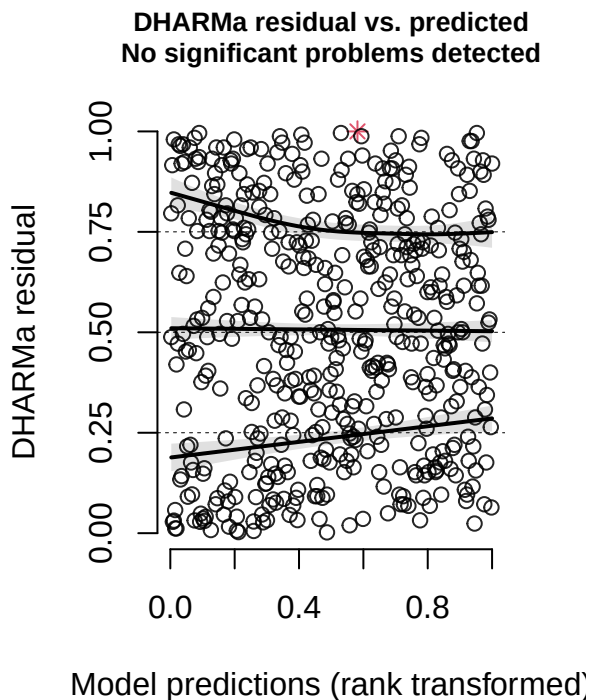
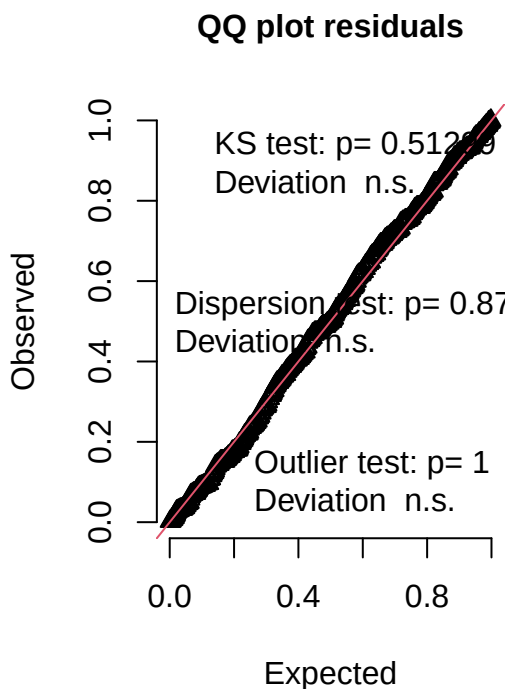
```
## Family: tweedie ( log )
## Formula:
## keystrokes ~ condition + domain + scale(num_characters) + scale(num_minor) +
##   scale(num_major) + scale(MT_Quality) + scale(Difficulty) +
##   (1 | PET) + (1 | text_name/Sent_id)
## Data: data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  4483.1   4546.2   -2226.6   4453.1     481
##
## Random effects:
##
## Conditional model:
## Groups      Name          Variance Std.Dev.
## PET          (Intercept)  0.24139  0.4913
## Sent_id:text_name (Intercept) 0.19421  0.4407
## text_name      (Intercept)  0.05942  0.2438
## Number of obs: 496, groups: PET, 8; Sent_id:text_name, 62; text_name, 8
##
## Dispersion parameter for tweedie family (): 4.71
##
## Conditional model:
```

```
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    3.4374626  0.2467990  13.928 < 2e-16 ***
## condition2     0.0105001  0.1316736   0.080 0.936442
## condition3     0.1644821  0.1225124   1.343 0.179410
## condition4     0.0001308  0.1220912   0.001 0.999145
## domainnews     0.0528307  0.2220476   0.238 0.811939
## scale(num_characters) 0.3696635  0.0693018   5.334 9.6e-08 ***
## scale(num_minor)    0.0440478  0.0532688   0.827 0.408296
## scale(num_major)    0.0680994  0.0424087   1.606 0.108320
## scale(MT_Quality)   -0.1621306  0.0437090  -3.709 0.000208 ***
## scale(Difficulty)    0.1057136  0.0550186   1.921 0.054679 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
reduced_model_noDiff <- update(reduced_model, . ~ . - scale(Difficulty))

res <- simulateResiduals(reduced_model_noDiff)
# Nella grafica base di R il titolo si assegna con 'main'
plot(res)
```

DHARMA residual



```
# now it works

summary(reduced_model_noDiff)

## Family: tweedie ( log )
```

```
## Formula:
## keystrokes ~ condition + domain + scale(num_characters) + scale(num_minor) +
##   scale(num_major) + scale(MT_Quality) + (1 | PET) + (1 | text_name/Sent_id)
## Data: data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##   4484.8   4543.7  -2228.4   4456.8      482
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## PET              (Intercept)  0.21555  0.4643
## Sent_id:text_name (Intercept)  0.19108  0.4371
## text_name        (Intercept)  0.05374  0.2318
## Number of obs: 496, groups: PET, 8; Sent_id:text_name, 62; text_name, 8
##
## Dispersion parameter for tweedie family (): 4.72
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    3.37090    0.23460  14.369 < 2e-16 ***
## condition2      0.05010    0.13052   0.384 0.701088
## condition3      0.19971    0.12176   1.640 0.100971
## condition4      0.04567    0.12028   0.380 0.704166
## domainnews      0.13091    0.21130   0.620 0.535580
## scale(num_characters) 0.36379    0.06910   5.265 1.4e-07 ***
## scale(num_minor)    0.04600    0.05342   0.861 0.389122
## scale(num_major)    0.06654    0.04252   1.565 0.117568
## scale(MT_Quality)   -0.14641    0.04308  -3.398 0.000678 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(reduced_model, reduced_model_noDiff)
```

```
## Data: data
## Models:
## reduced_model_noDiff: keystrokes ~ condition + domain + scale(num_characters) + scale(num_minor) + ,
## reduced_model_noDiff:   scale(num_major) + scale(MT_Quality) + (1 | PET) + (1 | text_name/Sent_id)
## reduced_model: keystrokes ~ condition + domain + scale(num_characters) + scale(num_minor) + , zi=~0,
## reduced_model:   scale(num_major) + scale(MT_Quality) + scale(Difficulty) + , zi=~0, disp=~1
## reduced_model:   (1 | PET) + (1 | text_name/Sent_id), zi=~0, disp=~1
##              Df      AIC      BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
## reduced_model_noDiff 14 4484.8 4543.7 -2228.4  4456.8
## reduced_model       15 4483.1 4546.2 -2226.6  4453.1 3.6909    1    0.05471
##
## reduced_model_noDiff
## reduced_model
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# clear indication to stick with the second reduced model (no difficulty)
```

Plotting the regressors' effects on keystrokes

```
# extract fixed effects table
coefs <- summary(reduced_model_noDiff)$coefficients$cond
df <- as.data.frame(coefs)

# add term names
df$term <- rownames(df)

# rename columns for convenience
df <- df %>%
  rename(
    estimate = Estimate,
    std.error = `Std. Error`,
    p.value = `Pr(>|z|)`
  )

# get confidence intervals
ci <- confint(reduced_model_noDiff, method = "Wald")

# keep only fixed effects
ci <- as.data.frame(ci)
ci$term <- rownames(ci)

df <- df %>%
  left_join(ci, by = "term") %>%
  rename(conf.low = `2.5 %`, conf.high = `97.5 %`)

# remove intercept
df <- df %>%
  filter(term != "(Intercept)")

# relabel terms
df <- df %>%
  mutate(
    term = recode(term,
      "condition2" = "H-QE",
      "condition3" = "H-APE",
      "condition4" = "S-APE",
      "domainnews" = "Domain: News",
      "scale(num_characters)" = "Text length",
      "scale(num_minor)" = "Minor errors",
      "scale(num_major)" = "Major errors",
      "scale(MT_Quality)" = "MT quality"
    ),
    significance = ifelse(p.value < 0.05, "Significant", "Not significant")
  )

df <- df %>%
  mutate(significant_flag = p.value < 0.05) %>%
  arrange(desc(significant_flag), desc(abs(estimate))) %>%
  mutate(term = factor(term, levels = rev(term)))
```

```

p <- ggplot(df, aes(x = estimate, y = term)) +
  geom_vline(xintercept = 0, linetype = "dashed", linewidth = 0.6, color = "gray50") +

  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high),
    height = 0.2, linewidth = 0.8, color = "gray40") +

  geom_point(aes(shape = as.factor(!significant_flag)),
    size = 3, stroke = 1.1, color = "black") +

  scale_shape_manual(
    name = NULL,
    values = c(16, 1),
    labels = c("Significant", "Not Significant")
  ) +

  labs(
    title = "Effects on Keystrokes",
    subtitle = "Estimates with 95% confidence intervals (log scale)",
    x = "Effect size",
    y = NULL
  ) +

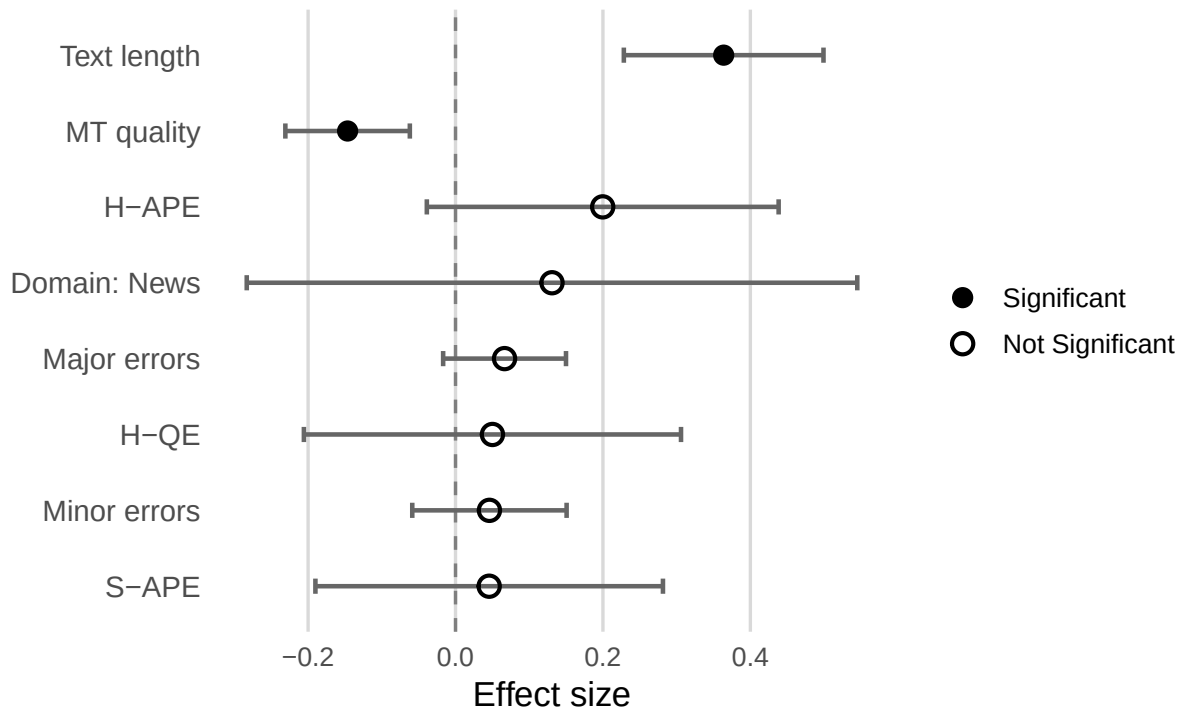
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    plot.subtitle = element_text(color = "gray30"),
    axis.text.y = element_text(size = 11),
    axis.text.x = element_text(size = 10),
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_line(color = "gray85"),
    plot.margin = margin(10, 15, 10, 10),
    legend.position = "right"
  )

```

p

Effects on Keystrokes

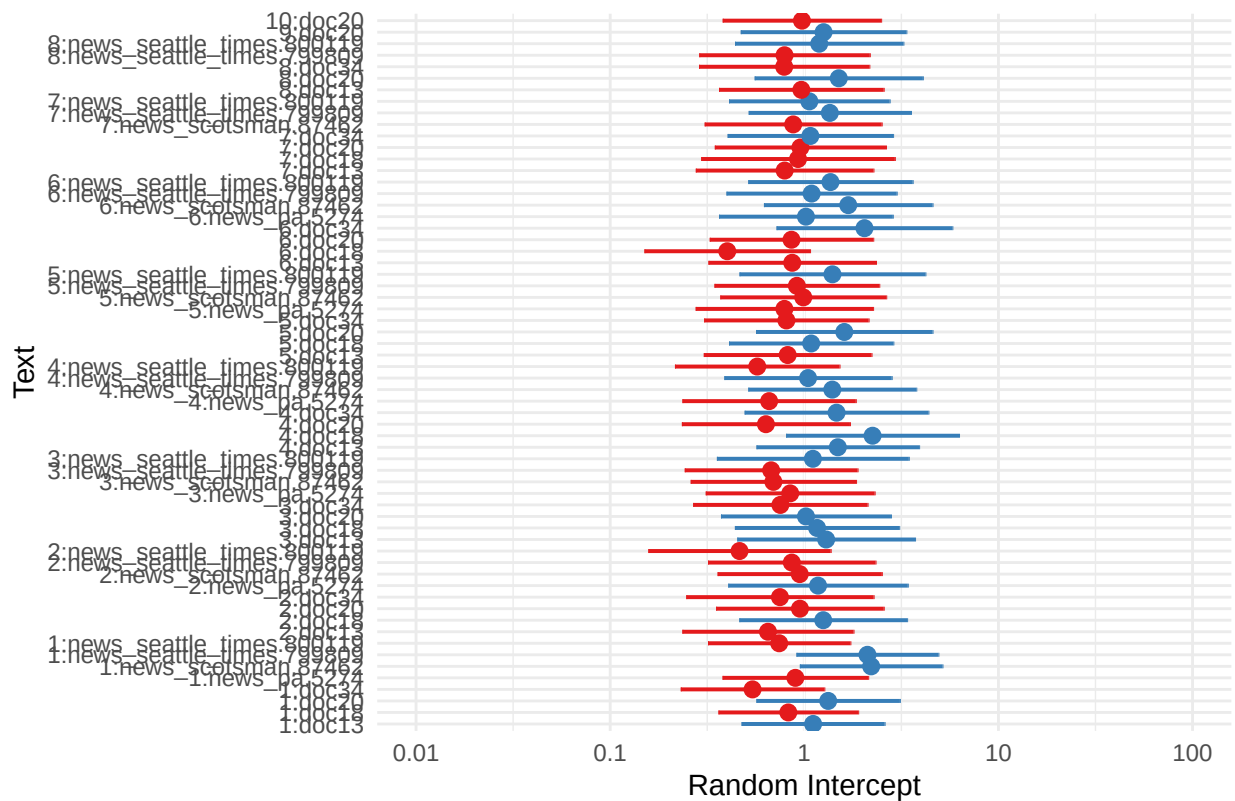
Estimates with 95% confidence intervals (log scale)



Random Intercepts plots

```
sjPlot::plot_model(reduced_model_noDiff, type = "re", grid = FALSE)[[2]] +  
  theme_minimal() +  
  labs(x = "Text", y = "Random Intercept", title = "Random Effects: Texts")
```

Random Effects: Texts



Comparing final model Vs other models

```
# Defining main formula
main_formula <- keystrokes ~ condition + domain + scale(num_characters) +
  scale(num_minor) + scale(num_major) + scale(MT_Quality) +
  (1 | PET) + (1 | text_name / Sent_id)

# Current Tweedie Model
mod_tweedie <- glmmTMB(main_formula,
  family = tweedie(link = "log"),
  data = data)

# Zero-Inflated Negative Binomial
# (Assumes some zeros come from a separate always-zero process)
mod_zinb <- glmmTMB(main_formula,
  ziformula = ~1,
  family = nbinom2,
  data = data)

# 3. Hurdle Model (Truncated Negative Binomial)
# (Assumes a process: 0 vs >0, then counts for the rest)
mod_hurdle <- glmmTMB(main_formula,
  ziformula = ~.,
```

```

        family = truncated_nbinom2,
        data = data)

comparison <- compare_performance(mod_tweedie, mod_zinb, mod_hurdle, metrics = "common")

## Random effect variances not available. Returned R2 does not account for random effects.

print(comparison)

## # Comparison of Model Performance Indices
##
## Name          |   Model | AIC (weights) | BIC (weights) | R2 (cond.)
## -----
## mod_tweedie | glmmTMB | 4484.8 (0.225) | 4543.7 (>.999) |      0.634
## mod_zinb    | glmmTMB | 4537.3 (<.001) | 4596.2 (<.001) |      0.966
## mod_hurdle  | glmmTMB | 4482.4 (0.775) | 4587.5 (<.001) |
##
## Name          | R2 (marg.) | RMSE | ICC
## -----
## mod_tweedie |      0.191 | 31.539 | 0.548
## mod_zinb    |      0.241 | 35.284 | 0.956
## mod_hurdle  |      0.263 | 34.485 |

```

The 'rho' (power parameter)

```

glmmTMB::family_params(mod_tweedie)

## Tweedie power
##      1.396782

```

Even though the Hurdle model has a slightly lower AIC, We would argue for the Tweedie model for three reasons: 1) The BIC weight for the Tweedie model is $>.999$. This is a massive statistical signal that the Tweedie model is the most parsimonious. 2) The Tweedie model has the lowest RMSE (31.539). This means its actual predictions are closer to actual data points than the other two models. 3) In the Tweedie model, the Itra Class Correlation (ICC) is 0.548, while in the ZI model, it's 0.956, which is high and might suggests that the model is struggling to separate the random effects from the zero-inflation logic.